

CFA Level 1 Notes

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1 Quantitative Methods

1.1 Introduction

Means. The arithmetic mean return assumes that the amount invested at the beginning of each period is the same, i.e. equally-weighted.

In general, the arithmetic return is biased upward unless each of the underlying holding period returns are equal. The bias in arithmetic mean returns is particularly severe if holding period returns are a mix of both positive and negative returns because the arithmetic mean will be skewed by the high value but the geometric mean will balance out high numbers using the low values. Geometric averages are convex (below the average) whether rising or falling.

Harmonic mean. E.g. P/E ratio of company A is 100 and for company B 5, calculate the P/E ratio of a fund with 10 percent of A and 90 percent of B:

$$HA = \frac{10 + 90 \text{ dollars}}{\frac{10 \text{ dollars}}{100} + \frac{90 \text{ dollars}}{5}}$$

See the AM-GM inequality, which only works for non-negative reals. If there are only two objects, then the means are related by:

$$\sqrt{AM * HM} = GM$$

Unless all the observations in a dataset are the same value,

$$AM > GM > HM$$

The harmonic mean is smallest because it apportions low weights to large ratios. Reduces the effect of outliers because each denominator unit (such as a stock) is weighted equally, and those with high ratios will have low numbers of units.

Holding period return = the percentage increase throughout the period(s).

Money-weighted rate of return is just the IRR over the whole period, which weights net cash inflows and outflows. It weights more for when there is more invested. However, it's hard to use it to compare different portfolio products, even two investors of the same stock if they invested different amounts at different times. To calculate, don't tabulate all the investment gains and losses; just tabulate what went into the fund and what came out of it for the cashflows.

Start of Year 1	One share purchased at \$100
End of Year 1	\$5.00 dividend/share paid and one additional share purchased at \$125
End of Year 2	\$5.00 dividend/share paid and both shares are sold for \$140 per share

The money-weighted rate of return (no dividend reinvestment) is

$$-100 + \frac{-120}{(1 + IRR)} + \frac{290}{(1 + IRR)^2} = 0$$

and solve for IRR. The calculation will be different with dividends. The time-weighted rate of return is the geometric average of

$$r_1 = \frac{5 + (125 - 100)}{100}$$

and throw away the dividend

$$r_2 = \frac{5 \times 2 + (2 \times 140 - 2 \times 125)}{250}$$

but the calculation can be different with the dividend. Typically the dividend is not reinvested.

Time-weighted returns. Follows 1 dollar through the fund. Even if dividends are not reinvested, and calculations don't include dividends being reinvested, they are conceptually reinvested. Break into subperiods and retrieve the percentage change by removing effects from cash inflows and outflows. String together the total by multiplication, then can take roots if want to calculate per period. More comparable and therefore meaningful.

There is a continuously-compounded rate r associated with the holding period R :

$$e^r = \lim_{n \rightarrow \infty} \prod \left(1 + \frac{r}{n}\right)^n = 1 + R = \frac{P_T}{P_0}$$

$$e^{rT} = (1 + R)^T$$

where the T is the period consistent with r , e.g. if rate is 0.5 percent per year and the period is 3 months, then $T = 0.25$. The rate r is:

$$r = \ln(1 + R) = \ln\left(\frac{P_T}{P_0}\right)$$

So the r is the rate if the holding period to get R is infinitely decomposed. Further,

$$\ln\left(\frac{P_T}{P_0}\right) = \ln\left(\frac{P_T}{P_3} \frac{P_3}{P_2} \frac{P_2}{P_1} \frac{P_1}{P_0}\right) = r_T + r_3 + r_2 + r_1$$

Gross fund returns include trading expenses but not management fees.

Taxes are paid before adjusting for inflation, not the other way around, as normally tax rates are expressed as nominal rates.

Leveraged returns. Use futures contracts. Or borrow money - but then need to subtract interest on the leveraged return.

$$R_{Leveraged} = \left(\frac{R_P(V_E + V_B) - (V_B r_D)}{V_E} \right) = R_P + \frac{V_B}{V_E}(R_P - r_D)$$

The latter form shows that the leverage really depends on whether $R_P > r_D$.

1.2 Time value of money

Coupon bond implied return. The YTM assumes an investor expects to receive all promised cash flows through maturity and reinvest any cash received at the same YTM. The YTM is basically calculating the IRR for a given bond's price (as distinct from its par value). For coupon bonds, this involves periodic interest payments only, while for level payment instruments such as mortgages, the calculation assumes both interest and amortized principal may be invested at the same rate. YTM has no closed-form solution, so iterate or use a calculator. Note: as the YTM changes throughout time, be careful to distinguish whether the original payments are reinvested at the original YTM or not.

Perpetuities $\frac{PMT}{r}$, where the perpetuity starts in one period (i.e., it doesn't start now). Note the coupon rate is usually the annualized rate already, if there are more periodic payments.

Annuities. Fixed constant payments, comprising changing proportions of interest and principal, e.g. mortgages. The number of periods of the annuity t , for an ORDINARY SIMPLE annuity only, where payments begin in one period (i.e. it doesn't start now):

$$A = \frac{r(PV)}{1 - \frac{1}{(1+r)^t}}$$

Note:

$$PV = \frac{A}{r} \left(1 - \frac{1}{(1+r)^t} \right) = \text{Perpetuity} \left(1 - \frac{1}{(1+r)^t} \right)$$

Multiply each side by $(1+r)^n$ to get FV:

$$PV(1+r)^n = \frac{A}{r}(1+r)^n \left(1 - \frac{1}{(1+r)^n} \right)$$

$$FV = \frac{A}{r} ((1+r)^n - 1)$$

Annuity types. Ordinary annuity = payment at end of the regular period (e.g. end of the month). Simple annuity = payment frequency coincides with compounding frequency; otherwise, it's a general annuity.

Equities. For constant dividend growth, where dividend begins in one period (i.e. it doesn't start now):

$$PV_t = \frac{D_t(1+g)}{r-g} = \frac{D_{t+1}}{r-g}$$

For two-stage dividend growth (short-term and long-term), note the n exponent instead of $n = 1$ in the last term because the $\frac{D_{t+1}}{r-g}$ discounts back to $t = 0$ instead of $t = 1$:

$$PV_t = \sum_{i=1}^n \frac{D_t(1+g_s)^i}{(1+r)^i} + \sum_{j=n+1}^{\infty} \frac{D_{t+n}(1+g_l)^j}{(1+r)^j} = \sum_{i=1}^n \frac{D_t(1+g_s)^i}{(1+r)^i} + \frac{\frac{D_t(1+g_s)^n(1+g_l)}{r-g_l}}{(1+r)^n}$$

where the numerator of the very RHS term is the terminal value.

Equity implied returns, i.e. with growing dividends. Since

$$PV = \frac{D_{t+1}}{r-g}$$

Then

$$r = \frac{D_{t+1}}{PV} + g$$

Price-to-earnings ratio (P/E). This is actually price per share divided by earnings per share. The above equations can be modified to get the P/E ratio:

$$Price = P_t = PV_t = \frac{D_t(1+g)}{r-g}$$

$$\frac{P_t}{E_t} = \frac{\frac{D_t(1+g)}{E_t}}{r-g}$$

where $\frac{P_t}{E_t}$ is the P/E ratio and $\frac{D_t}{E_t}$ is the dividend payout ratio. The equation can be instead divided by the forward earnings E_{t+1}

$$\frac{P_t}{E_{t+1}} = \frac{\frac{D_{t+1}}{E_{t+1}}}{r-g} = \frac{\text{Dividend payout ratio}}{r-g}$$

where the first term is the forward P/E ratio and the right-hand numerator is the expected dividend payout ratio.

Cash flow additivity. To compare two cash flows, can align the cash flows by time and then add/subtract them. Then get one cash flow, and find NPV to

determine which is better.

Binomial option pricing, which is applied cash flow additivity. In a call option, the investor takes a proportion of the underlying asset (the hedge ratio, h) and also SELLS (i.e. loses in value) a call option at price c_0 :

$$V_{call} = hU - c_0$$

which is then equated to the discounted future cashflows. The high and low case cashflows will be equal at the hedge ratio: from the high case, subtract the call that will be claimed by the buyer; the low case is as-is.

In a put option, the investor takes a proportion of the underlying asset and BUYS a put option p_0 :

$$V_{put} = hU + p_0$$

The high case is as-is, while the put claim is added to the low case.

The hedge ratio is calculated by looking at the binomial possibilities and equating the two branch returns. If exercise were certain, the investor would borrow $X(1+r)^{-T}$ at inception in the call, and lend $X(1+r)^{-T}$ for the put. However, since it is uncertain, the investor instead borrows or lends a proportion of $X(1+r)^{-T}$ based on the likelihood of exercise at time T , which is closely associated with the moneyness of an option. Then the return is discounted and equated to the V above.

1.3 Statistical measures of asset returns

Trimmed mean 5% means trim the top and bottom 2.5 percentiles. Winsorized 95% means replace the top and bottom 2.5 percentiles with the nearest values. In a continuous distribution, there is no value for the percentile, there are only ranges. Be careful when noting which bin extremum number to use - the highest or the lowest.

The interquartile range is the second and third quartiles. The box and whisker plot shows the IQR as the box. The whiskers can be the range extrema values or 1.5 times the IQR.

Mean absolute deviation. The sum of absolute deviations around the mean. Note MAD denominator is n , not $n-1$!

$$MAD = \frac{\sum |x - \bar{x}|}{n}$$

Standard deviation, semideviation, etc. are usually SAMPLE versions unless otherwise noted.

Sample covariance where there are n paired observations:

$$Cov(x, y) = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

Note the DOFs are $n-1$, not $n-2$ even though there are both $\bar{x}, \bar{y}$. There are ways to look at this. (1) One Constraint, Not Two: While the covariance formula uses both the mean of x and the mean of y , these are not entirely independent constraints. Calculating one mean indirectly influences the other. Essentially, we lose one degree of freedom for each mean calculation, but the second mean calculation is somewhat dependent on the first. (2) Visualizing Degrees of Freedom: Imagine a scatter plot of your x and y data. To calculate covariance, we center the data around the means. This effectively fixes the center of the data cloud. Now, only $n-1$ points can move freely; the last point is constrained to maintain the fixed center.

Sample target semideviation.

$$S = \sqrt{\sum_{X < B} \frac{(X_i - B)^2}{n-1}}$$

Note: n of the entire set, not just those which are below the target value

Coefficient of variation.

$$CV = \frac{s}{\bar{X}}$$

where s is the standard deviation. If $\bar{X} < 0$ then the CV does not have direct meaning.

Sample skewness for $n > 100$:

$$Skewness \approx \frac{1}{n} \frac{(X - \bar{X})^3}{s^3}$$

where s is the sample standard deviation.

Kurtosis. A normal distribution has a kurtosis of 3.0, a leptokurtic (thin-tailed) distribution has kurtosis greater than 3.0 and a platykurtic (fat-tailed) distribution is less than 3.0. Excess kurtosis is relative to the normal or 3.0. Sample excess kurtosis is:

$$K_E \approx \left(\frac{1}{n} \frac{\sum (X - \bar{X})^4}{s^4} \right) - 3$$

Sample covariance:

$$s_{XY} = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n-1}$$

Sample correlation coefficient which ranges from -1 to 1 and indicates the strength and direction of correlation:

$$r_{XY} = \frac{s_{XY}}{s_X s_Y}$$

The coefficient of determination R^2 is just the square of the correlation coefficient, which ranges from 0 to 1 because it is squared. The coefficient of determination is used to help determine how much of a linear regression is explained by the independent variable (compared to noise). The linear equation slope is slightly different:

$$\beta = \frac{\sigma_{xy}}{\sigma_x^2}$$

which reduces to rise over run, the slope.

Expected value of a random variable - this is the PROBABILITY-weighted mean. The variance of a random variable is the PROBABILITY-weighted variance.

$$\begin{aligned}\mu &= E(X) \\ \text{Var}(X) &= E[(X - \mu)^2]\end{aligned}$$

In a probability tree, each branch is the numerator of the conditional probability (conditional expectation, conditional variance).

Total probability:

$$P(B) = P(B|A)P(A) + P(B|\bar{A})P(\bar{A})$$

We can reverse it because of independence between events A, B :

$$P(A) = P(A|B)P(B) + P(A|\bar{B})P(\bar{B}) + \dots$$

Bayes formula. Since events A and B are independent, i.e.

$$P(B|A) = P(B)$$

$$P(A|B) = P(A)$$

So,

$$P(B|A)P(A) = P(A|B)P(B)$$

And so Bayes' formula:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

$P(B)$ is the prior probability (the initial guess which is updated based on seeing event A), $P(A|B)$ is the evidence - the probability of seeing A given B, and $P(B|A)$ is the posterior - the updated belief given the evidence A. The RHS $\frac{1}{P(A)}$ is the scaling factor. E.g. $P(B)$ is the probability of the coin being fair; $P(A)$ is the probability of 8 heads in 10 coin flips; $P(A|B)$ is the probability of 8H in 10 flips given the coin being fair, and $P(B|A)$ is the probability of the coin being fair given 8H in 10 flips. Another e.g. $P(B)$ is the probability of having a disease, like 1 in 1000; $P(A)$ is the probability of detecting the disease on anyone; $P(A|B)$ is the probability of detecting positive with a test given the person has the disease; and $P(B|A)$ is the probability of the person having the disease given the positive result.

1.4 Portfolio Mathematics

Covariance.

$$\begin{aligned}\sigma^2(R_p) &= E[(R_p - E(R_p))^2] \\ &= E[(w_1R_1 + w_2R_2 + w_3R_3 - E(w_1R_1 + w_2R_2 + w_3R_3))^2] \\ &= w_1^2\sigma^2(R_1) + w_2^2\sigma^2(R_2) + w_3^2\sigma^2(R_3) + \\ &\quad 2w_1w_2\sigma^2(R_1, R_2) + 2w_1w_3\sigma^2(R_1, R_3) + 2w_2w_3\sigma^2(R_2, R_3)\end{aligned}$$

Generalizing, and this can be assembled using a linear algebra matrix:

$$\sigma^2(R_p) = \sum_{i=1} \sum_{j=2} w_i w_j Cov(R_i, R_j)$$

Covariance can also be calculated if there is a joint probability function in a matrix $P(R_i, R_j)$:

$$Cov(R_i, R_j) = \sum_i \sum_j P(R_i, R_j)(R_i - ER_i)(R_j - ER_j)$$

Covariance of returns is negative if, when the return on one asset is above its expected value, the return on the other asset tends to be below its expected value (an average inverse relationship between returns).

If the covariances are 0, then just sum the variances. As the number of holdings increases, covariance becomes increasingly important. If they are net-positive, they add to risk. If they are net-negative, they reduce risk.

When dealing with percentages, they are 100% squared, 10,000%². So, 400%² is 0.04.

Correlation coefficient

$$\rho(R_i, R_j) = \frac{Cov(R_i, R_j)}{\sigma(R_i)\sigma(R_j)}$$

Can construct a correlation coefficient matrix, as for covariance matrices. The correlation coefficient matrix is constructed by calculating each entry using the formula above from the covariance matrix. The diagonals should be 1.00, as they are perfect correlations.

Applications to financial risk management.

Safety-first rules focus on shortfall risk, the risk that portfolio value (or portfolio return) will fall below some minimum acceptable level R_L . Roy's shortfall ratio (SFR), which is just the standard normal Z, creates an optimal portfolio which minimizes the probability that R_p will fall below R_L . Assuming R_L is normally distributed, this criterion minimizes:

$$SFR = \frac{E(R) - R_L}{\sigma(R)}$$

The SFR minimum R_L is a percentage, not a number. The portfolio with the highest SFR is the safety-first optimal portfolio. Substituting the risk-free ratio for R_L makes it the Sharpe ratio.

Value-at-risk (VaR) is the amount of expected losses given a period of time (e.g. a day, a year) and a risk level (e.g. 5 percent) using a given (e.g. normal) distribution.

1.5 Simulation Methods

Lognormality. Many stock prices, such as future stock prices, are lognormal, while stock returns are normal. Lognormal distributions are bounded left by 0 and right-skewed. If Y is normally distributed (e.g. stock returns), then X is lognormal (e.g. stock prices):

$$Y = \ln(X) \Rightarrow X = e^Y$$

If a stock price rises from 112 to 160 and then down to 120,

$$160 = 112e^{r_1}$$

$$120 = 160e^{r_2}$$

and adding $r_1 + r_2$ will give the total compounded rate, as if

$$120 = 112e^r$$

Note:

$$\begin{aligned}\mu_{LN} &= e^{\mu+0.5\sigma^2} \\ \sigma_{LN}^2 &= e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)\end{aligned}$$

If a stock's continuously compounded (i.e. using Euler's number) return r is normal, the future stock price is necessarily lognormal because:

$$P_T = P_0 e^r \Rightarrow \ln(P_T) = \ln(P_0) + r \Rightarrow \ln(P_T) = r$$

If r is not normal, because

$$\ln\left(\frac{P_T}{P_0}\right) = \ln\left(\frac{P_T}{P_3} \frac{P_3}{P_2} \frac{P_2}{P_1} \frac{P_1}{P_0}\right) = r_T + r_3 + r_2 + r_1$$

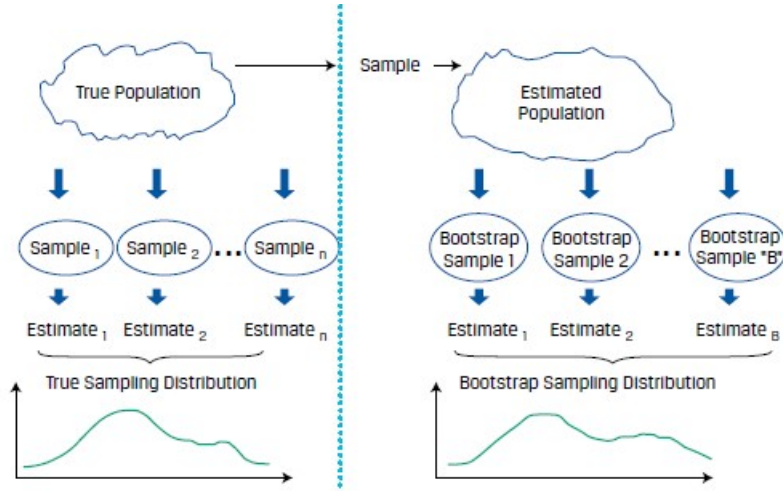
If the individual r_t are normal, then sum is normal. If r_t are not normal, the sum is approximately normal by the CLT. Note:

$$E(r_T) = E(r_0) + E(r_1) + \dots = \mu + \mu + \mu + \dots = \mu T$$

$$\sigma^2(r_T) = \sigma^2 T \Rightarrow \sigma(r_T) = \sigma\sqrt{T}$$

Monte Carlo simulations. Builds on generating random data with certain known statistical distribution of parameter values.

Bootstrapping. Sample from the sample of the population. Each resample size is the same as the sample size. With replacement. Simple procedure and accurate, so powerful.



Standard error of the sample mean is:

$$S_X^2 = \frac{1}{B-1} \sum_b^B (\theta_b - \bar{\theta})^2$$

where B is the number of resamples (each having the same number of members as the original sample, with replacement). The θ is the resample means, while $\bar{\theta}$ is the mean of the sample means. This is basically the sample variance applied to bootstrapping. Can estimate sample median and standard error of the sample median, in addition to the mean.

Jackknife. Bootstrap but leave out one observation each time, so each 'resample' size is $n - 1$, but they are NOT random resamples; they are calculated combinations - so for this reason, if one observation is always left out, then there must be n resamples. For e.g., a sample with 5 observations: [A, B, C, D, E] has jackknife 'resamples' [B, C, D, E], [A, C, D, E], [A, B, D, E], [A, B, C, E], and [A, B, C, D]. Can estimate sample median and standard error of the sample median, in addition to the mean.

Both Monte Carlo and bootstrapping are non-analytic. Analytical methods provide more cause-effect relationship insight.

1.6 Estimation and Inference

Random or quasi-random sampling methods.

- Simple random sampling.
- Systematic sampling - take every n -th member of the population.

- Stratified random sampling - divide populations into strata based on certain characteristics and sample based on the proportional size of those strata - guarantees subdivisions of interest are proportionally represented and smaller variances. Stratified random sampling is often used in bond indexing as pure bond indexing, or exactly replicating an index, is not possible as there are too many assets represented in the index.
- Cluster sampling. Subdivide populations, each cluster is a mini-representation of the population. Randomly select some clusters. One-stage cluster sampling: take all members of each selected cluster. Two-stage cluster sampling: sample from each selected cluster. Least accuracy because members of a cluster might not be representative (e.g. states in the US), but it's most efficient in time.

Non-random sampling methods.

- Convenience sampling. Sampling depending on whether or not the member is available.
- Judgment sampling. Pick members based on researcher or professional judgment e.g. auditing.

Do not mix two populations to create a larger population and randomly sample from it. Should randomly sample from the two separate sub-populations. Otherwise, get invalid (invalidly-weighted) statistics like means and standard deviation.

Central Limit Theorem. Given a population described by any probability distribution having mean μ and finite variance σ^2 , the sampling distribution of the sample mean $\bar{X}$ computed from random samples of size n from this population will be approximately normal with mean μ and variance σ^2/n when the sample size n is large.

Each mean is

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

So, the variance of the sample mean:

$$\sigma^2(\bar{x}) = n \frac{\sigma^2}{n^2} = \sigma^2/n$$

And the standard error of the sample mean:

$$\sigma/\sqrt{n}$$

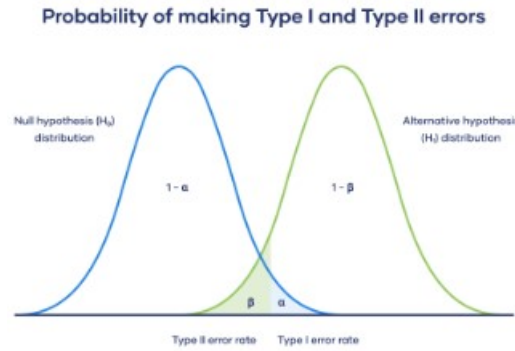
but in reality we usually use:

$$s/\sqrt{n}$$

where the n above is of the random sample size, not how many times the random sample is sampled.

1.7 Hypothesis Testing and Inference

Hypothesis testing. There is a correct and incorrect H_0 , and the decision to reject or not. The H_0 always has an equal sign - otherwise, can't form a null distribution. Type I and II errors both relate to H_0 . Type I: reject a true H_0 at the level of significance α with the confidence level $1 - \alpha$. Type II: fail to reject a false H_0 with probability β (and the complement $1 - \beta$ is the power of the test). The other two cases: correctly reject or correctly not reject H_0 . To decide error levels, consider costs - the only way to reduce both Type I and II errors is a larger sample size.



To decide whether to reject H_0 , compare test statistic versus critical values. To test whether a sampled MEAN is different from the population mean μ_0 :

$$t_{n-1} = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

Confidence interval:

$$\mu_0 = \bar{X} \pm t_{n-1} \frac{s}{\sqrt{n}}$$

Paired comparisons test. Test of the mean differences (paired comparisons over each time period), which does NOT assume samples are independent (DOF $n - 1$):

$$t_{n-1} = \frac{\bar{d} - \mu}{s_d/\sqrt{n}}$$

Test of the difference in means (DOF $n_1 + n_2 - 2$), which assumes samples are independent of each other, population variance unknown but assumed EQUAL - use the pooled variance

$$t_{n_1+n_2-2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}}$$

Pooled variance (assuming equal variances):

$$s_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Test of the difference in means, samples independent, which assumes unknown and UNEQUAL variances:

$$t_{DOF} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

with DOF

$$DOF = \frac{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)}{\frac{\left(\frac{\sigma_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{\sigma_2^2}{n_2}\right)^2}{n_2-1}}$$

The test of paired comparisons is more powerful than the test of the difference in the means because by using the common element (such as the same periods or companies), we eliminate the variation between the samples that could be caused by something other than what we are testing.

Complete list (note $s_{\bar{d}} = s_d/\sqrt{n}$ for the test of the mean of differences below):

What We Want to Test	Test Statistic	Probability Distribution of the Statistic	Degrees of Freedom
Test of a single mean	$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$	<i>t</i> -distributed	$n - 1$
Test of the difference in means	$t = \frac{(\bar{X}_{d1} - \bar{X}_{d2}) - (\mu_{d1} - \mu_{d2})}{\sqrt{\frac{s_{d1}^2}{n_{d1}} + \frac{s_{d2}^2}{n_{d2}}}}$	<i>t</i> -distributed	$n_1 + n_2 - 2$
Test of the mean of differences	$t = \frac{\bar{d} - \mu_{d0}}{s_{\bar{d}}}$	<i>t</i> -distributed	$n - 1$
Test of a single variance	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$	Chi-square distributed	$n - 1$
Test of the difference in variances	$F = \frac{s_{Before}^2}{s_{After}^2}$	<i>F</i> -distributed	$n_1 - 1, n_2 - 1$
Test of a correlation	$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$	<i>t</i> -Distributed	$n - 2$
Test of independence (categorical data)	$\chi^2 = \sum_{i=1}^m \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$	Chi-square distributed	$(r-1)(c-1)$

Non-parametric tests. Often ranked tests, converting data to ordinal, e.g. Wilcoxon signed-rank test, Mann-Whitney U test. Used (1) when the data do not meet distributional assumptions (e.g. normal distributions), (2) when there are outliers (which can skew parametric tests), (3) when the data are given in ranks or use an ordinal scale (non-parametric tests often convert data to ordinal anyways), or (4) when the relevant hypotheses do not concern a parameter (e.g. the question is about whether the sample is random or which distribution it has).

1.8 Parametric and Non-parametric Tests of Independence

Check first to see if normally distributed, etc. for whether to use a parametric or non-parametric test.

Parametric test. Testing Pearson's r , i.e. the correlation. Test statistic:

$$t_{n-2} = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Non-parametric test. Spearman's rank correlation, r_s (note: n is the number of PAIRS of data):

$$r_s = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)}$$

where the two data series are aligned, rank the data series ordinaly, then take differences of the ranks and square them d^2 . Just plug r_s into the Pearson parametric correlation test for $n > 30$ but for $n < 30$ need special critical values and tables.

Contingency (two-way) tables.

$$\chi^2_{(r-1)(c-1)} = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where

$$E_{ij} = \frac{(\text{Total row } i)(\text{Total column } j)}{Total}$$

The standardized residual to compare each cell is just the square of each RHS term:

$$\sqrt{\frac{(O_{ij} - E_{ij})^2}{E_{ij}}}$$

1.9 Simple Linear Regression

$$\hat{\beta}_1 = \frac{Cov(X, Y)}{Var(X)}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

$$r = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

$$\hat{\beta}_1 = r \frac{\sigma(Y)}{\sigma(X)}$$

This also means:

$$r^2 = \hat{\beta}_1 \frac{\sigma(X)}{\sigma(Y)}$$

Residuals differ from errors; residuals sum to 0 in regression because.

Assumptions: linearity, homoskedasticity (i.e. there can't be different regimes),

the observations pairs (x_i, y_i) are independent of each other (i.e. no autocorrelation like seasonality) - thereby implying that the residuals are uncorrelated, and the residuals are normal (look for outliers which can skew the residuals). For the latter assumption, if the sample size is large, test statistics valid even if not normal residuals.

According to CFAI, the assumptions of the linear regression model are that (1) the relationship between the dependent variable and the independent variable is linear in the parameters b_0 and b_1 , (2) the residuals are independent of one another, (3) the variance of the error term is the same for all observations, and (4) the error term is normally distributed; the dependent variable need not be normally distributed.

The sum of squares total (SST) of a regression can be decomposed into sum of squares error (SSE; unexplained errors) and sum of squares regression (SSR; explained errors; this is NOT the sum of squares of residuals):

$$SST = SSE + SSR$$

$$\sum (Y_i - \bar{Y})^2 = \sum (Y_i - \hat{Y}_i)^2 + \sum (\hat{Y}_i - \bar{Y})^2$$

where SSE is the explained part, from the estimated function value $\hat{y}$ to the average value $\bar{y}$, and SSR is from the observed value to the estimated function value. Coefficient of determination (R^2) is the unexplained part as a proportion of the total variation:

$$\begin{aligned} R^2 &= \frac{SSR}{SST} \\ R^2 &= \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2} = \frac{\sum (\beta_0 + \beta_1 Y_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2} = \frac{\sum (\bar{Y} - \beta_1 \bar{X} + \beta_1 X_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2} \\ &= \frac{\sum (\beta_1 X_i - \beta_1 \bar{X})^2}{\sum (Y_i - \bar{Y})^2} = \beta_1^2 \frac{\sum (X_i - \bar{X})^2}{\sum (Y_i - \bar{Y})^2} = \beta_1^2 \frac{\sigma(X)}{\sigma(Y)} \end{aligned}$$

but from above,

$$r^2 = \hat{\beta}_1 \frac{\sigma(X)}{\sigma(Y)}$$

So,

$$\begin{aligned} R^2 &= r^2 \\ R^2 &= \frac{SSR}{SST} = r^2 \end{aligned}$$

So,

$$\begin{aligned} \frac{SSR}{\sum (Y - \bar{Y})^2} &= \frac{[\sum (X - \bar{X})(Y - \bar{Y})]^2}{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2} \\ SSR &= \sum (X - \bar{X})(Y - \bar{Y})\beta_1 \end{aligned}$$

Hypothesis test for whether a slope is non-zero:

$$H_0 : b_1 = b_2 = b_3 \dots = 0$$

$$H_a : \text{At least one is non-zero}$$

The test statistic is F:

$$F_{k, n-k-1} = \frac{MSR}{MSE}$$

$$MSR = \frac{SSR}{k}$$

where k is the number of independent variables (k=1 for SLR).

$$MSE = \frac{SSE}{n - k - 1}$$

Hypothesis test for whether a slope in SIMPLE LINEAR REGRESSION is different from specified value B_1 :

$$t_{n-k-1} = \frac{\hat{b}_1 - B_1}{s_b}$$

where s_b is the standard error of the slope coefficient:

$$s_b = \frac{s_e}{\sqrt{\sum(X - \bar{X})^2}}$$

where s_e standard error of the estimate, the standard deviation of the residuals
- the standard error of the estimate is also called the root-mean square error:

$$s_e = \sqrt{MSE}$$

$$s_e = \sqrt{\frac{\sum(Y_i - \hat{Y}_i)^2}{n - k - 1}}$$

Note in SLR that to test whether a slope is equal to ZERO:

$$t^2 = F$$

which can be seen by looking at the F-ratio and t-statistic above. Fundamentally, both the t-statistic and F are variance ratios. Hypothesis test statistic (standard error of the intercept) on whether the SIMPLE LINEAR REGRESSION intercept is non-zero:

$$t_{n-k-1} = \frac{\hat{b}_0 - B_0}{\sqrt{MSE} \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{\sum(X - \bar{X})^2}}} = \frac{\hat{b}_0 - B_0}{\sqrt{\frac{\sum(Y_i - \hat{Y}_i)^2}{n-k-1}} \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{\sum(X - \bar{X})^2}}}$$

where the degrees of freedom are for the entire regression, $n - k - 1$ as for each coefficient in a regression, because all the coefficients - including the intercept -

are determined as a whole, simultaneously.

NOTE: If the t-statistic for the intercept is insignificant, but the t-statistic for the coefficient is significant, use the full equation - with the intercept in it.

For yes or no type variables (e.g. announcement or no announcement), use indicator variables (dummy variables) as x and do a hypothesis test on the slope of b_1 :

$$y = b_1x + b_0$$

To test if the yes and no populations are different, test the slope or use a test of differences in means.

Forecasted variable, use the below with a t-value having $n - 2$ DOF in the forecast interval, the variance (and standard error of the forecast:

The estimated variance of the prediction error, s_f^2 , of Y , given X , is

$$s_f^2 = s_e^2 \left[1 + \frac{1}{n} + \frac{(X_f - \bar{X})^2}{(n-1)s_X^2} \right] = s_e^2 \left[1 + \frac{1}{n} + \frac{(X_f - \bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \right],$$

and the standard error of the forecast is

$$s_f = s_e \sqrt{1 + \frac{1}{n} + \frac{(X_f - \bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2}}.$$

Note: the X_f is the INDEPENDENT VARIABLE, not the dependent variable!

Note: to get the forecast interval (or any confidence interval) based on the standard error, multiply by the two-sided critical t-value!!!

Sometimes, a log transformation is needed to make a non-linear relationship linear. Can use log-lin, lin-log, or log-log (elasticities model) transformation. The logged variable is now a relative change, and the lin variable is an absolute change. Look at the residuals and use the R^2 , F-statistic, and s_e to determine the transformational form. When looking at residuals, chart them versus the independent variable(s) and look for any patterns.

1.10 Big Data Techniques

Large dataset analysis, analytical tools. Big data - traditional and alternative. Alternative data generated by (1) individuals (e.g. social media), (2) business processes (e.g. corporate exhaust - banking records, supply chain information, transaction data) and (3) sensors (e.g. satellite imagery). Big data has 4V's: volume (high), velocity (speed and frequency), variety (more variety of sources), veracity (how good is the data).

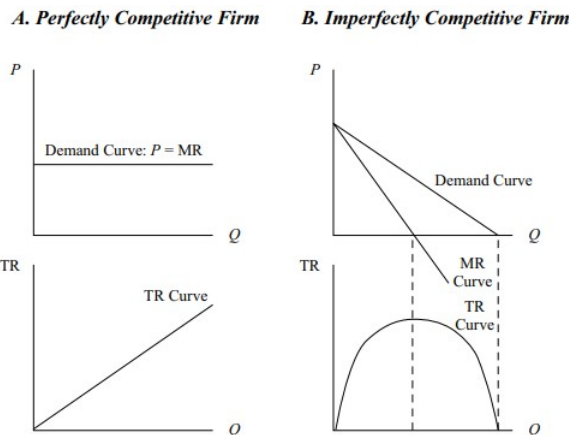
Machine learning (ML) use has increased due to big data. ML is 'learn the pattern, extract the rule, apply the rule'. It divides data into training set (perhaps 70% of the data), validation set (and then tune hyperparameters based on this; perhaps 20%), and test set. Still need human to understand and select data. Can be underfit (doesn't find enough fundamental patterns) or overfit (finds patterns in noise). Also a 'black box'. Supervised (input data are labelled to identify best variable or signal, e.g. regression), unsupervised (input data unlabelled, find appropriate new groupings, e.g. anomaly detection and clustering), and deep learning (supervised or unsupervised; looks at simple concepts first and then applies them to more complex concepts).

Data science. Data processing: capturing, curating (find errors), storing, searching, transferring. Data visualization. Text analytics and NLP.

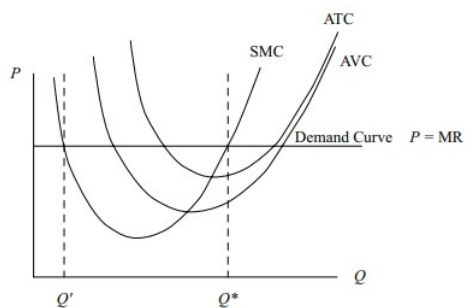
2 Economics

2.1 Introduction

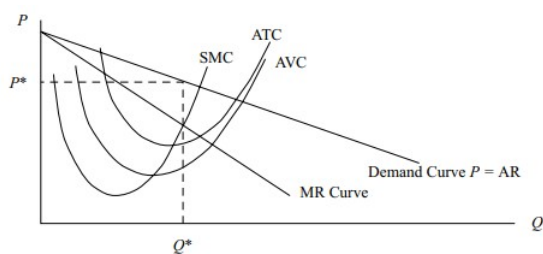
Perfectly competitive versus imperfectly competitive markets: demand curve flat versus demand curve downwardly-sloping. The firm in a PC market is so small that the demand curve 'looks' flat.



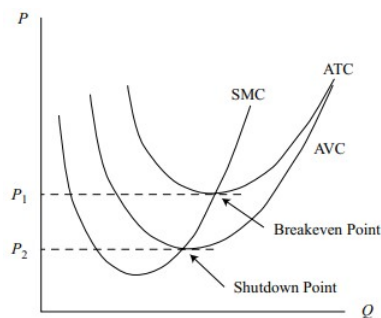
Profit-maximization is Q -star, where P is greater than ATC . The SMC always intersects the AVC and ATC at the minima for a firm in a PC market. In the long-run, competitors will enter the perfectly competitive market and price will decline.



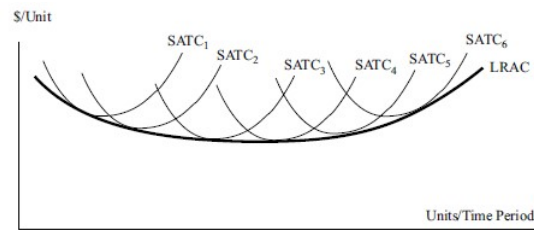
For monopolists, profit-maximization is also Q^* , where P is greater than ATC , but competitors won't enter.



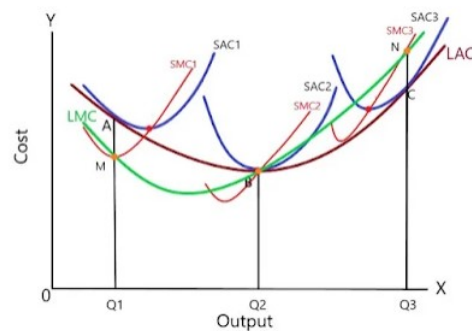
Economic profit = accounting profit + implicit opportunity costs (e.g. cost of capital). If the economic profit is 0, then earning normal profit (e.g. 5 percent by covering opportunity costs - e.g. factors of production like capital). A positive economic profit is beyond this. In the LR, perfectly competitive firms won't earn a positive economic profit. They will earn 0 economic profit. In the LR, if less than 0 economic profit, will exit but if P is greater than AVC (shutdown point), then better to continue for that short-run when FCs are sunk (e.g. lease for that time period - but quit upon lease renewal).



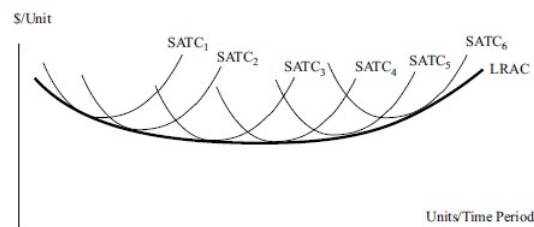
The short-run is when at least one factor of production is fixed; in the LR, all are variable.



The SRATC in scale economies is at a tangent to the LRAC because it is producing low levels - so, it should produce more to get scale economy efficiency.



Returns to scale and scale economies.



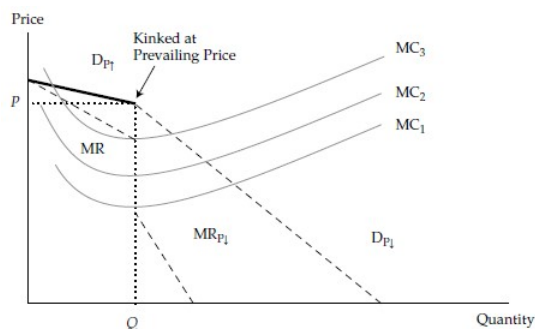
Market structures: perfect competition, monopolistic competition (differentiation), oligopoly (strategic moves), monopoly.

Monopolistic competition. Low barriers to entry. Similar to monopoly in the short run, but in the long run, demand curve falls as rivals join, leading to demand being tangential to ATC, resulting in zero economic profit.

Oligopoly. High barriers to entry. The three basic pricing strategies are pricing interdependence, the Cournot assumption, and the Nash equilibrium.

Kinked demand curve. Given a prevailing price, the price elasticity of demand will be much greater if the price is increased and less if the price is decreased. The firm's customers are more responsive to price increases because its rivals have lower prices. Alternatively, the firm's customers are less responsive to price

decreases because its rivals will match its price change. This means two demand curves (and two MRs) and therefore a kinked demand curve. The kinked demand curve shows why oligopolies can be stable, but they don't show what the equilibrium p and q are.

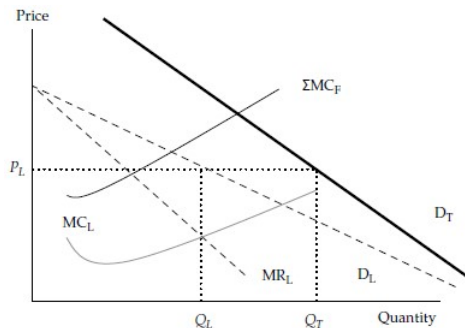


Cournot duopoly. Have a market demand curve, e.g. $Q = 450 - P$, then split $Q = q_1 + q_2$, and solve for P : $P = 450 - (q_1 + q_2)$. Assume constant MC, e.g. $MC = 30$. Setup TR_1 and TR_2 and differentiate each to get marginal revenue MR and equate to constant MC. Then solve simultaneous equations. The total quantity is less than in perfect competition, and the price is higher.

Stackelberg duopoly. First mover has advantage, as the first mover increases quantity so better for second mover to cut back to increase their profits.

Nash equilibrium. Use game theory strategies. Collusion and cartels possible.

Oligopoly price leader - often the largest and lowest-cost company. As price decreases, fewer of the smaller suppliers will be able to profitably remain in the market, and several will exit because they do not want to sell below cost. Therefore, the leader will have a larger market share as P decreases, and their demand curve is gentler in slope than the market. Price leader will produce at their own P_L which corresponds to a market quantity Q_T .



History shows gains in price wars are temporary. Over time, new entrants in oligopolies and this will dilute the market structure.

Measuring market power. The theoretical answer is to estimate the elasticity of demand and supply in a market. If demand is very elastic, the market must be very close to perfect competition. If demand is rigid (inelastic), companies may have market power. Use econometrics - but endogeneity is a problem (need both S and D curves) or lots of data needed, time series leads to changes in market structure over time, causality is unclear.

The other way to measure market power is to add market shares of the largest shares. But then if there's a merger, this doesn't change much, i.e. it's unresponsive. So, use HHI: add squares of shares.

2.2 Business cycles

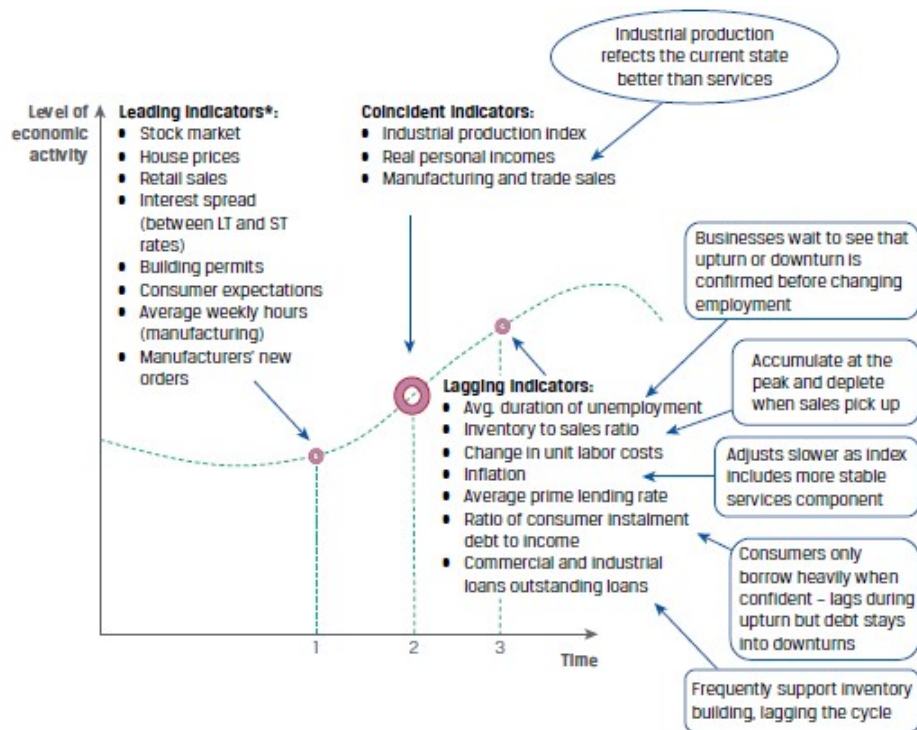
Emphasis on GDP. Classical cycles - GDP rises and falls, but this doesn't decompose into long-term trend and fluctuations around the trend. Growth cycle - compared to potential GDP.

Credit cycles. Private credit, especially loose policies, have been important in the past. Often, financial cycles boom ahead of recessions. Credit can help explain timing and severity of cycles, as well as construction and housing sector projections.

Looking at inventories, capital spending, wages, etc. to determine where in the business cycle. Leading, coincident, and lagging economic indicators. There are also composite indicators. A diffusion index shows macroeconomic activity BREADTH and combines the movements (e.g. q/q) of several indicators and averages them.

Defensive industries. Non-cyclical with stable earnings, e.g. retail and groceries, tend to be consumer staples. Safe or defensive for investors.

Indicators. Note: unemployment insurance claims are a leading indicator. Non-agricultural payrolls (because they're income!!!) and manufacturing and trade sales are coincident indicators.



In the slowdown phase, inflation accelerates, hiring still grows but at a slower rate, businesses operate near capacity.

Notes. (1) Large projects may not be cancelled in a contraction because they take a long time and cancellation costs are high. In contrast, equipment orders might be cancelled ahead of a contraction. (2) Inventories are a small part of the economy but highly influential because they change drastically. (3) Productivity is highest at the bottom of a recession, capital investment and borrowing is high and inflation is high at the peak of a boom, and qualified worker availability is low.

2.3 Fiscal policy

Keynesians, Monetarists, expansionary/contractionary fiscal policies, automatic stabilizers. Expansionary fiscal policy may crowd out private investment because governments will borrow to fund the fiscal expansion. High levels of debt can lead to expectations of higher taxes in the future, disincentivizing economic activity.

Fiscal multiplier. NT = net taxes; taxes after transfers. Disposable income $YD = Y - NT = Y - Yt = Y(1 - t)$. Marginal propensity to consume is c , marginal propensity to save is $1 - c$. Fiscal multiplier is then $\frac{1}{1 - c(1 - t)}$.

Balanced budget (spend G , raise taxes $T = G$) leads to an increase in output. Tax rise leads to reduction by MPC c , so $-Tc = -Gc$ but spending rises by G . So, output rises by $G - Gc = G(1 - c)$ and this is then fiscally-multiplied.

Ricardian equivalence. If a government reduces taxation by USD10 billion one year and replaces that revenue with borrowing of USD10 billion from the public, will it have any real impact on the economy? Whether RE holds is empirical. If it holds, fiscal policy is neutral, so favor monetary policy.

Budget deficits not always the best indicator of fiscal policy. During recessions, there will be automatic stabilizers which increase the deficit - but this is just automatic and not a policy decision.

Fiscal policy problems. Timing, information, uncertainty. Economy may already be at full employment - then fiscal expansion just leads to inflation. If the problem is not unused capacity but just shortage of labor, then fiscal policy to increase demand will just lead to inflation.

2.4 Monetary policy

Central banks influence monetary policy through open market operations, policy rates, and reserve requirements. The transmission mechanisms include market rates, asset prices, expectations/confidence, and exchange rates. So, there is a belief that money is not neutral, i.e. money affects the real economy.

Inflation targeting. Depends on central bank independence, credibility, and transparency. Most developed country central banks target inflation, but BOJ and US Fed don't - BOJ because deflation causes credibility conundrums, US Fed because it is also targeting maximum employment. Many developing countries have problems with these.

Exchange-rate targeting. Tying currency to a stable low-inflation economy (e.g. US or Germany) means using similar interest rates, and therefore 'importing' stable low inflation, e.g. if developing country inflation rises, its currency weakens, so the developing country central bank buys its own currency and sells the pegged currency, leading to monetary tightening and higher interest rates. But, exchange-rate targeting leads to more volatility in interest rates and forex because of frequent central bank interventions. Also need credibility and possibly independence and transparency.

Neutral policy rate - the average over the business cycle; neither contractionary nor expansionary in the steady state. Economist will differ on what is the neutral rate.

$$\text{Neutral rate} = \text{Real economy growth} + \text{Long-run inflation}$$

Problems in monetary policy transmission. (1) Long-run interest rate expectations aren't correct; sometimes, bond market vigilante activity might correct or destabilize. Liquidity trap - monetary loosening becomes ineffective; people won't borrow even if the rates are lowered. (2) Deflation. If deflation sets in, people defer purchases because prices will decline and people will save more because real debt values rise and they want to avoid rising debt. Here, sometimes quantitative easing (like OMOs but larger and used to affect long-run rates and the broader economy and involve purchases of private securities).

2.5 Geopolitics

Main spectra of dimensions: cooperation versus non-cooperation, globalization versus nationalism. Geopolitical tools (for cooperation or non-cooperation, globalization or nationalism): economic, financial, military.

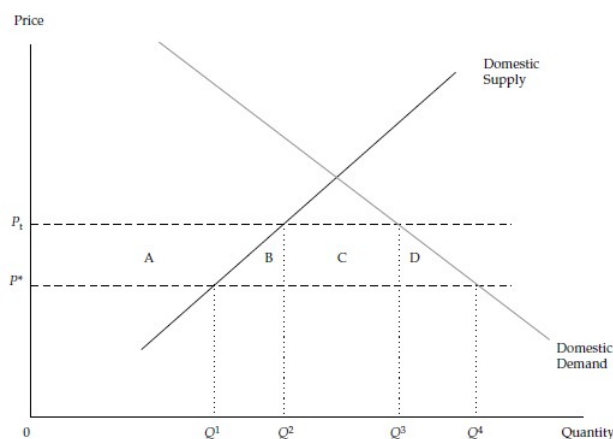
Event (calendar) risk (e.g. Brexit referendum), exogenous risk (sudden, unanticipated) risk, thematic risk (evolves over time; e.g. cybersecurity, climate change, ongoing civil war). Assess geopolitical threats based on likelihood, velocity of impact, and scale of impact.

A black swan risk is an event that is rare and difficult to predict but has an important impact. Investors with the appropriate time horizon and risk tolerance may make tactical changes to their investment choices as a result of these events. Long-term changes are unlikely to be necessary.

A non-cooperative economic tool is nationalization. Restrictions on FDI are a non-cooperative financial tool.

2.6 Trade

Tariffs. Below is from the importing country's perspective.



In the above, $A+B+C+D$ is consumer surplus lost; A is producer gain; C is government tariff revenue; $B+D$ is deadweight loss. If instead of a tariff there were a quota (import quota or VER), then C is the quota rent in the above. Export subsidies can be countered by CVDs; the exporter has an incentive to shift sales from domestic to international, resulting in welfare losses (market distortion) as domestic prices rise and international prices are pressured.

If the government of the country that imposes the quota can capture the quota rents by auctioning the import licenses for a fee, then the welfare loss under the quota is similar to that of a tariff, represented by areas $B + D$. The VER allows the quota rent resulting from the decrease in trade to be captured by the exporter (or exporting country), whereas in the case of an import quota, there is ambiguity regarding who captures the quota rents.

Large country can benefit from tariffs. In this context, a small country is not necessarily small in size, population, or GDP. Instead, a small country is one that is a price taker in the world market for a product and cannot influence the world market price. A large country, however, is a large importer of the product and can exercise some influence on price in the world market. When a large country imposes a tariff, the exporter reduces the price of the good to retain some of the market share it could lose if it did not lower its price.

This reduction in price alters the terms of trade and represents a redistribution of income from the exporting country to the importing country. So, in theory, it is possible for a large country to increase its welfare by imposing a tariff if (1) its trading partner does not retaliate and (2) the deadweight loss as a result of the tariff is smaller than the benefit of improving its terms of trade. If the exporter reduces its price by the same amount of the tariff, it will not export the same volume because the tariff value = tariff price $\times$ tariff VOLUME. So there will be deadweight losses no matter what! Global welfare would therefore still experience a net reduction—the large country cannot gain by imposing a tariff unless it imposes an even larger loss on its trading partner.

An export subsidy is a payment by the government to a firm for each unit of a good that is exported. Its goal is to stimulate exports. But it interferes with the functioning of the free market and may distort trade away from comparative advantage. Hence, it reduces welfare. The net welfare effect is negative in both the large and small country cases. Thus, export subsidies decrease rather than increase national welfare.

Regional trade agreements. Degrees of integration: FTA, customs union (common policy against non-members), common market (allows free movement of factors of production), economic union (institutions are integrated).

Notes: (1) Increasing supply-chain interdependence is a COST of globalization, not a motivation for it. (2) Geopolitical risk has a stronger effect if markets in

a downturn. (3) Geopolitical risk index (GPR) - high geopolitical risk leads to macroeconomic effects. (4) Event risks have the smallest degree of uncertainty - even compared to thematic risks.

2.7 Foreign exchange

Nominal exchange rates are what are traded.

$$\text{Direct nominal} = \frac{\text{Currency}_{\text{home}}}{\text{Currency}_{\text{foreign}}}$$

Real exchange rates are not traded or quoted; they are constructed by analysts to understand purchasing power. Real exchange rates are:

$$\begin{aligned} \text{Real} &= \frac{\text{Direct nominal}}{P_{\text{home}}} P_{\text{foreign}} \\ &= \frac{\text{Currency}_{\text{home}}}{\text{Currency}_{\text{foreign}}} \times \frac{P_{\text{foreign}}}{P_{\text{home}}} = \frac{\text{Currency}_{\text{home}}/P_{\text{home}}}{\text{Currency}_{\text{foreign}}/P_{\text{foreign}}} \end{aligned}$$

where the prices can be for e.g. CPI. If the nominal rate increases AND the foreign price increases faster than the domestic price, then the real exchange rate rose. A higher real exchange rate means the home currency has less purchasing power.

Theoretically, real exchange rates should guide future nominal exchange rates, e.g. if foreign prices rise faster than domestic prices, the domestic currency should appreciate - the nominal rate should decline (and therefore adjust to the current real exchange rate). But the timing is uncertain so there's no useful tendency to forecast nominal exchange rates using foreign exchange rates.

PPP-adjustments to GDP take into account price differences (based on baskets of goods) between countries and assumes the basket of goods should cost the same amount. GDP calculated using market exchange rates are misleading. So, calculate GDPs using PPP-adjusted exchange rates - PPP-adjusted exchange rates are as if the basket of goods in each country cost the same as in the US. Theoretically, PPP-adjusted exchange rates should forecast nominal exchange rates, but empirically the forecasting ability is poor.

Buy-side FX: Corporations, money and institutional investors, professional traders, retail accounts, governments, central banks, sovereign wealth funds. Sell-side FX: large banks, local banks.

Most FX trading is not related to international trade; around half is for FX swaps - for speculative positions and for hedging.

Conventions. Direct quote: price currency = home currency, base currency = foreign. Indirect: inverse of direct. For two-sided prices (bid and offers), the

dealer is buying or selling the base currency; the price shown is the price currency. For example, for a two-sided CHF/EUR: 1.1583–1.1585, the client will receive CHF1.1583 for selling EUR1 to the dealer and must pay CHF1.1585 to the dealer to buy EUR1.

History. Gold standard based on price-specie-flow mechanism, where a trade surplus leads to accumulation of gold, higher prices and therefore less export competitiveness. Destabilized in the 1930s due to trade wars, then gold standard with limited adjustments from the Bretton Woods agreements. In the 1970s, Bretton Woods replaced with Smithsonian Agreements to allow for flexible FX - which Milton Friedman called for much earlier because fixed FX would lead to speculative attacks. Exchange rate regimes. Ideal regime: credibly fixed FX, freely convertible, countries have freedom to pursue monetary and fiscal policies.

Dollarization, monetary unions, currency boards (legislated to use FX only to maintain a fixed rate and monetary base is backed by foreign reserves), fixed parity (fixed rate can be changed so credibility is an issue; no legislated target; nothing specified on how much is backed by foreign reserves), target zones (like fixed parity but with wider FX bands), passive and active crawling pegs (usually pegs move with inflation; rising inflation increases likelihood of foreign reserve run because of depreciation expectation, and to prevent this, central bank would depreciate the currency passively or actively - actively would allow some control over inflation expectations), fixed parity with crawling bands (fixed rate and then gradually loosen), managed floats, free floats.

Fixing FX rates leads to effectively importing inflation or deflation from the country to which the currency is pegged.

A trade surplus must lead to a net savings increase and/or increase in fiscal surplus (i.e. government 'savings'):

$$X + I + G = M + S + T$$

where the left side is household inflow, the right side is household outflows.

$$X - M = (S - I) + (T - G)$$

Capital flows are more important for FX rates in the short term, while trade is more important in the longer term.

Capital controls might be used in addition to monetary and fiscal policy to achieve objectives.

Forward FX rates (also called swap rates). If they are rising, the base currency is at a forward premium, and if they are falling, the base currency is at a forward discount. Forward FX rates are quoted in pips, e.g. +25.6 pips in 1 year, which means $25.6/10000 = 0.00256$ higher in one year (divide by 10,000

because, with the exception of JPY with 2 decimal places, FX rates are to 4 decimal places). They can alternatively be quoted in percentages (but note: forward percentages are quoted as positive OR NEGATIVE changes). Forward pips are theoretically related to interest rate differentials - the wider, the larger the forward pips.

$$(1 + r_d) = S_{f/d}(1 + r_f) \frac{1}{F_{f/d}}$$

Note that in the above, the numerator-denominator convention is reversed - it's f/d instead of d/f so it's indirect convention. Rearranging,

$$F_{f/d} = S_{f/d} \frac{1 + r_f}{1 + r_d}$$

$$F_{f/d} - S_{f/d} = S_{f/d} \frac{1 + r_f}{1 + r_d} - S_{f/d} \frac{1 + r_d}{1 + r_d}$$

$$F_{f/d} - S_{f/d} = S_{f/d} \frac{r_f - r_d}{1 + r_d}$$

Currency with higher interest rate will trade at a forward discount. If the time-to-maturity is changed, the equation changes, e.g. for $\tau = \frac{30}{360}$ days,

$$F_{f/d} = S_{f/d} \frac{1 + r_f \frac{30}{360}}{1 + r_d \frac{30}{360}}$$

$$F_{f/d} - S_{f/d} = S_{f/d} \frac{r_f - r_d}{1 + r_d \frac{30}{360}} \frac{30}{360}$$

Note also that the interest rate differential scales linearly with the forward rate increase.

3 Equity

3.1 Market organization and structure

Market regulation is about controlling fraud, agency problems (e.g. bad security dealers), fairness, ensuring long-term liabilities are funded (capital reserve requirements are OK).

Margin calls. For a futures contract, must pay an initial margin. If it drops below the maintenance margin, then must top up to at least the initial margin.

Swaps. Interest rate swaps - fixed for variable payments. Commodity swap - fixed payments in exchange for payments that depend on future prices of a commodity. Currency swap - payments denominated in different currencies, fixed or vary depending on future interest rates in the two countries. Equity swap, fixed payments for payments depend on the returns to a stock or a stock

index. Credit default swaps (CDS) are insurance contracts that promise payment of principal in the event that a company defaults on its bonds. Bondholders use credit default swaps to convert risky bonds into more secure investments.

Exchanges, brokers. Often indistinguishable, exchanges often regulated and require financial disclosures to reduce capital costs. Alternative trading systems (ATSS) are like exchanges, some are called dark pools as they don't show orders - institutional investors like these as large orders can move the market. Buying large orders often comes with premium and large sell orders at big discounts because of the trouble of finding counterparties. Dealers buy and trade and provide liquidity.

Dealers provide liquidity to buyers and sellers who arrive at the same market at different times. They move liquidity through time. Arbitrageurs provide liquidity to buyers and sellers who arrive at different markets at the same time. They move liquidity across markets.

Securitization, e.g. ETFs. Mortgage securitization - investors buy MBSs because risk is easier to understand than individual mortgages; mortgage-holders theoretically get lower rates. Mortgages appear on the bank's accounts as assets and the mortgage-backed securities appear as liabilities. Financial intermediary often avoids placing the assets and liabilities on its balance sheet by setting up a special purpose vehicle (SPV) or special purpose entity (SPE). Advantageous to investors because asset pool is better protected if the financial intermediary goes bankrupt. Often ABSs have several tranches. The senior tranches have first rights to the cash flow from the asset pool. The more junior tranches bear a disproportionate share of the risk of the pool. The complexity associated with slicing asset pools into tranches can make the resulting securities difficult to value.

Positions. Long position = benefit when asset or underlying asset value rises. Long position in futures or forwards = take delivery (benefit when asset value rises). Options = option holders are long (even for put options); option writer is always short. Swaps = arbitrary to say which is the long position; but usually the side that gains if the asset value rises. Long forward position in one currency is a short forward position in the other currency, e.g. Long the dollar against the yen means they have bought dollars and sold yen.

Buy on margin from broker using a margin loan. Interest on margin is the call money rate. Many regulations on initial margin requirement, but sometimes can borrow more if risk is modelled (portfolio margining).

Financial leverage ratio. Leverage return-on-equity. Total position divided by equity portion, e.g. 40 invested in 100 = 60 + 40 position means borrowing 60,

and ratio is 2.5; if the portfolio gains 10 percent, it is 10 percent x 2.5 because:

$$\frac{60 + 40}{40} = 2.5$$

$$\frac{60 + 40}{40} \times 1.1 = 2.5 \times 1.1$$

so that the gain is

$$2.5 \times 0.1$$

or 2.5 times 10 percent.

Maintenance margin requirement. A sign of how much 'slack' a borrower has relative to the portfolio value. Usually 25 percent of the CURRENT portfolio value, e.g. if 40 percent equity in 20 position, if price drops below 16 then margin call:

$$\frac{8 + (P - 20)}{P} = 0.25$$

Generally,

$$\text{Margin} = \frac{\text{No. shares} \times P - \text{Borrowed amount}}{\text{No. shares} \times P} = \frac{\text{Current equity}}{\text{Current price or value}}$$

For short sale, initial margin PLUS proceeds of sale deposited with broker. Initial margin covers the loss (i.e. gain in price) of the underlying.

Note: "Buying on 75 percent margin" means buyer contributes 75 percent of the equity, and 25 percent is provided by the lender.

The MAINTENANCE margin is different from the ordinary margin calculation, e.g. if buying a stock on 55% margin, then it's 55% equity and 45% debt, so the leverage ratio is 1.82.

Open-ended funds and closed-ended funds. When investors want to sell their shares, investors of an open-end fund sell the shares back to the fund whereas investors of a closed-end fund sell the shares to others in the secondary market. Closed-end funds are available to new investors but they must purchase shares in the fund in the secondary market. The shares of a closed-end fund trade at a premium or discount to net asset value, whereas open-end fund shares can be sold back at the net asset value.

Execution instructions indicate how to fill the order, validity instructions indicate when the order may be filled, and clearing instructions indicate how to arrange the final settlement of the trade. Market order (buy or sell asap at current price), limit order (buy or sell now but only at specific price or price range; otherwise don't fill order), stop order (once a certain price is met, it triggers a market order), marketable limit order, many different kinds of orders, including

stop orders (i.e. stop-loss for a long sell or stop-buy orders for a short buy).

A stop-buy order is usually if you want to buy at a HIGHER price; if you want to buy at a lower price, it would usually be a limit order at a lower price.

Marketable limit orders. If you make your limit order marketable by setting it at 10.70 when Stock A is trading at 10.65, you may get filled at 10.68 because of slippage - allows immediate execution but with buffer.

Stop-loss and take-profit orders. For example, if you bought Stock A at 10.65, you can set a Stop Loss at 10 and a Take Profit at 12. If a trader wants to completely stop a loss below a certain price in a falling market, use a put option at that price. Stop-sell orders become valid when prices fall below a certain level, stop-buy orders become valid when prices rise above a certain level (so, stop-sell and stop-buy orders destabilize the market!). A limit price places a limit on what trade prices will be acceptable to the trader; stop price indicates when an order can be filled.

Behind the market, makes the market, makes a new market (or inside the market). If match, take the market (can be buy or sell).

Primary market. Investment banks underwrite and book build. Can have underwritten offerings (bank buys shares if no one else buys) or best-effort offering.

Investment banks have a conflict of interest with respect to the offering price in underwritten offerings. As agents for the issuers, they generally are supposed to select the offering price that will raise the most money. But as underwriters, they have strong incentives to choose a low price. If the price is low, the banks can allocate valuable shares to benefit their clients and thereby indirectly benefit the banks. If the price is too high, the underwriters will have to buy overvalued shares in the offering and perhaps also during the following month if they must support the price in the secondary market, which directly costs the banks. These considerations tend to lower initial offering prices. Seasoned offerings are not like this; the secondary market has already discovered the price.

In IPO, the amount raised is the total but the gross proceeds are for the newly-issued shares as opposed to the founders' shares. Moreover, the gross minus net proceeds as a proportion of gross proceeds is the percentage, generally around 7%, which is the fee for the investment bank.

Shelf registration - go through offering procedure once, then trickle release into secondary market. Rights offerings - corporation distributes call options with low strike prices in proportion to shareholder holdings, so they will exercise and buy stock but dilute their shares.

Execution mechanisms. Quote-driven (price-driven) markets - traders trade

with dealers; order-driven - exchanges; broker-driven - bespoke.

3.2 Security market indices

Even with identical stocks and weights, can have (1) price index differing from (2) total return index. Total returns include all reinvestment and interest. Example:

$$Index = \frac{\sum nP}{D} = \sum wP$$
$$Index = \frac{\sum n(P + Inc)}{D} = \sum w(P + Inc)$$

Divisor is chosen arbitrarily; often adjusted when constituent securities are shuffled.

Return indices. For a security market index, price and total return can be calculated in two ways (give same answers): either the percentage change in value of the price or return return index, or the weighted average of price or total returns of the constituent securities. The weights depend on the kind of index used.

Weightings. Price-weighting (averages or like averages; readjust divisor upon stock split), equal-weighting (equal values; e.g. \$10, \$20, \$30 requires scale-up ratios of 6, 3, 2), market-cap weighting (also called value-weighting; weights based on total market cap of shares outstanding), float-adjusted market market-cap (exclude privately-held and controlling shares and possibly others which aren't free for trading), fundamental-weighting (use earnings, book value, cash flow, revenues, earnings, dividends, and number of employees).

Rebalancing - changing weights. Equal-weighted indices often rebalanced on rebalancing dates, e.g. each equarter. Price and market-cap weightings often aren't changed unless merger or acquisition or other corporate actions. Reconstitution - changing constituent securities.

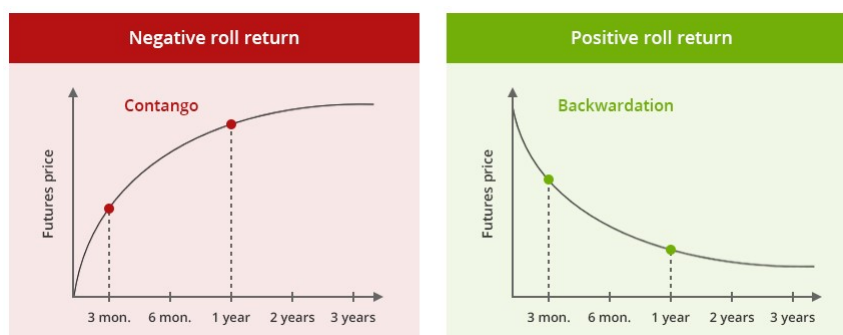
Sector indexes provide a means to determine whether a portfolio manager is more successful at stock selection or sector allocation. Style indexes represent groups of securities classified according to market capitalization, value, growth, or a combination of these characteristics.

Equity and fixed-income indexes. An aggregate fixed-income index can be subdivided by market sector (government, government agency, collateralized, corporate), style (maturity, credit quality), economic sector, or some other characteristic to create more narrowly defined indexes.

Alternative investment indices. Real estate indexes can be categorized as appraisal indexes, repeat sales indexes, and real estate investment trust (REIT)

indexes. Because commodity indexes do not have an obvious weighting mechanism, such as market capitalization, commodity index providers create their own weighting methods. Due to different weightings, unlike commodity indexes, broad equity and fixed-income indexes that target the same markets share similar risk and return profiles. Commodity indexes consist of futures contracts on one or more commodities. The performance of commodity indexes can be quite different from that of the underlying commodities because the indexes consist of futures contracts on the commodities rather than the actual commodities. Index returns are affected by factors other than changes in the prices of the underlying commodities because futures contracts must be continually “rolled over” (i.e., replacing a contract nearing expiration with a new contract). Commodity index returns reflect the risk-free interest rate, the changes in future prices, and the roll yield.

Roll yields. Positive in backwardation as short-dated futures sold at higher price than buying the long-dated futures, negative in contango.



Fundamental weighting has a value tilt. Fundamentally weighted indexes generally will have a contrarian ‘effect’ in that the portfolio weights will shift away from securities that have increased in relative value and toward securities that have fallen in relative value whenever the portfolio is rebalanced. Fundamentally weighted indexes weight companies based on factors like revenue, earnings, book value, or dividends. This often leads to higher weights for undervalued companies with strong fundamentals. Securities that have increased significantly in price relative to their fundamentals will see their weights reduced, as their relative weight in the index will have decreased. Securities that have fallen in price relative to their fundamentals will see their weights increased, as their relative weight in the index will have increased.

3.3 Market efficiency

In an efficient market, a passive investment strategy (i.e., buying and holding a broad market portfolio) that does not seek superior risk-adjusted returns can be preferred to an active investment strategy because of lower costs (for example, transaction and information-seeking costs).

Market efficiencies.

Forms of Market Efficiency	Market Prices Reflect:		
	Past Market Data	Public Information	Private Information
Weak form of market efficiency	✓		
Semi-strong form of market efficiency	✓	✓	
Strong form of market efficiency	✓	✓	✓

Abnormal returns are returns in excess of those expected given a security's risk and the market's return. Technical analysts assume weak form inefficiency; fundamental analysts assume semi-strong form inefficiency. If a market is semi-strong efficient, passive portfolio management should produce better returns than active ones.

Value stocks that have below-average price-to-earnings (P/E) and market-to-book (M/B) ratios, and above-average dividend yields. However, Fama and French developed a three-factor model to predict stock returns with market returns as specified by the capital asset pricing model (CAPM), the size of the company as measured by the market value of its equity and the company's book value of equity divided by its market value of equity, which is a value measure. The Fama and French model captures risk dimensions related to stock returns that the CAPM model does not consider. Fama and French find that when they apply the three-factor model instead of the CAPM, the value stock anomaly disappears.

Behavioral theories of loss aversion can explain observed overreaction in markets, such that investors dislike losses more than comparable gains.

Finding abnormal return strategies could just be a statistical artifact, so therefore a market anomaly. It doesn't necessarily imply a market inefficiency.

3.4 Overview of equity securities

Common shares. Voting - statutory (regular) or cumulative (favors shareholders with a smaller number of shares - e.g. if a shareholder has 100 shares, 4 director positions, 400 votes and can allocate to a single director position). Cumulative voting can occur even with common shares - don't need cumulative shares.

Common shares, compared to putable and callable PREFERENCE shares, are the riskiest for investors, so have the highest yields. Preference shares have priority in liquidation.

Preferred stock shareholders generally do not share in the operating performance of the company and do not have any voting rights, unless explicitly allowed for at issuance. Preference shares have characteristics of both debt securities and common shares. Similar to the interest payments on debt securities, the dividends on preference shares are fixed and are generally higher than the dividends on common shares. However, unlike interest payments, preference dividends are not contractual obligations of the company.

Dividends on cumulative preference shares accrue so that if the company decides not to pay a dividend in one or more periods, the unpaid dividends accrue and must be paid in full before dividends on common shares can be paid. In contrast, non-cumulative preference shares have no such provision. This means that any dividends that are not paid in the current or subsequent periods are forfeited permanently and are not accrued over time to be paid at a later date. However, the company is still not permitted to pay any dividends to common shareholders in the current period unless preferred dividends have been paid first.

Participating preference shares entitle the shareholders to receive the standard preferred dividend plus the opportunity to receive an additional dividend if the company's profits exceed a pre-specified level. In addition, participating preference shares can also contain provisions that entitle shareholders to an additional distribution of the company's assets upon liquidation, above the par (or face) value of the preference shares. Non-participating preference shares do not allow shareholders to share in the profits of the company. Instead, shareholders are entitled to receive only a fixed dividend payment and the par value of the shares in the event of liquidation. The use of participating preference shares is much more common for smaller, riskier companies where the possibility of future liquidation is more of a concern to investors.

Convertible preference shares - popular for venture capital financing. Higher dividends than common shares, benefit from a rise in the price of the common shares through conversion, and price is less volatile than the underlying common shares because the dividend payments are known and more stable, can convert to common shares for IPO.

Private equity. Through private placements; CANNOT be sold on secondary markets and have no market-determined quoted prices. Three primary types of private equity investments: venture capital, leveraged buyouts, and private investment in public equity (or PIPE). Venture capital - illiquid, may take 3-10 years to IPO to sell interests. LBOs (if existing management LBO, this is MBO) - often large amounts of undervalued assets (which can be sold to reduce debt) and generate high levels of cash flows (which are used to make interest and principal payments on the debt. Going private - reduce regulatory costs and not feel pressured to meet earnings expectations in the short-term; they

can look at the long term.

Non-domestic equity. Domestic control, stabilise currency flows. Depository receipts - foreign company deposits its shares in a local bank, local bank issues DRs on exchange which are at a ratio to the deposited equity shares. Depository receipt can be sponsored (foreign company gets involved and the DRs are equivalent to common shares; greater regulatory oversight) or non-sponsored (foreign company has not involvement; depository bank buys shares from foreign company and bank retains voting rights). ADRs (American depository receipts) - Level I is OTC (SEC reg); Levels II and III trade on US exchanges (SEC reg; can raise capital with this; listing fees are high); Level IV is private (no SEC reg); GDRs (global depository receipts) - outside US. ADSs (American depository shares) and GDSs (global depository shares) are the actual shares and more flexible.

The price of each depository receipt (and, in turn, returns) will be affected by factors that affect the price of the underlying shares—such as company fundamentals, market conditions, analysts' recommendations, and exchange rate movements. The number of depository receipts issued affects their price but does not affect the returns.

Risk. Puttable common or preference shares are often lower risk (and therefore pay lower dividends) because investor can sell and know the minimum price for equity sale. Callable common or preference shares are riskier (and therefore pay higher dividends) because the issuer has the option to redeem the shares at the strike price.

Accounting return on equity. It measures the total amount of net income available to common shareholders generated by the total equity capital invested in the company. It is computed as net income available to ordinary shareholders (i.e., after preferred dividends have been deducted) divided by the average total book value of equity (BVE). The measure depends on accounting conventions.

$$ROE = \frac{NI_t}{(BVE_t + BVE_{t-1})/2}$$

or

$$ROE = \frac{NI_t}{BVE_{t-1}}$$

Use book value of equity. Using market value, the ratio becomes earnings yield, which is just the inverse of P-to-E. Use price-to-book ratio to compare market value of equity to book value of equity. Book value is different from market value which is different from intrinsic value (which is NPV of cashflows).

Note: total book value of company is simply assets minus liabilities.

Investors require minimum return rates. For debt, this is interest. For equity, harder - it's when the cost of equity is equal to the minimum return rate. Companies issue both so calculate WACC which is the minimum return rate for the company.

Share repurchases. A share repurchase is viewed as equivalent to the payment of cash dividends of equal value in terms of the effect on shareholders' wealth, all other things being equal. Company managements have expressed several key reasons for engaging in share repurchases—namely, (1) signaling a belief that their shares are undervalued (or, more generally, to support share prices), (2) flexibility in the amount and timing of distributing cash to shareholders, (3) tax efficiency in markets where tax rates on dividends exceed tax rates on capital gains, and (4) the ability to absorb increases in outstanding shares because of the exercise of employee stock options.

3.5 Company analysis: present and future

Determine business model. Business model: what is it selling to who and how.

Determine revenue drivers. Bottom-up or top-down. Financial modelling of revenues. Forecast revenues.

Costs. Operating costs are for generating current period revenue. Investing costs for future revenue. Financing costs are payments for debt and equity.

Cost	Intuition	IFRS and US GAAP Classification: Income Statement	IFRS and US GAAP Classification: Statement of Cash Flows
Research and development	Investing	Operating	Operating
Depreciation and amortization*	Investing	Operating	Non-cash add-back to Operating <i>Capital expenditures</i> in Investing
Interest expense	Financing	Other (not operating)	US GAAP: Operating IFRS: Operating or Financing
Income taxes	Separate	Other (not operating)	Operating

Cost of equity and debt depend on the riskiness of the business.

Operating profit.

$$\pi = [Q \times (P - VC)] - FC$$

The P-VC is the contribution margin. For positive profit, contribution margin must be positive and Q high enough to offset fixed costs (FC). Degree of

operating leverage:

$$DOL = \frac{\text{percent change operating profit}}{\text{percent change sales in money terms}}$$

Increase DOL by increasing the fixed costs and decreasing the variable costs - but if comparing DOL between two different time periods (e.g. past and future), where the future has two different sales (e.g. high and low cases), then DOL will differ even if no differences in VC and FC - in fact, lower sales will have higher DOLs because of leveraging.

Degree of financial leverage. Increased by higher interest expenses that are fixed with respect to operating income. Analogous to the DOL but is determined by financing costs rather than fixed operating costs.

$$DFL = \frac{\text{percent change net income}}{\text{percent change operating income}}$$

Compare WACC (required rate of return) with ROIC = (net income - dividends) / (debt + equity).

Use DuPont ROE decomposition. ROE = Net profit margin (= net income divided by revenue) $\times$ Asset turnover $\times$ Equity multiplier. Alternatively, ROE = DTL $\times$ Asset turnover $\times$ Equity multiplier. The equity multiplier is leverage. DTL is degree of total leverage = DOL $\times$ DFL. Note: revenue in DTL is essentially sales.

$$DTL = \frac{\text{percent change net income}}{\text{percent change sales (revenue)}}$$

3.6 Industry and competitive analysis

McGahan and Porter (1997) estimated the relative importance of industry versus company-specific effects on both the emergence of economic profits (ROIC in excess of WACC) and their sustainability over time, finding that industry was the most important factor in the sustainability of economic profits. Importantly, they found that company-specific effects were much larger for low performers than for high performers in an industry, suggesting that industry and competitive forces can act as a ceiling on a company's returns but not as a floor.

Industry classification. Commercial databases, e.g. Refinitiv. However, they are often strict, i.e. a company even if it sells multiple products is in one classification - but this isn't useful if looking at a different company and want to know all substitutes. Moreover, some industries, e.g. healthcare, are more local.

Industry size is typically measured by total annual sales from the product or customer perspective, which is not necessarily all sales of each company. This

is hard to measure, it's changing through time and tiny company performances are hard to measure. Mature industries move in line with GDP growth.

Studying industrial structure is looking within an industry; PESTLE (political, economic, sociological, technological, legal, environmental) looks to factors outside of industry.

Competitive strategies. Look for whether they defend against Porter's five forces, consider PESTLE, and whether the company can execute the strategy. Some successful strategies: cost leader (e.g. utilities), differentiated (e.g. BMW), or focus (on a specific target segment).

	Cost Leadership	Differentiation	Focus
Means of executing strategy	▪ Economies of scale from fixed costs ▪ Favorable access to raw materials ▪ Culture of strict cost control ▪ Aggressive pricing to gain high volume ▪ Low-cost distribution ▪ Economies of scope	▪ Investments in advertising, brand, customer service, proprietary distribution channels ▪ Protection using trademarks, copyright, patents ▪ Superior quality, unique features ▪ Culture of strong customer experience ▪ Premium pricing ▪ Integration of services, software, and hardware	▪ Proximity to customers and strong understanding of their needs ▪ May incorporate elements of strategy from both cost leadership and differentiation, but focused on particular group
Which of the Five Forces it defends against (why it works)	▪ Threat of new entrants: Capital requirements and scale advantages deter entrants ▪ Bargaining power of customers: Customers can only bring prices down to the costs of the marginal producer, leaving margin for the cost leaders ▪ Industry rivalry: Rivals may not be able to compete on price with cost leaders	▪ Threat of new entrants and of substitutions: Customer loyalty to unique product can deter switching, protect market share ▪ Bargaining power of customers: Customers may be unable or unwilling to comparison shop or switch ▪ Bargaining power of suppliers: The company may have the ability to pass along price increases to customers and/or margin to absorb cost increases	▪ Threat of new entrants and of substitutes: Customer loyalty to unique product can deter switching, protect market share ▪ Bargaining power of customers: Customers may be unable or unwilling to comparison shop or switch
Industry appropriateness	▪ Capital intensive ▪ Price-conscious customers ▪ Customers do not value or notice product differences ▪ Minimal innovation in industry	▪ Price is not foremost concern for customers ▪ Customers value distinctiveness ▪ Innovation in industry, with products varying in features and forms	▪ Difficult (or uneconomical) to serve customer group, product, or geography for other players
Risks to the strategy	▪ Cost inflation, loss of discipline ▪ Technological change that results in loss of cost leadership or market share ▪ Desire for premiumization among customers	▪ Imitation by competitors ▪ Buyers become sophisticated, no longer demand level of service ▪ Pricing premium becomes too high for customers to bear ▪ May preclude high market share, as customers value exclusivity	▪ Larger competitors outcompete on price ▪ The differences in demand between the narrow group and industry as a whole narrow ▪ Buyers become sophisticated, no longer demand level of service

3.7 Forecasts

Historical (assume precedent is future), historical base rates and convergence (e.g. for mature industries), management guidance, analyst judgment (including ad hoc judgments for e.g. lawsuits, and new technologies).

Management guidance tends to be most accurate for in-company variables but investors tend to have a better forecast of macro. A key focus of an analyst is to understand management's assumptions embedded in guidance and to scrutinize their plausibility. It is not uncommon for companies to suspend guidance altogether in periods of high uncertainty. While the midpoint of a guidance range might be intuitively interpreted as management's "true" expectations, management may provide a range specifically intended to create a hurdle that is more easily cleared, for which management can be rewarded.

Analyst judgment. Note to take into consideration economic and business cycles in forecasts. Note to include/exclude non-recurring items. Include risk factors, e.g. competition, changes in the business cycle, inflation and deflation, and technological developments.

Capex. Estimate useful life by fixed asset total divided by depreciation and amortization expenses. Analysts must also make projections about a company's future capital structure. Leverage ratios such as

- debt to capital
- debt to equity
- debt to EBITDA

are often used as the forecast object to project future debt and equity levels.

Scenario analysis. Hard to find cost decompositions of companies. One approach is to assume variable costs are

$$\%VC \approx \frac{\%\Delta (\text{cost of revenue} + \text{operating expenses})}{\%\Delta \text{ revenue}}$$

and then $\%FC = 1 - \%VC$. This works best when total costs are close to and just under revenues. Another approach which can be used to corroborate the first approach is to decompose each operating cost component - e.g. cost of revenue, depreciation, R&D, SG&A as % of costs and then further estimate what % of these are fixed, and then add the fixed cost %s.

3.8 Equity valuation

Market values can be overvalued, undervalued, or fairly valued compared to analyst intrinsic valuations.

Valuation models. (1) DCF (PV) models. Cashflows can be modelled with dividend-discount or FCF-to-equity models. (2) Market multiplier models, e.g. P-to-E. Enterprise value (EV) multiples have the form (Enterprise value)/(Value

of a fundamental variable). Two possible choices for the denominator are earnings before interest, taxes, depreciation, and amortization (EBITDA) and total revenue. Enterprise value, the numerator, is a measure of a company's total market value from which cash and short-term investments have been subtracted (because an acquirer could use those assets to pay for acquiring the company). (3) Asset-based valuation models. Estimate intrinsic value of a common share from the estimated value of the assets of a corporation minus the estimated value of its liabilities and preferred shares. The estimated market value of the assets is often determined by making adjustments to the book value (synonym: carrying value) of assets and liabilities.

Dividends. Many companies give out regular dividends but others, extra or special dividends. Companies in cyclical industries and companies undergoing corporate and/or financial restructuring are among those observed to use extra dividends. Stock dividends (bonus share issues) don't change the size of the 'pie' so aren't relevant for DDM. To pay dividend, a company has a 'declaration date', then next comes the 'ex-dividend date' when share prices no longer have the upcoming dividend (and so price falls), then 'holder-of-record' date when the shareholder who officially owns the shares is recognized for dividends, then 'payment date'.

Dividend discount model - NPV with forecasted dividends. Hard to forecast dividends, so use FCFE.

FCFE - NPV with dividend-paying capacity. Cashflow from operations (CFO) = NI + non-cash expenses - investment in working capital. Non-cash expenses reduce NI but aren't paid out - e.g. depreciation and amortization, stock compensation, impairment. Investment in working capital is increase in inventory, increase in AR, decrease in AP. FCFE is cash flow available for distribution to common shareholders.

$$FCFE = CFO - FCInv + \text{Net borrowing}$$

FCInv is fixed capital maintenance (usually reported capex). Net borrowing is borrowing minus repayments.

Required rate of return for DCF models - use CAPM.

Required return on share = Current expected risk-free return + β_i [Market equity risk premium]

The β is non-diversifiable risk. The market equity risk premium is a broad stock market portfolio. Can be hard to estimate β , so might just add a risk premium for the company to the risk-free rate.

Preferred stock. Non-callable, non-convertible with no maturity date (no redeemable date):

$$V_0 = \frac{D_0}{r}$$

Non-callable, non-convertible with maturity date, like a bond:

$$V_0 = \sum \frac{D_t}{(1+r)^t} + \frac{F}{(1+r)^n}$$

Retractable term preferred shares. Puttable at par value on retraction date; before retraction date, issuer can call. Calculate implied value of retractable option by DCF calculation minus market value of share.

Gordon growth model. Dividend grows at constant rate - particularly suited for companies insensitive to business cycles. The g can be calculated by:

$$g = ROE \times \text{Earnings retention rate} = ROE \times (1 - \text{dividend payout ratio})$$

Can have multi-stage DDMs where the dividend growth rate changes.

Multiplier models. Often don't consider the future, but can include future values by first forecasting them. Often don't have any connection to DCF models, but multiples can be justified by them, e.g. Gordon growth model turns into a P-to-E ratio, which indicates dividend growth rate is inversely proportional to P-to-E, but note high growth rate can be related to low dividend payout ratio (higher payout ratio may imply a slower growth rate as a result of the company retaining a lower proportion of earnings for reinvestment) - the dividend displacement of earnings.

Method of comparables. Based on law of one price. Allow for relative comparisons, both cross-sectional (versus the market or another comparable) and in time series. Calculate multiple and compare it to a benchmark. Benchmark = close match of a stock, or industry average. Can use various multiples.

P-to-E ratios. Cyclical industries have cyclical earnings. Earnings are very sensitive to cycles; but sales are less sensitive because earnings leverage on fixed costs. So, use P-to-S for cyclical industries.

Enterprise value.

$$EV = \text{market cap} + \text{market value of preferred stock} + \\ \text{market value of debt} - (\text{cash} + \text{investments})$$

Cash and investments are cash equivalents and short-term investments (subtract this because it's already part of market value as part of treasury). EV is the cost of a takeover: acquiring company assumes the acquired company's debt but also receives its cash. Useful when comparing companies with significant differences in capital structure. Often use EV/EBITDA or EV/OI or something else.

Note: EBITDA is useful as it's like operating cashflows - operating profit excludes depreciation and amortization which aren't actually paid out in cash. Moreover, EBITDA is usually positive even when earnings are negative - and negative earnings can cause problems for P-to-E ratios.

Asset-based valuation. Often used for private companies. An asset-based valuation of a company uses estimates of the market or fair value of the company's assets and liabilities. Thus, asset-based valuations work well for companies that do not have a high proportion of intangible or "off the books" assets and that do have a high proportion of current assets and current liabilities. If a company has hard to value property, including plants and IP and goodwill, asset-based valuation is difficult.

4 Fixed Income

4.1 Features of fixed income securities

Bond principal = par value = face value. Maturity = final date; tenor = time to maturity. If tenor < 1 year, then it is a money-market security (e.g. T-bills, commercial paper). If tenor > 1 year, capital market security.

Variable interest rates = floating-rate notes = market-reference rate (MRR; lowest default risk; this reflects top-down factors like interest rates!) + issuer-specific spread (credit spread). Zero-coupon bonds = pure discount bonds.

Seniority. Senior debt versus junior = subordinated debt. Secured debt requires liens or pledges while unsecured debt against just cash flows. There can be senior and junior unsecured debt. Secured debt relies primarily on operating cashflows for repayment and secondarily on legal claims on assets.

Contingency provisions. Embedded options = call, put or conversion to equity options. These resemble option contracts but cannot be traded separately from the bond itself. However, the value of these embedded options may be established by comparing the value of a bond with a contingency provision with that of an otherwise similar standard bond from the same issuer.

$$\text{Current yield, \%} = \frac{\text{Coupon value}}{\text{Bond price}}$$

YTM is the IRR using the bond's price. IRR = YTM if all cash is reinvested at the bond's YTM.

Yield curves. Most fixed-income issuers have many debt instruments outstanding. A useful way of evaluating the YTM on one issue is to graph all an issuer's debt instruments with identical features by their YTM and times to maturity. Usually upward-sloping. Can compare to sovereign bonds, which are vertically

lower - that indicates the risk level for that issuer.

Indenture - legal contract and conditions of the bond. Covenants agreed at time of issuance, e.g. voting rights, premium if issuer is acquired. Affirmative covenants require issuer to do something, negative covenants prohibit them from doing something (and may include incurrence tests, e.g. financial ratios that must be met before some action can be taken; negative pledge clause which prohibits disposal of assets for instance). Pari passu clause says the debt is treated equally as issuer's debt of similar seniority, while a cross-default clause (affirmative covenant) specifies issuer is in default if they default on another debt instrument.

An incurrence test might be a solvency ratio such as debt-to-EBITDA.

4.2 Fixed-income cashflows

Bullet bond = standard fixed coupon bonds. Bond issuer receives principal, pays fixed interests, then repays principal.

Amortizing debt = part of the principal is retired before repayment. Lower investor risk because borrower's liability is reduced over time. Calculation is an iterative process.

Partially-amortizing debt (with balloon payment at the end). To calculate, find the PV of the balloon payment. Subtract it from the PV (i.e. the principal). Then fully amortize the remainder BUT CALCULATE INTEREST PAYMENTS ON THE ENTIRE PRINCIPAL, not just the remainder after subtracting the discounted balloon payment (interest on the principal minus discounted balloon payment means lower interest portions in the annuity payments and faster repayment of the principal and then separately paying the interests for the balloon payment in parallel, but this is the same as ending up with the balloon payment as the final unpaid principal, with higher interest portions and slower repayments). While each mortgage payment is smaller than the fully-amortized version, there is less principal repaid each mortgage payment, leading to higher interest rates than the fully-amortized version!

Sinking funds. Issuer sets aside money to periodically retire parts of the bond early, as per indenture covenants.

Waterfalls. Bond repayments divided into tranches, with some receiving before others.

Bond interest rates can change. MRRs can change. Coupon rates can change if credit quality deteriorates, e.g. leveraged loans and credit-linked notes. Payment-in-kind (PIK) terms allow a bond principal to increase instead of paying interest or coupon. Step-up bonds either (1) have specified increases in coupon rates

to protect investors against rate rises, or (2) depend on a condition, e.g. credit quality, CO2 emissions target - specified margins at specified dates.

Index-linked bonds. Inflation-linked bonds, usually issued by governments. There are capital-indexed bonds, e.g. Treasury Inflation-Protected Security (TIPS), where each payment period, the bond principal is adjusted upwards (or downwards) in accordance with e.g. CPI, and then the fixed coupon percentage is applied. Other inflation-linked bonds are interest-linked bonds, where only the interest rate is linked to inflation - this is basically an FRN.

Zero-coupon bonds. Unlike fixed-coupon bonds, protected from falling interest rates, which reduces reinvestment (and therefore reduces below original YTM).

Deferred-coupon bonds. Coupons only paid later in term. Usually for construction projects or capex projects where cashflows generated later. Discounted because of 'missing' interest payments.

Callable bonds - right to redeem early, e.g. if interest rates fall. Usually have a call protection period where can't be called, and then subsequently a call period. The call price may be specified in the indenture. Investors need to be compensated for call risk. Whether call will be exercised depends on YTM (prevailing market rate) versus coupon rate; if YTM is higher than coupon rate, issuer won't call because refinancing is more expensive. Exercise if interest rates fall. Make-whole call bonds are callable bonds but with high penalties to issuers (e.g. buyback priced at sovereign bond, i.e. expensive).

Puttable bonds. Bondholders can sell at specified put price. Usually exercise put if YTM (interest rate) rises above coupon rate.

A make-whole call is a contingency feature under which issuers can buy bonds back at a price usually based on the yield-to-maturity of a sovereign bond of similar maturity plus a predetermined spread. Makewhole calls are rarely executed and have little economic impact on issuers or investors.

Convertible bonds. Often used by growth companies. Bondholders have the right to buy the issuer's shares at the specified conversion price. Bondholders usually accept low YTM's to buy convertible bonds. Conversion ratio is the number of shares per bond on conversion, calculated by bond par value divided by conversion price. To decide whether to convert, compare bond price with conversion value ($= \text{conversion ratio} \times \text{share price}$).

Warrants are not embedded; they can be traded separately. Allow bondholders to buy issuer's stock at fixed exercise price until bond matures. Call provisions allow issuers to redeem early to limit bondholder gains.

Contingent convertible bonds (CoCos), usually used by banks to protect against

liquidity problems, specify that upon certain events (e.g. issuer's equity ratio drops below regulatory minimum) that the bonds are automatically converted to equity. This can happen if issuer's share price declines. YTM's here are higher to compensate bondholders.

Types of bonds: domestic, foreign, Eurobonds, global bonds (latter three are international bonds). A global bond is a bond issued simultaneously in the Eurobond market and in at least one domestic bond market, ensuring sufficient demand for large bond issues and access to all fixed-income investors regardless of location. Sukuks are sharia-compliant (can't invest in gambling, alcohol) and pay profit based on asset owned by issuer. Foreign bonds are bonds sold in a country in that country's currency by a foreign entity. Eurobonds are bonds issued outside the jurisdiction of any single country - issued in domestic currency outside that country (bond is issued outside of the borders of the currency's home country, e.g. US-denominated Eurobonds in Japan). Eurobond's currency has greatest influence on its price. Eurobonds used to be bearer bonds but now mostly registered.

Taxes. Bonds are interest expenses, so tax-deductible for corporate issuers. For individual investors, taxed at ordinary rate like income. Original issue discount (OIDs) are par value minus discounted bond price - a prorated portion is recognized as interest income in the US annually but no capital gains taxes (while in many other jurisdictions, no OIDs but capital gains taxes). Note dividends are not capital gains.

4.3 Fixed income issuance and trading

Corporate issuers issue more types of different debt instruments than equities. Fixed income usually categorized along issuer (sector), credit quality, and time to maturity.

Credit quality - credit ratings. BBB- and higher are investment grade; BB+ and lower are speculative/junk.

Commercial paper is a short-term bond to finance short-term or seasonal working capital and may or may not be secured (if secured, called asset-backed commercial paper ABCP). Usually less than 3 months, often used for bridge financing. Usually issued in variance currencies to hedge and broaden investors, usually issued by large financial institutions, governments. International CP = Eurocommercial paper. Depending on terms, some CP can be rolled over by paying with new CPs, but this introduces rollover risk to the investor, who usually require the borrower to have backup lines of credit.

Credit facilities, which are loan agreements with pre-specified terms and limits but with fluctuating balances based on borrower-specific needs at different points in time, analogous to a credit card.

Bond indexes. Many more bonds than equities, and bond indices reflect this. Also, more frequent rebalancing because of more new issues and limited maturities compared to equities.

Secondary fixed-income markets consist mostly of quote-driven or over-the-counter markets, while secondary trading for listed equity markets primarily occurs on an electronic exchange.

Distressed debt is traded until either the corporate issuer has liquidated its assets or its outstanding bonds have been restructured. In contrast, an equity security may be removed or delisted from secondary trading on an exchange if it fails to meet the listing requirements of the exchange, which often include minimum share prices, net worth, and free float. By the time an issuer's debt has become distressed, its equity securities will likely have already been delisted.

Reopening or tapping an existing bond (can be at price very different than before) - to save money, issue additional bonds at the same maturity date and conditions as an existing bond.

Secured bonds typically have higher yields as their issuers are more risky.

The Bloomberg Barclays Global Aggregate Index is a flagship measure of global investment grade debt from twenty four local currency markets. This multi-currency benchmark includes treasury, government-related, corporate and securitized fixed-rate bonds from both developed and emerging markets issuers. There are four regional aggregate benchmarks that largely comprise the Global Aggregate Index: the US Aggregate, the Pan-European Aggregate, the Asian-Pacific Aggregate, and the Canadian Aggregate Indices.

4.4 Fixed income markets for corporate issuers

Main short-term financing: (1) uncommitted lines of credit - least reliable for borrower - borrower has no upfront costs, just pays interest and usually unsecured but bank monitors their account, usually requires minimal reserves at bank, and can renege if conditions worsen, so borrower can't use this primarily but bank uses to keep customers; (2) committed lines of credit - committed up to a certain amount but there is a fee for the full or unused amount, usually requires higher capital reserve at bank, borrower faces renewal risk upon maturity; (3) revolving credit agreement (operating lines of credit) - most reliable for borrower - usually multiyear agreements but with restrictive covenants.

Secured loans. Lender files security interest. Can pledge accounts receivable to a lender (called a factor), and factor can even take over the collection of the AR (factoring agreement).

Banking. Checking accounts usually pay little interest as they are for transactions, while savings accounts pay more - sometimes Certificate of Deposits (CDs) pay more. Banks sell CDs.

Interbank market. Secured or unsecured. Central bank requires some reserves from banks; if below minimum, banks can borrow at central bank funds rate, which is set by the central bank by OMO and deposit interest rates. Banks can borrow from central bank at the discount window lending rate, but this is usually higher than the central bank funds rate and can result in greater oversight from the central bank.

ABCPs. In many cases, banks create a special purpose entity (SPE) and give short-term loans (and backup line of credit) to the SPE in exchange for cash. The SPE issues ABCPs, using the loans. This is off-balance sheet financing because the bank's balance sheet no longer includes the risk from short-term loans. Up to the GFC, the SPEs with MBS's led to misplaced confidence and opacity. The cash was used for the bank's long-term assets but with the GFC, this cash dried up, the backup lines of credit were issues, and caused long-term asset financing to fail. So now, back to using this cash primarily for bank's short-term assets.

A Special Purpose Entity (SPE) primarily serves as a conduit for securitization, a process of converting illiquid assets (like accounts receivables) into more liquid securities (like asset-backed commercial paper or ABCP).

Repurchase agreements (repos). Secured short-term financing, using e.g. US treasuries. Because it's secured with high-quality collateral usually, low interest rates. Sale of a security where the seller will buy the same (or a similar) security back at the repurchase price (based on a repo rate - which can be negative if there's high borrowing demand for the underlying security; also influenced by repo term, initial margin, etc.) on the repurchase date. From the perspective of cash lenders, it is a 'reverse repo'. The security seller in effect borrows money. Security seller still owns the security and gets the coupons. There's an initial margin on which the repo agreement is based (usually more than 100 percent to protect the lender from declining collateral value), where the security price at the time of the agreement will be higher than the loan amount (=purchase price, the amount the cash lender 'buys' the security for):

$$\text{Initial margin} = \frac{\text{Security price}_0}{\text{Purchase price}_0}$$

The security price is the value of the security; the purchase price is the value of the loan. If the initial margin 100%, then it is overcollateralised and the cash lender (security buyer) is better protected.

The repo agreement is based on a haircut which the borrower must find elsewhere. The lender of money, because it is lending the more liquid asset, receives

margin. To provide that margin, securities used as repo collateral are priced at market value minus a haircut; the size of the haircut varies depending on the maturity, quality, scarcity value, and price volatility of the underlying collateral, on the term of the repo, and on the creditworthiness of the customer. The standard haircut is generally between 1% and 3%.

$$\text{Haircut} = \frac{\text{Security price}_0 - \text{Purchase price}_0}{\text{Security price}_0}$$

The 'purchase price' changes through the repo term at the repo rate. For instance, for a 30-day repo term, the purchase price at 5 days is 5/30 of the repo rate. At the end of the repo term, the purchase price is the 'repurchase price'. If the security price changes, then the margin changes. If it becomes over-collateralized or undercollateralized, parties can request within the terms of the repo agreement, additional collateral or releases of collateral, using the variation margin at time = t :

$$\text{Variation margin} = (\text{Initial margin} \times \text{Purchase price}_t) - \text{Security price}_t$$

The repurchase price is the purchase price x repo rate, because the purchase price is the loan value. The purchase price increases through the repo term by the repo rate (pro-rated if in the middle of the repo term). The repo rate is not necessarily the discount calculated from the initial margin; it is set by the market. If the variation margin is negative, then the cash borrower (security seller) can request some collateral be released.

Repos used for (1) short-term financing by cash borrowers, (2) short-term financing by cash lenders, (3) cash lenders can short the security by borrowing it (profit if short position gain exceeds repo rate), (4) in addition to OMO and policy rates, central banks use repos for very short-term effects.

Repo risks: default, collateral (is it correlated with counterparty risk?), margining (ability to margin call quickly), legal, netting/settlement risk. Repos have low liquidity and default risk.

Investment-grade versus high-yield debt instruments. IG is more like YTM of sovereign bonds, and usually have few restrictive covenants. HY is more like equity because cashflow is less certain, and to minimize investor loss, have more restrictive covenants (so they often are more likely to issue equity instead). HY issuers might use callable bonds and other means to increase financial flexibility. Fallen angels are rated HY but their debt is usually IG-like but prices would have deteriorated a lot because they lack investor protections for HY-grade debt.

Sovereign debt. Fiscal policy determines the level of sovereign debt, while government debt management determines the composition of this debt. Ricardian equivalence: a government's choice of debt maturity is irrelevant in determining the present value of future tax cash flows (analogous to Modigliani-Miller for

sovereign financing). Taxpayers expect government debt to be offset by higher future taxes, assuming consumers smooth their spending, rational expectations of the future, perfect capital markets, taxpayers pass on their wealth to descendants without selfishness. If issued debt is too short, rollover risk and if too long, maturity risk (cost), so strike a balance and issue different maturities.

Sovereign debt issuance. Sold at public auctions. Single price auction (lower buyer costs and therefore greater distribution of buyers) and multiple-price auctions (latter leads to more yield volatilities because different people pay different prices). Non-competitive bidders always get securities. Competitive bidders pay either their bid if they win or the lowest cut-off price to sell the issuance. On-the-run securities are more liquid and therefore used to price bonds. Sovereign bonds are used for non-economic purposes other than for profit, e.g. banks must hold them for regulation, many types of bonds are off-limits for financial institutions (too much risk), the central bank may use them for monetary policy, other governments may hold them for reserves, etc.

Lower-level governments and government agencies may also issue debt. Might not get full national low-rate though. Some have taxation powers, some have fees, government transfers, etc. Can issue general-obligation (GO) bonds for general purposes (based on general tax, transfers, fees) or revenue bonds for specific projects (and fees and revenue from that specific project).

Public sector financial accounting standards vary widely and are often prepared using cash, rather than accrual-based, principles, typically excluding such items as the depreciation of fixed public goods, such as federal highways, or the accrual of unfunded liabilities, such as government pension obligations.

4.5 Fixed income valuation

Market discount rate = required yield = required rate (of return). If market discount rate = coupon rate, bond PV = par value. If bond PV < par value, then we say the bond trades at a discount; it trades at a premium otherwise.

Between coupon payment periods,

$$\begin{aligned} PV^{full} &= \frac{PMT}{(1+r)^{1-t/T}} + \frac{PMT}{(1+r)^{2-t/T}} + \dots + \frac{PMT}{(1+r)^{N-t/T}} \\ &= \left(\frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \dots + \frac{PMT}{(1+r)^N} \right) (1+r)^{t/T} \\ &= PV \times (1+r)^{t/T} \end{aligned}$$

Between coupon payments (assuming there are any), use linear time elapse for the accrued interest AI (not affected by interest rate changes) added to the flat price (affected by interest rate changes). The full price is the 'invoice' or 'dirty'

price:

$$PV^{full} = PV^{flat} + AI$$

AI is calculated using actual/actual (government equivalent) or 30/360 (corporate bond) conventions. These are 'street conventions' and don't include actual payment dates (e.g. on weekdays following any weekend or holiday payment dates), i.e. 'true yield'.

Note that $PV^{flat} \neq PV$. PV is just the price at the beginning of the coupon period - it usually appreciates as t/T time passes. To calculate PV^{flat} , FIRST calculate PV^{full} , THEN calculate AI:

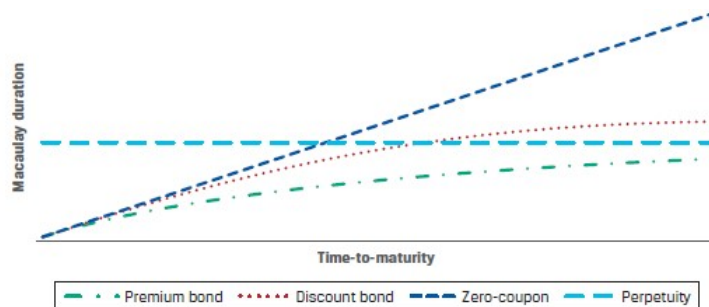
$$AI = \frac{t}{365} \times PMT$$

and then SUBTRACT AI from PV-full. Don't calculate PV-flat, because that was PV-flat IN THE PAST!!!

Bond dealers will quote the flat price (they can't keep changing the bond price as time moves forward), but the buyer must pay the full price because there is accrued interest (the buyer is 'closer' to the next coupon payment date).

For the same time-to-maturity, a lower-coupon (or zero-coupon) bond has a greater percentage price change than a higher-coupon bond when market discount rates change by the same amount. This is because the FV has a bigger weight for zero-coupon bonds.

Maturity effect. Longer-maturity bonds are generally affected more by interest rate changes. For instance, the maturity effect always holds on bonds priced at a premium above par value. Exceptions to the maturity effect are rare and occur ONLY for low-coupon (but not zero-coupon) long-term bonds trading at a discount - see Macaulay durations; for long-dated bonds trading at a discount to par, interest rate risk could be lower than that of a shorter-term bond:



Price-yield relationships. Constant yield price trajectory. All bonds approach par value with time. Any bond trading at a discount will increase in price because of the pull-to-par effect. the price-yield relationship generally becomes

less convex as a bond approaches maturity because it's more like a short-term bond. A high-coupon bond trading at a premium will decrease in price as it approaches maturity.

Convexity. Graph yield versus bond price. As yield declines, prices rise but in non-linear way. A linear decline/rise in yield leads to a percentage price increase being larger than percentage price decline. The relationship is positively convex. As yields approach 0, price will go to infinity. To get price to fall to 0, yields are at infinity.

Bond pricing. To price a bond, look at known prices for bonds with similar maturities, coupon rates, and credit quality. Most important to look at yields (not bond prices) and average them based on maturity times. Then interpolate based on maturity times. Use linear interpolations, averages to estimate bond price. Most bonds are not traded, so prices on BBG and elsewhere are usually estimates. On-the-run bonds have lower YTM's because they have higher prices.

4.6 Yields and yield spreads

Periodicity of bond, e.g. 4 for quarterly. Effective annual rate ('real' rate) depends on periodicity, e.g. 5 percent compounded semi-annually means compounding twice a year at 2.5 percent, which is actually 5.0625 effective annual. So, the effective annual rate actually has a periodicity of 1.

Bond yields. Bond basis or equivalent yields are the periodic rates multiplied by the periodicity ('fake' rate). E.g. semiannual bond basis yield is the semi-annual yield times two. Can use the equation below to convert periodicities:

$$\left(1 + \frac{APR_m}{m}\right)^m = \left(1 + \frac{APR_n}{n}\right)^n$$

where effective rates can be converted to basis or equivalents as well as between effective rates and basis or equivalent rates.

To change bond yields between different timing conventions, 30/360 to actual/actual, just multiply the bond basis yield by 365/360. This is an approximation and assumes the 30/360 bond yield is based on 360 days and the actual/actual on 365.

Simple yields. Sum of coupon payments plus the straight-line amortized share of the gain or loss, divided by the flat price. It is used mostly to quote Japanese government bonds (JGBs).

Bonds with embedded options. Callable bonds will specify in their indenture date ranges for call prices. Calculate yield-calls, and assemble into a yield schedule based on this, i.e. assuming the bond is called at the earliest time of each date range (e.g. take now to 3 years if the first call date range is 3-4 years).

Make sure to include the CALL PRICE as the FV. Calculate YTM's based on the SAME PV. E.g.

Yield-to-first call	5.149%
Yield-to-second call	5.247%
Yield-to-third call	5.313%
Yield-to-fourth call	5.362%
Yield-to-maturity	5.374%

Yield-to-first will often have the lowest yield and yield-to-maturity often the highest. Yield-to-worst is the worst YTM of these, and it can be anything, it's the worst and used for quotes. While the investor will get a higher yield overall because of the embedded call option, the yield can be worse than the yield-to-maturity because it can be called. So, the callable bond yield is often:

$$Yield - to - worst < Yield < Yield - to - maturity$$

The callable bond can be valued more precisely by taking the flat value and adding an option-adjusted price using an option pricing model and volatility assumptions - from this, an option-adjusted yield can be calculated.

Spreads. Yield spread is over a particular government security which may or may not have the same time to maturity, so it's just a benchmark. G-spread is over an interpolated government security. I-spread is over the swap rate (an indication of the short-term interest rate). Issuers will use the I-spread to determine the relative cost of fixed-rate bonds versus floating-rate alternatives, such as a bank loan or commercial paper, while investors use it as a measure of a bond's credit risk. Whereas a standard interest rate swap involves an exchange of fixed for floating cash flows based on a floating index, an asset swap converts the periodic fixed coupon of a specific bond to a market reference rate (MRR) plus or minus a spread. If the bond is priced close to par, this conversion approximates the price of a bond's credit risk over the MRR index.

Q. An analyst is analyzing a three-year, 2.25% annual coupon bond issued by QWE Company. Currently, the bond's yield-to-maturity is 2.707%. The three-year swap rate is 1.840%. The government spot rates are presented in the table.

Maturity	Government Spot Rate
1Y	0.899%
2Y	1.260%
3Y	1.904%

The G-Spread (in basis points) for the QWE bond is closest to:

- A. 80.
- B. 87.
- C. 135.

Solution

A is correct. The G-spread is a yield spread above that of a government bond with the same maturity date. The yield-to-maturity for the corporate bond is 2.7070%. The yield-to-maturity for the government benchmark bond is 1.9036%.

$$\text{G-spread} = 2.707\% - 1.904\% = 0.8034\% = 80.3 \text{ bps.}$$

Z-spread (zero-volatility spread) is just a constant spread above the government benchmark rate. Z-spread uses the entire government yield curve, in contrast to the G-spread which just takes a single YTM. The Z-spread is solved from:

$$PV = \frac{PMT}{(1 + z_1 + Z)} + \frac{PMT}{(1 + z_2 + Z)^2} + \dots + \frac{PMT + FV}{(1 + z_N + Z)}$$

Option-adjusted spread = Z-spread - option value (of a callable bond). OAS is calculated using an option pricing model.

4.7 Yields for floating-rate instruments

FRN. Bond prices are much more stable, reducing investor risk. Most bonds are floating. Payment is usually 'in arrears', i.e. interest rate is set at the beginning of the term, and interest is paid at the end of the term.

MRR + quoted margin (spread). Quoted margin is specified in the bond indenture, but the required margin is set by the market and depends on e.g. issuer credit risk, bond liquidity, tax status. For instance, if the issuer credit rating declines, the required margin > quoted margin, so the bond will be priced at a discount equal to the difference in margins. Required margin is the spread such that the FRN is priced at par value on a rate reset date (so, if the bond is issued at par value, then the quoted margin = required margin).

Between coupon dates, its flat price will be at a premium or discount to par value if MRR goes down or up.

$$PV = \frac{\frac{MRR+QM}{m} FV}{\left(1 + \frac{MRR+DM}{m}\right)} + \frac{\frac{MRR+QM}{m} FV}{\left(1 + \frac{MRR+DM}{m}\right)^2} + \dots + \frac{\frac{MRR+QM}{m} FV}{\left(1 + \frac{MRR+DM}{m}\right)^N}$$

where DM is the discount margin = required margin.

Money market instruments. Bond yields-to-maturity are calculated using the standard time value of money. Money market instruments are often quoted using non-standard interest rates, such as discount rates or add-on rates. Money market instruments with different times-to-maturities have different periodicities. However, bond yields-to-maturity are given for a common periodicity for all times-to-maturity.

Money market interest rate can be stated as add-on (AOR) or discount rate (DR). The AOR is forward-looking from the PV to the FV:

$$PV = \frac{FV}{1 + \frac{\text{days}}{\text{year}} AOR}$$

so

$$AOR = \frac{\text{year}}{\text{days}} \frac{FV - PV}{PV}$$

whereas the discount rate DR is backward-looking from the FV to the PV:

$$PV = FV \times \left(1 - \frac{\text{days}}{\text{year}} DR\right)$$

$$DR = \frac{\text{year}}{\text{days}} \frac{FV - PV}{FV}$$

The discount rate understates the interest rate because it is backward-looking.

Conventions and Instruments	Description	
Quote Basis	Discount Rate (DR)	Add-on Rate (AOR)
Equation	$DR = (\text{Year}/\text{Days}) \times [(FV - PV)/FV]$	$AOR = (\text{Year}/\text{Days}) \times [(FV - PV)/PV]$
Quoted Amount	Face Value at Maturity (FV)	Price at Issuance (PV)
Typical Instruments	Commercial Paper, Treasury Bills, Bankers' Acceptances	Bank Certificates of Deposit, Repurchase Agreements (Repos), Market Reference Rates (MRRs)

Taking the discount rate (usually on actual/360) and calculating the PV on 100 par value and then solving for the equivalent AOR (on actual/365) is the 'bond equivalent yield' (or 'investment yield'). Bond yields-to-maturity are annualized and compounded, but yield measures in money markets are annualized but not compounded

Note: bond YTM's can be annualized and compounded (or annualized on well-defined periodicities like 2 or 4), but money market yields are only annualized (as seen from the equations above, which use the days/year ratio) - this makes them non-standard interest rates. So, money market yields have VARYING PERIODICITIES and are not compounded.

4.8 Term structure of interest rates

Spot rates - government zero-coupon bond YTM's. Assemble on-the-run freshly issued (not off-the-run which is less liquid) bonds of different maturities for these bonds into a spot curve. Usually increases with higher times to maturity; if it declines, it is an inverted yield curve. However, sometimes can't get zero-coupon or other problems so interpolate. Yield curve should have the same periodicity.

The spot curve has ZERO-coupon bond rates, e.g. the spot rate r_5 for 5 years from now means the bond $PV = \frac{FV}{(1+r_5)^5}$. However, to price a coupon bond, price each of the coupons using the appropriate spot rates - e.g. for a 5-year coupon bond, price the first PMT with the spot curve rate 1, the second PMT with the spot curve rate 2, etc. The principal is priced at the spot curve rate 5.

Given a spot curve, use the rates to calculate the single PMT (for each bond calculation) such that the $PV = FV$, i.e. price the bond with rates across the spot curve. This PMT is EFFECTIVELY the par rate, because the par rate is the YTM which makes $PV=FV$. It is essentially a YTM calculated using the numerator cashflows.

Implied forward rate IFR ('forward rate' or the 'breakeven reinvestment rate') for zero-coupon bond beginning at $t = A$ and $tenor = B - A$:

$$(1 + z_A)^A (1 + IFR_{A,B-A})^{B-A} = (1 + z_B)^B$$

The IFR is $Ay(B-A)y$ rate. If the investor believes in A years the rate will be higher than $(B-A)$, then buy the A -maturity bond and reinvest. If the investor believes it will be lower, then just buy the B -maturity bond. Can calculate spot rates from forward rates.

If the spot curve is rising, the forward curve is above the spot curve and vice versa.

Spot Curve Shape	Par Curve	Forward Curve
Upward Sloping	Below spot curve	Above spot curve
Flat	Equal to spot curve	Equal to spot curve
Downward Sloping (Inverted)	Above spot curve	Below spot curve

If the spot curve is rising and positive, the par rate will be below as the lower rates in the near-term (near-term 'weighs' more in a NPV calculation) mean

the overall bond price is buoyed and to reduce this back to $FV = PV$, reduce the PMT. If the spot curve is rising but negative, then par rate is above the spot curve and rising because the short-term PMT's need to be 'large enough' to outweigh the negative short-term spot rates.

Note: when calculating bond PVs, e.g. 5-year bond with semiannual coupons, if a semiannual interest rate (e.g. rate divided by 2) is used for the interest (but coupon is a value so it must always be divided by 2 for semiannual coupons), then the times have to be $t=1, 2, 3, \dots, 10$ (not 0.5, 1, 1.5, ..., 5). If we use EFFECTIVE annual rates (e.g. for 3.2 percent, divide by 2 and square it to get 3.226 percent) then use $t=0.5, 1, 1.5, \dots, 5$.

4.9 Average timing versus average completion time

If a person completes 50% of a job in year 1 and then 50% of the job in year 2, then the weighted average timing is 1.5 years. The sum of percentage-years is 150%-years because $50\% \times 1 \text{ year} + 50\% \times 2 \text{ years}$. This can be divided by $50\% + 50\% = 100\%$ to get 1.5 years.

If a person completes a job in 1 year, 3 years, and 5 years in three different occasions, the average completion time is 3 years.

The average timing, e.g. weighted average timing, provides information about the distribution or contribution of job progress. It's progress-weighted average time; where the centerpoint of the work progress is. The average completion time is about the total time to compete the job.

The average timing of 1.5 years can be used to replace the 1 year and 2 years in the first example. If the 50% of year 1 was instead now in year 1.5 and the 50% of year 2 also now in year 1.5, then the sum is $50\% \times 1.5 + 50\% \times 1.5 = 300\%$ again. This is the average of the timings.

4.10 Duration

If a bond is purchased at PV and after T years sold (i.e. the holding period is T), the return rate or horizon yield is

$$\text{Horizon yield} = r = \left(\frac{FV}{PV} \right)^{1/T} - 1$$

where $FV = \text{Coupons reinvested to time T} + \text{new bond PV at time T}$, i.e. carrying value. This is the IRR between the total return and bond purchase price. It's not specific to a particular time. Horizon yield rules can be seen from:

A portfolio manager gathers the following information about an option-free bond that was held to maturity:

Realized horizon yield	4.2%
Reinvestment rate	4.1%

Q. The yield to maturity at purchase was *most likely*:

- A. less than 4.2%.
- B. equal to 4.2%.
- C. greater than 4.2%.

Solution

- A. Incorrect because, since the reinvestment rate is less than the horizon yield, the yield-to-maturity is greater than the horizon yield of 4.2%.
- B. Incorrect because, since the reinvestment rate is less than the horizon yield, the yield-to-maturity is greater than the horizon yield of 4.2%.
- C. **Correct** because the realized horizon yield matches the original yield-to-maturity if (1) coupon payments are reinvested at the same interest rate as the original yield-to-maturity, and (2) the bond is sold at a price on the constant-yield price trajectory, which implies that the investor does not have any capital gains or losses when the bond is sold. Since the reinvestment rate is less than the horizon yield and the bond is held to maturity, the yield to maturity is greater than the horizon yield.

If the interest rate rises, the coupons are reinvested at a higher rate, so the bond becomes more valuable. However, a rise in the interest rate will also reduce bond value. The present values at each time change based on the sum of these forces. At the Macaulay duration, they are balanced because that's the average time of the present values. The FV constant for infinitesimally small interest rate changes but since the PV will change, the horizon yield will change. For larger changes, add convexity.

Carrying value. Price at which the bond was purchased plus (minus) the adjustment for amortization of the discount (premium) if the bond is purchased at a price below (above) par value. Any point on the constant-yield price trajectory.

Reinvestment risk - if interest rates rise, they affect the reinvestment returns of coupons; usually more of a risk with long-term investment horizons. Price risk - if interest rates change, the bond price changes (gives rise to capital gains/losses, no price risk if bond is held to maturity); usually more of a concern with short-term investment horizons (as the bond is sold off before maturity).

Macaulay duration is the time-weighted average of discounted PVs. It represents the weighted average time it takes to recover the initial investment in a bond. It is a fictitious idea; nothing calculated at the MD means anything. It is the 'weight' of time of the bond. As with weighted averages (e.g. 60 percent of people finish a job in 10 years, 40 percent in 2 years means the time-weighted average is 6.8 years but nobody finishes the job in exactly 6.8 years and it doesn't

mean anything; it's a 'weight').

$$MacDur = \sum_n^N (n - t/T) \left(\frac{\frac{PMT}{(1+r)^{n-t/T}}}{PV^{full}} \right)$$

Derivation. If C is the coupon payment and M is the principal, the bond price at the beginning is:

$$P_0 = \sum_{t=1}^T C e^{-rt} + M e^{-rT}$$

At some point $t = t_d$, with τ being points at which a coupon was paid,

$$P_{t_d} = \sum_{t=\tau+1}^T C e^{-r(t-t_d)} + M e^{-r(T-t_d)}$$

$$P_{t_d} = e^{rt_d} \sum_{t=\tau+1}^T C e^{-rt} + M e^{-rT}$$

$$P_{t_d} = e^{rt_d} P_0$$

and reinvested coupons being (up to time $t = \tau$):

$$R_{t_d} = \sum_{t=1}^{\tau} C e^{r(t_d-t)}$$

If interest rates rise, reinvested coupons rise and bond prices decline at time $t = t_d$ and vice versa:

$$\frac{dP_{t_d}}{dr} = -\frac{dR_{t_d}}{dr}$$

Product rule on LHS:

$$t_d e^{rt_d} P_0 + e^{rt_d} \left(\sum_{t=\tau+1}^T t C e^{-rt} + T M e^{-rT} \right) = - \sum_1^{\tau} t C e^{-rt}$$

Combine the summation operator terms:

$$t_d P_0 = \sum_1^T t C e^{-rt} + T M e^{-rT}$$

$$t_d = \frac{1}{P_0} \left(\sum_1^T t C e^{-rt} + T M e^{-rT} \right)$$

but notice that:

$$MacDur = t_d = \frac{1}{P} \left(\sum_{t=1}^T t C e^{-rt} + T M e^{-rT} \right) = -\frac{1}{P} \frac{dP}{dr}$$

which will be useful later.

MD is only for infinitesimally small interest rate changes. For larger changes, it's approximate. This is the same as the weighted average of discounted cash-flows (where C is an annual coupon payment for a T-year bond with principal M):

$$P_0 = \sum_{t=1}^T P_t$$

$$= Ce^{-r} + Ce^{-2r} + \dots + Ce^{-rT} + Me^{-rT}$$

And a time-weighted average divided by PV is:

$$MacDur = \sum_{t=1}^T w_t t = \sum_{t=1}^T \frac{P_t}{P_0} t = \sum_{t=1}^T \frac{tP_t}{P_0} = \sum_{t=1}^T \frac{1}{P_0} \left(\sum_{t=1}^T tCe^{-rt} + TMe^{-rT} \right)$$

Increasing C and r increase the pace at which reinvested coupon values rise, reducing MD. Closed form (using series formulas):

$$MacDur = \frac{1+r}{r} - \left(\frac{(1+r) - N(c-r)}{[c(1+r)^N - 1] + r} \right) - \frac{t}{T}$$

Further,

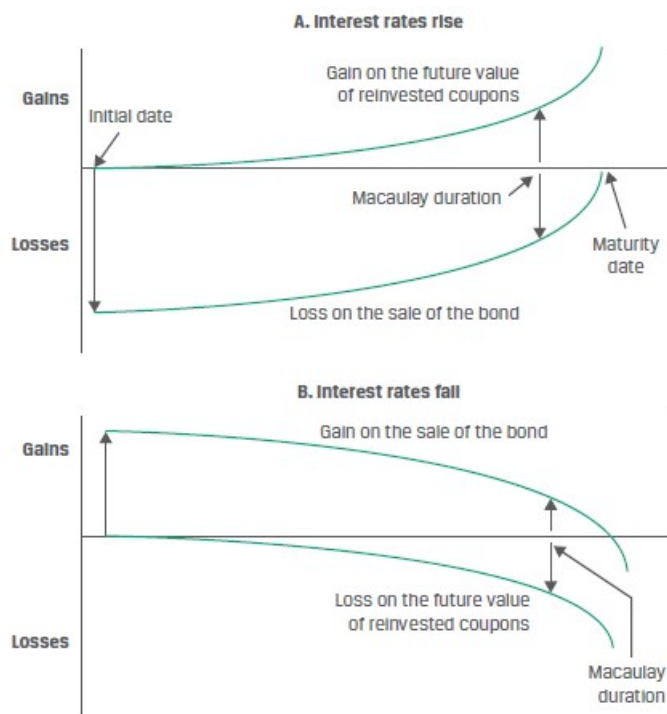
Duration gap = Macaulay duration – Investment horizon

	Dominant Risk	Source of Interest Rate Risk
Investment Horizon > Macaulay Duration	Reinvestment Risk	Falling Interest Rates
Investment Horizon = Macaulay Duration	Price Risk = Reinvestment Risk	—
Investment Horizon < Macaulay Duration	Price Risk	Rising Interest Rates

The MD indicates the bond's interest rate (and therefore, yield) risk. A bond with a higher Macaulay Duration has greater sensitivity to interest rate changes (because it takes longer to recover the bond price loss by reinvested coupons. If the investment horizon matches the Macaulay Duration, the bond is nearly hedged against interest rate risk. Any losses due to rising interest rates (price risk) would approximately be offset by gains from the reinvestment of coupons (reinvestment risk) and vice versa. MD of a zero-coupon bond is the time to maturity.

Longer bond durations mean lower reinvestment risk because more payments are weighted to later years so if the interest rate declines after purchase, more of that reinvestment happens later compared to a shorter-duration bond. Longer

bond durations mean higher price risk because it takes longer for the price to recover from higher interest rates. Both reinvestment risk and price risk relate to changes in interest rates.



MD calculation points. (1) Remember to annualize the MD by dividing by the appropriate factor if doing semiannual calculations, e.g. first half year is 1, full year is 2, next time period is 3, etc. (2) If the MD is available, e.g. 5.0 years, and it is 91 days into 30/360 in the first year (but NOT for FRNs!!!), then the MD is simply $5.0 - (91/360)$. (3) As times passes during the coupon period, the Macaulay duration declines smoothly and then jumps upward after the coupon is paid. The usual pattern is that longer times-to-maturity correspond to higher Macaulay duration statistics. This pattern always holds for bonds trading at par value or at a premium above par.

4.11 Yield-based bond duration measures and properties

Modified duration. Slope of bond price as a fraction or percentage of PV, versus YTM. Full, not flat, bond prices to changes in yields.

$$ModDur = \frac{MacDur}{1 + r}$$

The modified duration appears to remove one period of compounding from the Macaulay duration. Derivation.

$$PV = \frac{PMT}{(1+r)} + \dots + \frac{PMT + FV}{(1+r)^N}$$

Take derivative to get $\frac{dPV}{dr}$, factor out $-\frac{1}{1+r}$, then divide by PV for:

$$\frac{dPV}{PV} \frac{1}{dr} = -ModDur = -\frac{MacDur}{1+r}$$

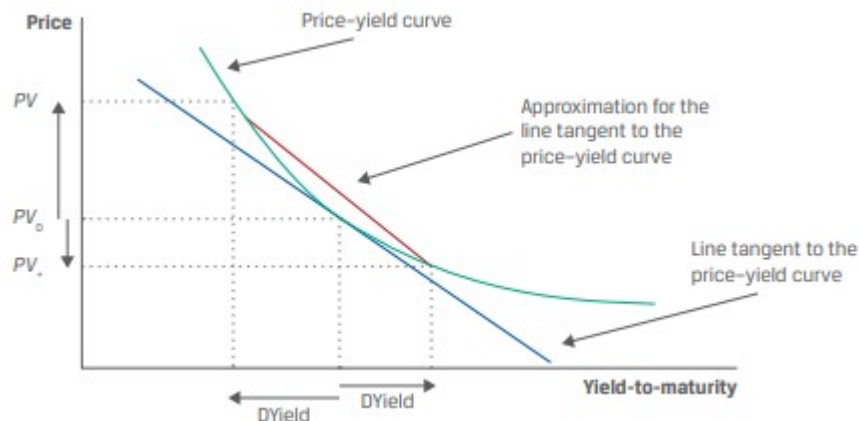
Note. For semiannual coupon bonds, can calculate the annualized MacDur first, then divide by $1+r$ where r is $YTM/2$. Or can calculate by taking semi-annual MacDur = 2 x annualized MacDur and then divide by $1+r$ where r is $YTM/2$, then divide the whole thing by 2. They're the same. They are both ANNUALIZED even if divided by $YTM/2$ in both cases.

Although ModDur appears to be MacDur divided by rate (or yield), and therefore time divided by yield, it's just that algebraically ModDur can be manipulated to include MacDur. MacDur is not actually 'inside' ModDur; ModDur is derived by differentiating PV by rate; since rate is already the denominator, the numerator is actually CHANGE IN PV. So there's nothing inconsistent, though there is a logical connection because MacDur is about interest rate sensitivity and the longer the MacDur, the more sensitive it is to rates. Multiply both sides by dr to get

$$\% \Delta PV = -ModDur \times \Delta r$$

Approximate modified duration. Below, divide by PV_0 (if unsure, let $YTM =$ coupon rate, or $PV_0 = 100$) because ModDur is a fraction:

$$ModDur \approx \frac{PV_- - PV_+}{2\Delta r} \frac{1}{PV_0}$$



Money duration is just the currency unit equivalent of ModDur:

$$MonDur = ModDur \times PV^{full}$$

where PV^{full} is the initial investment amount, e.g. 50 million USD. This means money duration changes with bond age. **USE MONEY DURATION INSTEAD OF MACDUR OR MODDUR WHEN COMPARING DECREASES IN ABSOLUTE PRICE.** It effectively neutralizes the PV_0 in ModDur. So,

$$\Delta PV^{full} \approx -MonDur \times \Delta r$$

Price value of a basis point (PVBP), which is the change in the full price of the bond by 1bp (01bp) - this is calculated MANUALLY (don't rely on money or modified duration). The 1 bp is based on the annual YTM (e.g. semiannual 7% would be semiannual 6.9-7.1%):

$$PVBP = PV01 = \frac{PV_- - PV_+}{2}$$

It is essentially the money duration with a 1bp change, so $\Delta r = 0.0001$, or 0.01%.

MacDur of non-callable perpetuities:

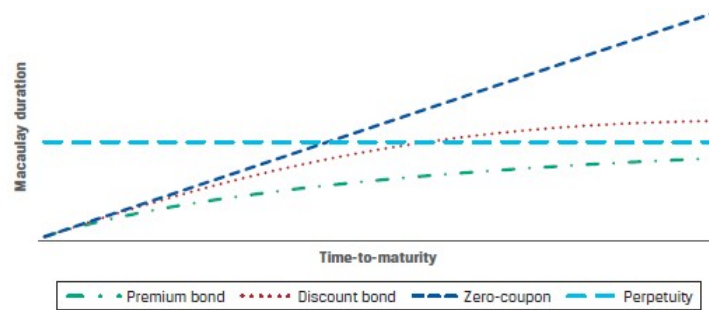
$$MacDur = \frac{1+r}{r}$$

MacDur of FRN's reset at next coupon period, so there's only interest rate risk BETWEEN coupon periods.

$$MacDur = \frac{T-t}{T}$$

This means if a bond starts with a MacDur of e.g. 10 years but and halfway into the first coupon period (which might be 1 year, 6 months, etc. into the 1st coupon period, 2nd coupon period, etc.) then the MacDur is always 0.5, which is very counterintuitive. This happens because the MacDur is always changing with FRNs.

Note for discount bonds that the MD initially rises with increasing T but eventually it begins to fall (if the graph below is extended to longer maturities) as the later payments and the principal payment become 'smaller'. This means long-dated bonds with low coupon payments can have shorter durations than short-dated bonds.



Duration properties.

	Effect on duration (interest rate risk) from an increase in feature
Coupon rate, c	↓ (Inverse relationship)
Yield to maturity, r	↓ (Inverse relationship)
Time-to-maturity, T	↑ (Direct relationship)
Fraction of current coupon period that has elapsed, t/T	↓ (Inverse relationship)

Bond durations decline smoothly as they progress towards maturity (if they're not FRNs). However, they will rise slightly upon coupon payments because the next coupon payment will now be 'further away'.

Modified duration assumes a FLAT yield curve. On the %PV/PV change versus YTM graph, the change in YTM assumes that all the yields (spot rates) are at the YTM chose on the x-axis, that the spot rates are all - every single one - equal to YTM.

4.12 Convexity

Convexity - compared to the modified duration, most bonds increase in value more with a decline in YTM and decrease less with a rise in YTM (so the more convex, the less risky the bond, and it will be priced higher - meaning its YTM will be lower). So add a convexity adjustment, which is the second term of the Taylor series. Start again with PV:

$$PV = \frac{c}{(1+r)} + \frac{c}{(1+r)^2} + \dots + \frac{c}{(1+r)^N} + \frac{M}{(1+r)^N}$$

For the first derivative,

$$\frac{d(PV)}{dr} = \frac{-c}{(1+r)^2} + \frac{-2c}{(1+r)^3} + \dots + \frac{-Nc}{(1+r)^{N+1}} + \frac{-NM}{(1+r)^{N+1}}$$

$$\frac{d(PV)}{dr} \frac{1}{PV} = \frac{-MacDur}{(1+r)} = -ModDur$$

Convert to Taylor polynomial:

$$\% \Delta(PV) = -ModDur \times dr + \dots$$

Add the second derivative:

$$\% \Delta(PV) = -ModDur \times dr + \frac{1}{2} \frac{d^2(PV)}{dr^2} \frac{1}{PV} dr^2 + \dots$$

$$\frac{d^2(PV)}{dr^2} = \frac{(2)c}{(1+r)^3} + \frac{(2 \times 3)c}{(1+r)^4} + \dots + \frac{N(N+1)c}{(1+r)^{N+2}} + \frac{N(N+1)M}{(1+r)^{N+2}}$$

$$Convexity = \frac{d^2(PV)}{dr^2} \frac{1}{PV} = \sum_{t=1}^N \frac{t(t+1)}{(1+r)^2} w_t$$

So,

$$\% \Delta(PV) \approx -ModDur \times dr + \frac{1}{2} Convexity \times dr^2$$

To calculate the Convexity, use a spreadsheet, an approximation or calculus.

$$Convexity = \sum_{t=1}^N w_t \frac{t(t+1)}{(1+YTM/n)^2}$$

where w_t is the proportion of discounted cashflows, t is the time period (e.g. year 1, year 2) and n is the periodicity. For semiannual YTM, $n=2$, and when manually calculating, divide by $2^2 = 4$, and for quarterly coupons, divide by $4^2 = 16$ because of the 'parabolic', i.e. power of 2, curvature of the Taylor series.

Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6 = Col. 2 × Col. 5	Col. 7 = Col. 2 × (Col. 2 + 1) × Col. 5 × (1 + YTM/2) ⁻²
Period	Time to Receipt	Cash Flow	Present Value	Weight	Time to Receipt × Weight	Convexity of Cash Flows
1	1.0	1.6	1.5748	0.0157	0.0157	0.0305
2	2.0	1.6	1.5500	0.0155	0.0310	0.0901
3	3.0	1.6	1.5256	0.0153	0.0458	0.1774
4	4.0	1.6	1.5016	0.0150	0.0601	0.2909
5	5.0	1.6	1.4779	0.0148	0.0739	0.4295
6	6.0	1.6	1.4546	0.0145	0.0873	0.5919
7	7.0	1.6	1.4317	0.0143	0.1002	0.7767
8	8.0	1.6	1.4092	0.0140	0.1127	0.9829
9	9.0	1.6	1.3870	0.0138	0.1249	1.2093
10	10.0	101.6	86.6875	0.8669	8.6688	92.3766
100.0000				1.0000	9.3203	96.9558
Annualized Macaulay Duration and Convexity					4.6601	24.2389

An approximation which is useful for bonds with uncertain cash flows, such as those with contingency features and default risk:

$$ApproxConv \approx \frac{(PV_-) + (PV_+) - 2PV_0}{dr^2 PV_0}$$

A fixed-rate bond will have greater convexity if it has more 'unbalanced' or 'end-heavy' cashflows, i.e. a longer maturity time, lower coupon rate, lower YTM, greater cashflow dispersion. For the same decrease in yield-to-maturity, the more convex bond appreciates more in price, while for the same increase in yield-to-maturity, the more convex bond depreciates less in price. Consequently, the more convex bond outperforms the less convex bond in both falling and rising yield environments, so it is less risky for investors. So, the more convex bond should be more valuable.

Multiply by PV to get the money convexity:

$$MoneyCon = Convexity \times PV^{full}$$

$$\Delta(PV) \approx -MoneyDur \times dr + \frac{1}{2} MoneyCon \times dr^2$$

The duration and convexity of a bond PORTFOLIO are calculated by aggregating cashflows. However, this is cumbersome so use the current bond portfolio values in a value-weighted average (PAR amount x CURRENT PRICE/PAR VALUE, e.g. 16 M of par amount at current price of 109 is 16 M x 1.09) of the individual bond durations and convexities. However, this implicitly assumes that yields of all maturities change by the same amount in the same direction, a parallel shift in the yield curve. For individual bonds, the duration and convexities only apply to that one bond's particular YTM which is for that bond's maturity time. However, a bond portfolio has bonds of different maturity times, so the dr is for all bonds of those different maturities - meaning we essentially assume a parallel yield curve shift instead of what is actually observed, e.g. steepening, flattening, or twisting yield curve.

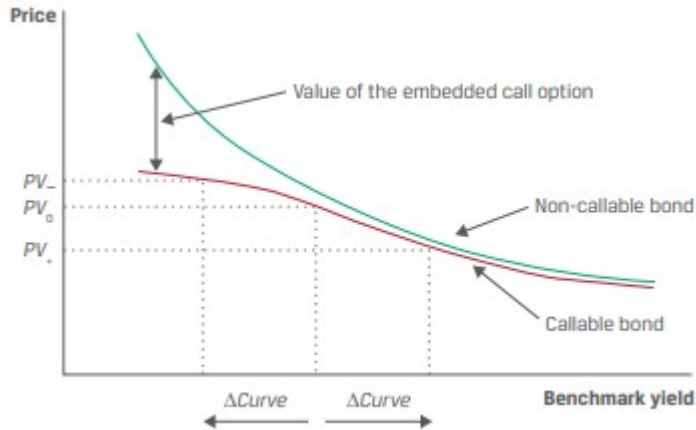
4.13 Curve-Based and Empirical Risk Measures

Note: curve durations are not classified as yield durations.

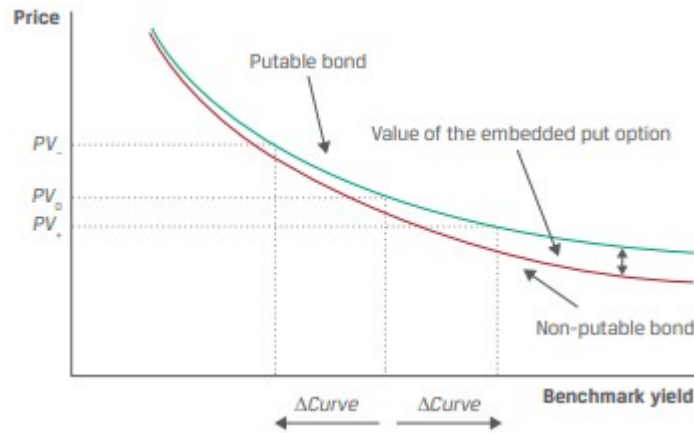
Bonds with embedded options. If bond has embedded option, there's no well-defined maturity time, so not appropriate to use standard YIELD-based durations and convexities for assessing interest rate risk; use GOVERNMENT CURVE-based measures, i.e. EFFECTIVE durations and convexities based on PARALLEL shifts of government yield curves.

Callable bonds. Have negative convexity at lower YTM's, so below the normal curve. Negative effective convexity implies that the bond price will decline

by a greater amount with an upward shift in benchmark yield than it will increase with a decline in the benchmark yield. If the yields decline, bond issuer can call the bonds and refinance at lower rates.



Putable bonds. Above the normal curve. Always positive convexity.



Use effective duration and effective convexity based on benchmark government yield curves - so the changes are in the curves, not individual YTM's. Important for MBSs, as refinancing often occurs when yields decline.

$$EffDur \approx \frac{(PV_-) - (PV_+)}{2\Delta Curve \times PV_0}$$

$$EffConv \approx \frac{(PV_-) + (PV_+) - 2PV_0}{(\Delta Curve)^2 \times PV_0}$$

$$\% \Delta PV^{full} \approx (-EffDur \times \Delta Curve) + \frac{1}{2} EffConv (\Delta Curve)^2$$

Further, the PV_- , PV_+ are determined from options pricing models which consider (1) the length of the call protection period, (2) the schedule of call prices and call dates, (3) an assumption about credit spreads over benchmark yields (which also includes any liquidity spread), (4) an assumption about future interest rate volatility, and (5) the level of market interest rates (e.g., the government par curve). Note: a parallel shift in the government par curve produces a non-parallel shift in spot (zero) rates and one-year forward rates.

Key rate duration (partial duration). Accounts for non-parallel government yield curve change. Each key rate duration is defined for a SPECIFIC maturity k because of non-parallel yield curve shifts:

$$KeyRateDur_k = \frac{-1}{PV} \frac{\Delta PV}{\Delta r_k}$$

$$EffDur = \sum_{k=1}^n KeyRateDur_k$$

Use the benchmark tenor that best matches the bond tenor, e.g.

Bond	Tenor	Position Size	Key Rate Duration
A	1 year	\$200,565,245	0.645
B	5 years	\$201,042,132	1.483
C	10 years	\$202,673,298	2.158
D	20 years	\$202,588,801	2.982

Information on the forecasted benchmark government par curve changes at the various maturities are as follows:

Maturity	1 year	5 years	10 years	20 years
Expected change	-23 bps	-25 bps	-18 bps	-10 bps

If we take the product of the key rate duration and curve shift at a certain maturity, we can forecast the change in the price of the bond. For example, for the one-year maturity, this would be $-0.645 \times -0.0023 = 0.148\%$. The calculation for all the bonds is shown below:

Maturities	1 year (Bond A)	5 years (Bond B)	10 years (Bond C)	20 years (Bond D)
% change in bond price	0.148%	0.371%	0.388%	0.298%

Empirical duration. Use statistics and history to derive YTM-price relationship. The equation-based durations are analytical durations - analytical durations assume government benchmark yields and yield spreads are independent

of each other. Empirical durations are based on statistics and history. The yield depends on government bond yields and credit spreads, and we've assumed they're independent of each other, but in reality they're not - and empirical durations can attempt to address, e.g. during stressed market conditions credit spreads widen as corporate debt is sold and government bonds are purchased. Analytical durations are appropriate for government bonds as the policy rates effectively govern bond prices, but empirical durations are better for corporate debt.

Duration comparisons. Analytical durations assume government benchmarks and yield spreads are independent of each other - overlay the modified duration for a government bond with another modified duration - this time for corporate bonds. Empirical duration of corporate bond is generally LOWER (i.e. less steep) than the government version. The same macroeconomic factors driving government bond yields lower in a market stress scenario will cause high-yield bond credit spreads to widen because of an increase in expected default risk. Since credit spreads and benchmark yields are negatively correlated under this scenario, wider credit spreads will partially or fully offset the decline in government benchmark yields, resulting in lower empirical duration estimates than analytical duration estimates. Analytical duration is therefore higher (less negative, shallower slope).

4.14 Credit risk

Default probability judged on C's of credit. Top-down: country, conditions, currency. Bottom-up: capacity, capital, collateral, character, covenants. Illiquid borrower can't meet obligations as they become due; insolvent borrower's assets are less than liabilities.

While the likelihood of default for investment-grade borrowers is typically well below that of a high-yield issuer, the investor's loss in the event of a default for secured debt is usually lower than unsecured debt due to the secondary source of repayment for secured bonds.

Measuring credit risk. Expected loss EL is equal to probability of default POD (typically, over the next 12 months) times loss given default LGD. LGD (=loss severity) is expected exposure EE times 1 minus the recovery rate RR of the debt.

$$EL = POD \times LGD = POD \times EE(1 - RR)$$

The G-spread or credit spread is approximated by EL since the government bond is risk-free.

The POD depends on higher profitability (as measured, for example, by EBIT margin) and higher coverage (EBIT to interest expense) coupled with lower leverage (debt to earnings or cash flow ratios, such as debt/EBITDA or cash

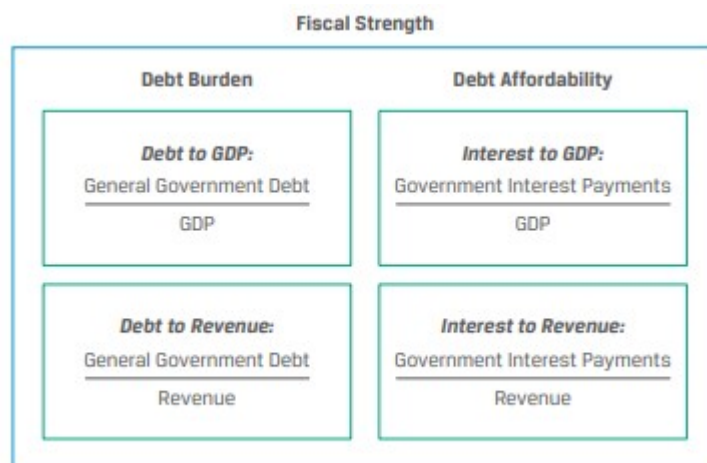
flow to net debt) are associated with lower POD and higher credit quality. The LGD depends on degree of security. Credit ratings. Moody's, Fitch, and S&P are the largest - there are smaller and regional ones too. Baa3, BBB-, BBB- are the lowest IG bonds. Credit migration risk is the risk of lower credit ratings in the future. Credit ratings tend to lag market conditions, though credit rating outlooks do a better job. Credit ratings seek primarily to assess the expected loss, whereas for distressed debt, pricing is focused more on default timing and expected recovery rates.

Yield spreads. Credit spread risk - lowest at top of macroeconomic cycle. High-yield bonds: have low correlation with sovereign bonds and IG bonds, so use for diversification and returns; more capital appreciation during macro cycle; equity-like returns with lower volatility. The credit risk of a corporate bond will reflect credit risk, market liquidity risk (which depends on credit rating and how much debt outstanding and how much of it is traded), and tax implications. For a given YTM, subtract off the benchmark yield (e.g. German bond yields), then take the credit spread and subtract out the liquidity yield (offer YTM minus bid YTM).

4.15 Government issuers

Sovereign immunity. Unlike corporates, can't usually force government to default. In practice, IMF often involved to help restructure debt. Unlike corporations, rely on economic data - but this can be thin and subject to politics, so use broader indicators like debt burden, debt affordability, economic indicators, trade balances, etc.

Fiscal strength.

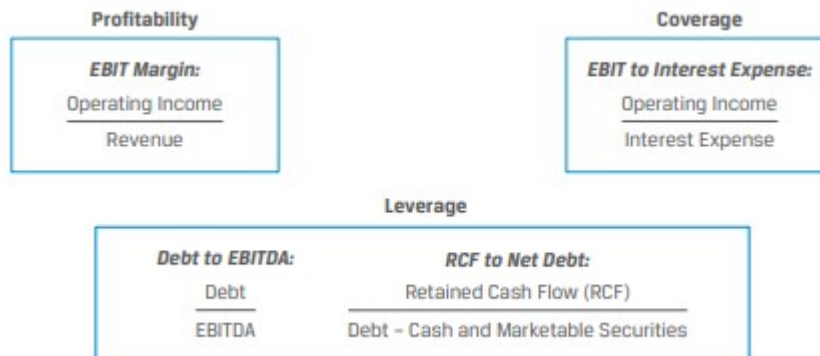


External stability.



Government agencies, intergovernmental organizations, local governments can also issue debt.

4.16 Corporate issuers



Note:

$$\text{Debt} : \text{Equity} = \frac{\text{Debt}}{\text{Equity}}$$

$$\text{Debt} : \text{Capital} = \frac{\text{Debt}}{\text{Equity} + \text{Debt}}$$

Priority of claims: mortgage (specific property) or first lien (specific property and intangible property), senior secured, junior secured, senior subordinated, subordinated, junior subordinated. Unsecured debt is non-dilutive. If the value of the collateral falls short of the total amount of pari passu secured debt, the balance will rank pari passu with unsecured creditors in insolvency or restructuring. All creditors at the same debt seniority level are treated as one class;

thus, a senior unsecured bondholder whose debt is due in 30 years has the same pro rata (pari passu) claim in bankruptcy as one whose debt matures in six months. In a bankruptcy proceeding, there is an incentive often to negotiate and compromise due to time and competitors gaining market share - and it can be expensive and complicated with legal and accounting issues - senior creditors may partially compensate junior creditors to expedite this process, so that junior credits may get more than they are entitled to.

Credit ratings. There may be corporate family rating (CFRs) or issuer ratings, in contrast to corporate credit ratings (CCRs) which are issue ratings. Issuer ratings are usually for senior unsecured debt (the most common form) and reflect POD for general company creditworthiness. Often, these ratings have the same POD but different LGDs, so the issue ratings are notched from the issuer ratings.

Structural subordination. If a subsidiary issues bonds and another subsidiary of the same company issues bonds, they are structurally subordinated. The cashflows from the subsidiaries first service their bonds before cashflow flows up to the head company.

Moody's defines credit risk as "the risk that an entity may not meet its contractual financial obligations as they come due and any estimated financial loss in the event of default or impairment." Moody's concentrates on both POD and LGD, so it looks at EL. However, S&P and Fitch look primarily at POD.

4.17 Fixed income securitization - securitized debt

Reference portfolio or collateral is put into an entity and then cashflows sold off in tranches. Frees up capital that would otherwise be used to backup the debt and can get a lump-sum payment for 'selling' those loans. Some examples. Covered bonds: not a separate legal entity, on the balance sheet. Pass-through securities: Separate legal entity, off the balance sheet. Bonds with structural enhancements: pass-through securities with extra features.

Securitized ABSs are like bonds, NOT like equities.

Securitization allows bank to remove the illiquid loan from their balance sheets, reducing capital requirements and lending risks, but also generating fees from loan origination. Investors don't have to buy bank shares or lend to the bank; they don't need to underwrite loans (which requires expertise and is costly) - they can just buy from the SPE where the loans are pooled. Investors also get a wider choice of products. So, securitized debt increases the liquidity of the loan. Individual companies can pool their assets and offer securitized debt as well, increasing debt liquidity and removing the illiquid debt from their balance sheets. Cashflows from the securitized debt also become steadier and less volatile.

Covered bonds have low risk because of at least 3 elements. (1) Eligibility Criteria. Strict rules govern what types of assets can be used as collateral for covered bonds. This ensures the underlying assets are of high quality and low risk, reducing the likelihood of losses for bondholders. (2) Dynamic cover pool. The pool of assets backing the covered bonds isn't fixed. The issuing bank regularly replaces weaker assets with stronger ones, maintaining the overall quality of the collateral. (3) Redemption Regime in Event of Sponsor Default. If the issuing bank (the "sponsor") goes bankrupt, there's a clear plan for how bondholders will be repaid. This often involves an independent trustee taking over and using the collateral to repay bondholders, minimizing losses.

Risky asset pools can be enhanced using overcollateralization (internal credit enhancement), cash collateral (external enhancement), letter of credit (external credit enhancement).

Risks. Cashflow timing risk - extensions and contraction risk. Risk of the collateral. GFC is related to these risks.

Parties. Seller of collateral is the depositor (the original lender). SPE, which is independent of any bankruptcy of the depositor ('bankruptcy-remote' but this can be legally challenged). Servicer of the loan - can be the SPE. Trustee (disinterested trustee) holds accounts and provides fund reports.

4.18 Asset-backed securities

Covered bonds. Bonds are not transferred to SPE; they are ringfenced. Covered investors have recourse to the covered pool of bonds first, and the second the issuing institution (e.g. bank). Usually lower credit risk, fixed payments (unlike ABSs, which are usually floating and higher credit risk); allow the bank to issue credit with higher repayment than simply issuing bank debt. Usually rules to replenish if there is a default in the asset pool, monitored by third party, often regulated, higher-quality and overcollateralized (e.g. higher LTVs, so that it is higher than face value of the bonds) to absorb losses. Rules for covered bonds usually, e.g. hard-bullet covered bonds (if default payment, then payments accelerated), soft-bullet covered bonds (if default payment, then payments postponed and default postponed), conditional pass-through securities (turn into pass-through securities if not all payments made, i.e. third party takes payments and passes onto investors) if past maturity date. Usually only one tranche.

Credit enhancements. Absorbs defaults in the collateral pool. Internal credit enhancements: Overcollateralization, excess spread (e.g. underlying loan coupon is higher than the ABS coupon), subordination (credit tranching), rapid amortization (e.g. if credit risk of collateral pool declines suddenly). External credit enhancements: e.g. financial guarantees.

Non-MBS. Amortizing ABSs have lockout or revolving periods (when investors only receive financing charges and fees) are when capital is replenished if loans are paid off early, but then after that an amortization period. If there isn't enough capital to replenish during the revolving period or if macroeconomic conditions change, it might trigger a rapid amortization provision where investors get paid early - this can be helpful for investors if the macro environment is uncertain. Prefunding periods allow the trust to buy more pool assets for diversification and theft.

Collateralized debt obligations (CDOs). Similar to ABS, except broad range of debt in the asset pool, actively managed by a collateral manager. Collateral pool can keep changing, even after the ramp-up period. CDOs are a generic term for a diversified pool of assets backed by one or more debt obligations: ABS, RMBS, CMBS, structured finance CDOs and others.

- CBOs. Collateralized bond obligations. Corporate and emerging market bonds.
- CLOs. Collateralized loan obligations. Secured - collateralized - leveraged bank loans, i.e. loans to highly-leveraged companies.

The equity tranches (i.e., the bottom tranches) are like leveraged transactions. Different overcollateralization ratios for each tranche, if triggered then the senior tranches get payments and the CDO is essentially deleveraged.

Mortgage ABSs usually gradually decline and aren't replenished. Non-mortgage ABSs, CDOs, and covered bonds have distinct end dates for dissolution.

Note: non-mortgage ABSs and CDOs aren't usually amortizing. Covered bonds can include residential and commercial mortgages.

4.19 Mortgage-backed securities

Prepayment risk. Prepayment risk is the risk that the borrower does not repay the principal (or a portion of the principal) as scheduled. Contraction risk occurs when the borrower repays the principal faster than anticipated, while extension risk occurs when the borrower repays the principal more slowly than planned.

Contraction risk. Often happens when interest rates are lower, when borrowers refinance at lower rates. Investors have to (1) reinvest proceeds, often at lower rate than in ABS, and (2) reduces potential price appreciation.

Extension risk. Often happens when interest rates are higher, when borrowers delay or do not refinance or buy new homes. So, this reduces the payment values, which are furthermore extended, which are still further discounted at higher rates.

Prepayment risk can be addressed by time-tranching - first tranche receives (with a short maturity period), then second (with longer maturity period), etc., so get different maturities for different tranches. Senior tranches have less prepayment risk overall, as well as less default risk.

Mortgages. Look at loan-to-value (LTV) and debt-to-income (DTI; use monthly values) to determine rate. Residential MBSs (RMBSs) may or may not be guaranteed by the government or other agencies (agency RMBS) versus government-sponsored entity guarantee or private entity guarantee (non-agency RMBS), but non-agency RMBS largely disappeared after the GFC. Fannie Mae and Freddie Mac are GSEs, they provide credit risk guarantees (a credit enhancement) but are not backed by the state. Ginnie Mae is however federally-related and therefore backed by the US government; Ginnie Mae guarantees RMBSs.

Recourse loans - where the lender can claim against the outstanding mortgage balance through personal assets and property; non-recourse - lender can only go after the property. Negative equity (underwater mortgage) when equity less than loan value. If mortgage is non-recourse, often strategic defaulting but less likely if recourse loan. Prime loans to borrowers with low default probabilities; subprime loans to those with high default probabilities.

Mortgage pass-through securities. Pass-through MBSs. Pass-through rates (coupon rates) received by investors is lower than the weighted average coupon (WAC). WAC is calculated by

$$WAC = \sum_i^n \text{coupon rate}_i \frac{\text{current balance}_i}{\text{total current balances}}$$

where i is the i-th mortgage in the mortgage pool of n assets. The weighted average maturity is

$$WAM = \sum_i^n \text{time left}_i \frac{\text{current balance}_i}{\text{total current balances}}$$

Collateralized mortgage obligation (CMO). MBS with collateral manager. Typical is sequential-pay CMO (which is just time tranching). Actual payments and timing not known, only generally known. Senior tranches protected against extension risk (lower tranches provide this protection as lower tranches have longer maturities) but face contraction risk, lower tranches protected against contraction risk (borne by senior tranches) but face extension risk.

Example time tranches:

1. Z-tranche (also called accretion or accrual bonds) - no interest or principal payments until a start date (instead, they get coupon during accrual)

period), often the maturity date is over 20 years but face no or little reinvestment risk if market yields decline. Usually last tranche.

2. Principal-only (PO) securities - no interest, when market yields decline PO bond values rise; hedge against interest risk.
3. Interest-only (IO) securities - interest only; no face value. Often companion to PO securities. Use this to hedge against interest risk.
4. Floating-rate tranches. Interest rate follows a benchmark or is MRR. Often have floor and cap. Can be used to hedge against interest rate changes.
5. Planned amortization. Very steady interest and maturity dates and values. Prepayment risk absorbed by other tranches.
6. Residual tranche. Risky. Banks avoid due to capital requirements.

CMBS. Commercial mortgages are similar to residential, except (1) call protection (so acts more like a corporate bond instead of an MBS) and (2) balloon payments. Call protection can be at MBS-level, where early tranches are protected at cost of junior tranches; or can be at individual mortgage level, where lenders can (a) use prepayment lockout periods, (b) prepayment penalties, or (c) defeasance (can prepay but lender must pay the issuer for assembling replicating portfolio of government securities). Commercial mortgages may not be fully amortizing; they are often partially amortized and have a balloon payment at the end. Commercial mortgagees may fail to pay the giant balloon payment, so they often have a 'workout-period' extension - which is an extension risk. Further, CMBs are (3) more risky because there are only a few large commercial mortgages, and (4) only non-recourse loans. Use the LTV and DSC (debt service charge coverage ratio):

$$DSC = \frac{\text{Net operating income NOI}}{\text{Debt service}}$$

Debt service is the annual amount of interest payments and principal repayments. NOI is defined as the rental income reduced by cash operating expenses and a non-cash replacement reserve reflecting the depreciation of the property over time.

$$NOI = (\text{Rental income} - \text{Cash operating expenses}) - \text{replacement reserves}$$

NOI excludes principal and interest payments on loans, capital expenditures, depreciation, and amortization.

5 Corporate issuers

5.1 Organizational Forms, Corporate Issuer Features, and Ownership

Three general forms of organization: companies (businesses), non-profit non-governmental organizations, government.

Businesses: sole trader/proprietorship, partnership, LLs. SPs are most in number but problems with access to financing and big risk-taking.

Partnerships: general partnerships (like many sole proprietorships banded together, unlimited liability, don't necessarily need a written agreement, everyone manages), limited partnerships (at least one general partner with unlimited liability and others are limited partners), LLPs - limited partnerships without a general partner (in the US, only allowed for certain professions and have restrictions on equity, etc.). Partnerships are generally pass-through businesses, i.e. taxed at individual and not business level - regardless of whether the income was passed to partners or retained in the business for reinvestment. Each partner's liability is INDIVIDUALLY limited - actions of one partner insulated from others.

LLs. Private limited companies (Ltd, LLCs, KKs, GmbH) - like limited liability partnerships - but owners are shareholders. Shareholders vote for board of directors (who appoint managers and authorize dividends); distinguishes between owners and managers. Private LCs - limits on number of shareholders, have to vote to transfer ownership (i.e. shares), and pass-through businesses (in the US; different in the UK - where difference between ltd and LLP depends on personal and corporate tax rates). Cannot offer shares to public. Public LCs (corporations, PLCs, Inc) - no restrictions on number of shareholders, voting to transfer ownership not necessary. Public LCs (both privately-held and widely-held) are disadvantaged in that profits are taxed at business level AND shareholder level (but in some countries shareholders get a tax credit and in other countries corporate tax is zero), but if profits reinvested in company then no shareholder taxes (so corporations are good for investment). Corporation separates ownership-management, so that it can get a wider pool of capital - from those who don't want to manage. Further, shares can be divided into very small amounts so that it's easier for people to purchase. A general partner in a limited partnership can itself be a public LC, leading to a dual-structure which allows profit-sharing AND partnership control AND limited liability.

Despite double-taxation, corporations are advantageous because they can access greater capital pool - less management expertise needed and small capital can be used to purchase a small number of shares. Moreover, corporate taxes can sometimes be lower than personal taxes.

5.2 Public v private corporate issuers

Private corporations are unlisted; public corporations are listed (at least some of its shares are listed). Note this distinction is different from the private versus public limited companies. Actively-traded shares are 'free float' (actively traded; not held by insiders, strategic investors, or sponsors), comprising a proportion of the total outstanding shares. Private corporations issue 'private placements' to sophisticated or accredited investors. Private corporations can go public in 3 ways: IPO, direct listing, acquisition (by a listed company or SPAC).

A direct listing does not involve an underwriter and no new shares are issued, so no capital is raised. Instead, the company simply lists existing shares on an exchange at a price determined by the market, and shares become available to the public as they are sold by existing shareholders. Direct listings, compared with IPOs, may take place more quickly and at a lower cost.

SPACs. Like reverse mergers (dormant listed company acquires private company, makes it public). Usually a shell company made to acquire a company (but investors and managers don't know which companies will be acquired but generally know the kind of targets). Investor proceeds from listing are kept in trust, to be used only for acquisition and time-limited (e.g. 18 months) to spend proceeds.

Going private. Companies can go private, usually by borrowing and buying shares at a premium. More control, less costs from regulatory disclosures.

Developing countries - more going public with money flowing in. Developed countries - more going private or staying private because of mergers and acquisitions and more private equity.

5.3 Investors and other stakeholders

Debt and equity. Debt is temporary and more secure; equity is residual and risky. Debtholder and shareholder interests can clash. Company issuing the debt and making payments can use interest payments as expense and therefore tax-deductible; can't do with dividends.

D/E is a SOLVENCY measure. Higher D/E ratio indicates greater chance of insolvency because of more debt payments.

Debt v equity. If the return on assets ($ROA = NI$, or NI before interest costs, or $EBIT$ divided by total assets; total assets = debt + equity) or return on incremental investment is higher than interest rate on debt (debt cost divided by debt), then ROE will be higher with more debt - because need less equity for the same amount of profit. But cash returns are usually highest (cash from company treasury if any) as it doesn't dilute and has no interest cost.

From an issuer's perspective, debt financing is less costly but involves greater risks than equity financing because of leverage. From an investor's perspective, stocks are riskier than bonds, also because of leverage but also because of unsteady payments and no security. If there is too much financial leverage, share prices will decline and shareholders use vote to reduce. While debt financing adds risk to shareholders, shareholders might prefer it because additional equity issuances mean dilution. The downside of dilution may be offset by an expectation that the firm will generate enough incremental profit to compensate.

Equity owners seek to maximize firm value (to increase their payoff), so could seek risk and demand dividends which increase leverage and risk to bondholders. Bondholders seek to ensure stability and low-risk projects with steady cashflows; they have no voting rights so request covenants such as limits on financial leverage.

Stakeholders - shareholders, debtholders, government, managers, employees, creditors, NGOs, etc. Shareholder theory of corporate governance - stakeholders are considered only to the extent they increase shareholder value. Stakeholder theory of corporate governance - all stakeholders are considered and balanced.

Private debtholders (more highly specialized and have more knowledge like banks; also called private lenders; often have much more influence over debt issuers) and bondholders ('public' debtholders like institutional investors and asset managers).

Board of directors. Represent shareholders. Appoint managers. Some managers can be directors; others are independent directors - independent directors often better represent minority shareholders. Some countries like Germany have a 2-tier structure with a supervisory board overseeing the BOD.

5.4 Corporate governance

Principal-agent relationships result in agency costs. So, shareholders ask for higher returns and debtholders ask for higher interest/risk premia.

Long-term debtholders impose more conditions than short-term debtholders.

Shareholder activists. Crystallizes company's strategic direction. Often use shareholder derivative lawsuits. A shareholder (stockholder) derivative suit is a lawsuit brought by a shareholder or group of shareholders on behalf of the corporation against the corporation's directors, officers, or other third parties who breach their duties.

A tender offer requires a majority of the shareholders to accept. A proxy fight aims to replace a good portion of the target's uncooperative board members. An

acquirer may choose to simply buy enough company stock in the open market to take control (hostile takeover by a tender offer) or by proxy fight. However, a company might adopt a staggered board or poison pill (shareholder rights plans) which discourage the unwelcome accumulation of company stock above a threshold by diluting an activist buyer's stake with discounted share sales to the other shareholders.

Committees. Creditor committees for bankruptcy, ad hoc committees before bankruptcy. BODs often have (and are often required by exchanges to have) (1) audit, (2) governing (also called nominating for nominating BODs), and (3) compensation (of BOD; so often comprised of INDEPENDENT directors) committees to help it. There are also risk committees. Governance committees watch out for laws and regulations - it's not the risk committee.

5.5 Working capital and liquidity

This is short-term assets and liabilities. Cash cycle - company has an operating cycle of buying materials, making with labor, and then producing, selling and collecting cash.

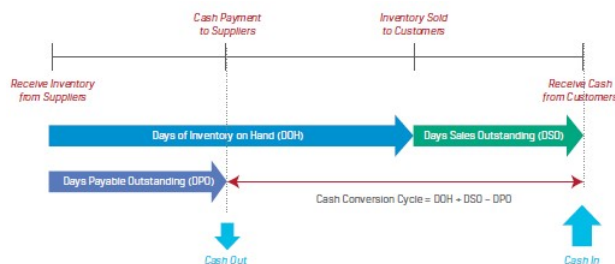
The amounts of time that accounts payable, inventory, and accounts receivable are outstanding on the BS are days payable outstanding (DPO), days of inventory on hand (DOH), and days sales outstanding (DSO).

$$DOH = \frac{\text{Average inventory} \times \text{No. of days}}{\text{COGS}} = \frac{\text{No. of days}}{\text{Inventory turnover}}$$

$$DSO = \frac{\text{Average AR} \times \text{No. of days}}{\text{Revenue}} = \frac{\text{No. of days}}{\text{Receivables turnover}}$$

$$DPO = \frac{\text{Average AP} \times \text{No. of days}}{\text{COGS}} = \frac{\text{No. of days}}{\text{Payables turnover}}$$

Cash conversion cycle: the amount of time between an issuer's paying its suppliers and receiving cash from customers (i.e., the time between derecognition of accounts payable and derecognition of accounts receivable). Short or negative CCCs are desirable. Cash conversion cycle = DOH + DSO - DPO. A negative cash conversion cycle means that inventory is sold before you have to pay for it - in other words, your vendors are financing your business operations.



Operating cycle = DOH + DSO. So, CCC = Operating cycle - DPO.

Trade credit. Supplier terms, e.g. 2/10 net 30 means pay the supplier within 10 days to receive a 2% discount, or simply forgo the discount and pay the supplier in 30 days. This is the same as borrowing from the supplier the discount, e.g. if a business takes \$100 of goods from a supplier at 2/10 net 30, the cost is effectively \$98 if paid in 10 days, or \$100 in 30 days - so it's a 2% loan for 20 days. The discount is:

$$100/98 = 1.0204$$

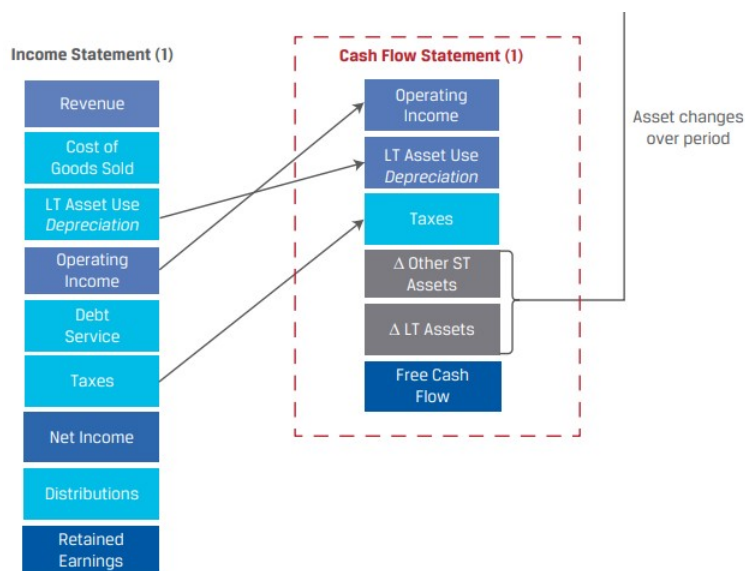
which is for 20 days, so the EAR is 44.6%:

$$(1.0204)^{365/20} - 1 = 0.446$$

which is compared to the EAR of a bank loan, e.g. 8%. Since here the bank loan is lower, borrow from the bank and pay early to get the discount.

Total working capital = current assets - current liabilities. Net (operating) working capital (removes items less related to operation cycle) = (operating) current assets (AR, inventory, prepaid expenses; excluding cash and marketable securities) - (operating) current liabilities (APs; excluding short-term and current debt). Usually normalize by dividing by sales. Short CCCs associated with low WCs (e.g. low DOH means low inventories which are assets).

Liquidity.



Primary liquidity sources, e.g. cash and sell securities. Secondary liquidity, e.g. issue securities and renegotiate supplier contracts or file for bankruptcy. Liquidity can be negatively affected: drag on liquidity (slowing cashflows) and pull

on liquidity (faster cash outflows).

The main difference between primary and secondary sources of liquidity is that using a primary source is not likely to affect the normal operations of the company, whereas using a secondary source might result in a change in the company's financial and operating positions. Secondary sources used by companies include negotiating debt contracts, such as relieving pressures from high interest payments or principal repayments, waiving debt covenants or suspending dividends.

Obsolete inventory is a drag on liquidity (as it has become not usable and therefore not sellable), while reduced credit limits (e.g. trade finance) is a pull on liquidity.

5.6 Capital investments and allocation

In contrast to WC, this is about long-term.

Long-term assets: capital investments (capital projects), more than 12 months.

(1) Going concern - capital for maintaining current business, e.g. replacing old assets, and this capital is low-risk. Match capital funding with project life, e.g. a utility company may issue a 30-year bond to finance replacement power generation equipment with an expected useful life of 30 years. This so-called match funding approach reduces financing risk, because funding long-term assets with shorter debt obligations introduces rollover risk, or uncertain financing cost or availability during the project before the capital investment reaches the end of its useful life. A company that borrows for longer than necessary may either pay a higher long-term rate of interest or face the cost of buying back debt in the future that is no longer needed. Analysts often estimate that annual maintenance capital expenditure is equal to the amount of depreciation and amortization expense reported on the income statement. The accuracy of this estimate depends on how closely the expected useful life of assets approximates actual useful life and whether the historical cost of an asset approximates its replacement cost. (2) Addressing regulation. (3) New business lines. (4) Expansion of existing business.

Capital allocation process: (1) idea generation, (2) investment analysis, (3) planning, (4) monitoring and review.

NPVs and IRRs. IRRs calculated with $NPV=0$. IRRs need to be compared to the required rate of return. If IRR is higher, then possibly take the project; if lower, then reject. USE NPV OVER IRR! Note IRR must be higher than the required return which is WACC.

Note: if cashflow sign changes more than once, don't use IRRs (especially on

calculators) because there is more than one root to the polynomial equation! Instead, in these cases, use NPVs. If two projects exist with one project being higher in NPV but lower IRR, go with the NPV. This could happen if: (a) A larger project with a higher total investment may have a higher NPV, even if it has a lower IRR; (b) A project with a higher NPV may have a lower IRR if it generates most of its cash flows in the later years of its life; (c) If one project is perceived as riskier than the other, it may have a higher required rate of return (IRR) to compensate for the increased risk. Using NPV is the most theoretically sound. However, don't look at project NPVs in isolation; may need to look at overall context, e.g. ESG.

Return on invested capital (ROIC; also return on capital employed ROCE). Use ROIC instead of NPV or IRR which require granular information. However, ROIC is aggregated.

$$ROIC = \frac{\text{After-tax operating profit}}{\text{LT liabilities} + \text{equity}}$$

$$ROIC = \frac{\text{After-tax operating profit}}{\text{Sales}} \times \frac{\text{Sales}}{\text{LT liabilities} + \text{equity}}$$

The left is a margin and the right side is a turnover. ROIC has disadvantages compared to NPV or IRR: (1) backward-looking, (2) ROIC, unlike NPV and IRR, is an accounting, not cash-based, measure. Operating profit and cash flows can differ materially, owing to the recognition of certain items and the difference between depreciation and capital expenditures.

Consider cognitive errors (e.g. inertia (increasing capital allocation despite declining returns), ignoring costs of internal financing and inflation) and behavioral biases (e.g. inertia in maintaining capital budgets, using short-term or compensation-based measures such as EPS) in analyzing capital allocation decisions by companies.

Proceeding with a project having an IRR which is lower than ROIC will reduce the ROIC.

Real options - company's capital allocation options will change with time. Similar to financial options, such as derivatives, real options grant a firm the right, but not the obligation, to take an action in the future. A company should only pursue (or exercise) a real option if it is value enhancing. So, project NPV (current value) = NPV (without options) + option costs (i.e. downside cost) - option values (i.e. upside value).

- Timing options - when to invest.
- Sizing options - abandonment and growth options
- Flexibility options - price-setting and production
- Fundamental - depends on e.g. commodity prices

5.7 Capital structure

Cost of capital. WACC. Cost of debt (after tax because it's an expense), and cost of equity - each is dependent on the investor required rates of return. The weightings of debt and equity are either their market value proportions or target weights provided by management, which are typically on a book value basis. Market value weights are more commonly used because book values reflect historical prices of debt and equity, while investors' opportunity costs are based on current market prices of debt and equity.

Issuers look to reduce WACC and match duration of assets. The total amount and type of financing needed (thus, the weightings in the WACC calculation) generally depend on the issuer's business model and its position in the corporate life cycle.

Capital-intensive businesses often need a lot of equipment, like PPE. They often lease this equipment. Implicit in the lease is an interest rate that is often much lower than what the lessee would have been lent cash for, because the leased asset is collateral and because lessors are often large companies that can borrow cheaply. Fungible equipment like office equipment can be instead bought and used for secured debt very cheaply, though this technique can't be used for more specialized equipment of little value to investors. For some industries, like banks, they have regulatory capitalization requirements. Capital-light businesses such as Uber and other services can reduce their capital requirements even further by stock compensation.

Feature / Phase	Startup	Growth	Mature
Revenue Growth	Initial	Rising	Slowing
Free Cash Flow	Negative	Rising	Stable
Business Risk	High	Medium	Low
Debt Availability	Little or None	Rising	High
Debt Type	Convertible	Secured	Unsecured

Companies often start with lots of equity funding then as they mature transition to debt funding.

Operating leverage.

$$\text{Operating leverage} = \frac{FC}{FC + VC}$$

Note this is different from degree of operating leverage. High operating leverage means variable costs are less influential; so sales has limited effect. Firms with high operating leverage experience a greater change in operating profits for a given change in revenues. This is more risky for debt. Financial leverage and interest coverage are also used to evaluate debt risk. Interest coverage = Profit

before interest and taxes/Interest expense.

Note: tax shield is

$$\text{Tax Shield} = \text{Value of Tax-Deductible Expense} \times \text{Tax Rate}$$

5.8 Modigliani-Miller

Theory: choice of capital structure doesn't affect firm value under assumptions. Assumptions: investors agree on the cashflows of a company, perfect capital markets (no transaction, tax, bankruptcy costs, no asymmetric info), risk-free rate (everyone has access to it), no agency costs (agency costs can lead to debt overhang where managers become risk-averse with lots of debt and perquisite consumption where managers consume perks because there are few shareholders with high debt), independent decisions (financing and investment decisions are independent).

There is MM1 and MM2.

MM1 (MM without taxes). Value of a company is just its discounted cashflows, so capital structure can't affect it. If it could, there would be arbitrage and/or a shareholder can borrow money and buy shares to get that optimal balance. So,

$$V = V_L = V_U = D + E$$

Since

$$r_{WACC} = \frac{D}{D + E}r_d + \frac{E}{D + E}r_e$$

and the cost of equity is equal to the unlevered WACC (r_0), i.e. with $D=0$,

$$r_{WACC} = \frac{E}{0 + E}r_e = r_e = r_0$$

then

$$r_0 = \frac{D}{D + E}r_d + \frac{E}{D + E}r_e$$

and rearranging,

$$r_0(D + E) = Dr_d + Er_e$$

$$E(r_e - r_0) = D(r_0 - r_d)$$

$$r_e = r_0 + (r_0 - r_d)\frac{D}{E}$$

so, increasing debt must be perfectly offset by an increase in equity cost (if debt interest is high, then return on equity is low; and vice versa).

MM2 (MM with taxes). Cost of equity capital linearly rises with increasing

leverage. Taxes (only corporate taxes; not personal taxes) lead to interest payments being shielded from income (tax shield) by tD . So, leveraged value is higher by the tax shield:

$$V_L = V_U + tD$$

and the tax shield value gets eaten into the leveraged equity value:

$$V_L = V_U + tD = E_L + D$$

where the right side is from the perspective of the investors. Rearranging,

$$V_U = E_L + D(1 - t)$$

The entire $r_0 - r_d$ is multiplied by $1 - t$ because reducing the cost of debt reduces the need for equity to compensate for that cost as well. We know in reality r_0 will not be the same as r_{WACC} , so we use the above transformation, substitution, and equation to pull out the new r_e . It's a way of not using $r_d(1 - t)$ and transferring the tax shield benefit into $D(1 - t)$ instead. But debt isn't weighted like this in the actual, conventional WACC calculation. So it's a 'virtual' equation. So insert E_L as equity and $D(1 - t)$ as debt into the r_0 equation:

$$r_0 = \frac{D(1 - t)}{D(1 - t) + E_L} r_d + \frac{E_L}{D(1 - t) + E_L} r_e$$

$$r_e = r_0 + (r_0 - r_d) \frac{D(1 - t)}{E_L}$$

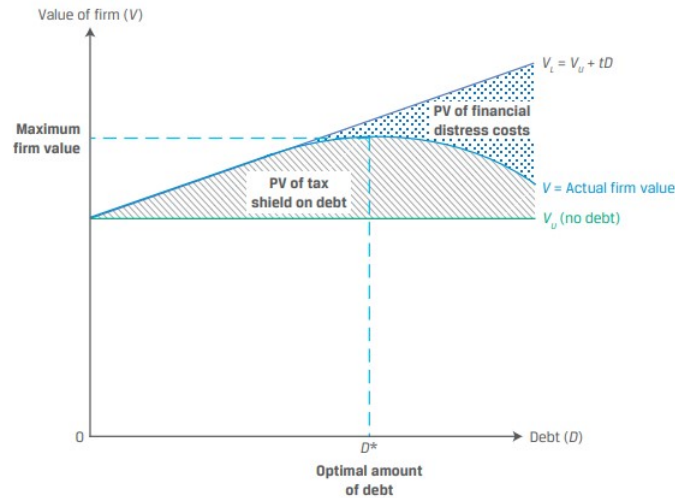
The leveraged WACC, using the conventional equation weighing debt properly (instead of virtually), is then:

$$r_{WACC,L} = \frac{D}{D + E_L} r_d(1 - t) + \frac{E_L}{D + E_L} r_e$$

Static tradeoff theory of capital structure. With PV = financial distress cost, which rises as debt increases.

$$V_L = V_U + tD - PV(\text{Financial distress})$$

The optimal capital structure can be theoretically found, but managers go for a 'target' structure based on book values (market values fluctuate; if share price rises, companies usually issue more so doesn't change anything; third parties like lenders use book value):



Asymmetric information. Managers know more than debtors and shareholders, who ask for more information to address this.

Pecking order theory. Concerned with information asymmetry. Cash (existing shareholders should favor most), then debt, then equity (existing shareholders should favor least). Public equity offerings often raise skepticism among investors because existing owners would seem less likely to want to share ownership of a company with strong future prospects. As a result, managers prefer internal financing, and if external financing is needed, they prefer private debt to public debt and prefer equity issuance least of all. Further, issuers tend to issue more equity when stock overvalued (and/or distressed), so equity issues are treated by investors with skepticism. Debt issuances look like managers have more confidence to meet these payments.

Free cash flow hypothesis. Concerned with agency costs. Agency cost is a concern with managers. Higher debt levels discipline managers by forcing them to manage the company efficiently and use cash wisely so the company can make its interest and principal payments.

5.9 Target capital structure

Infer capital structure by one of 3 approaches: (1) assume the company is at target capital structure and use market values; (2) infer from management; (3) use comparable companies.

Firm managers usually express target capital structure in ranges of BOOK values.

Transaction costs and minimum deal sizes make it impractical for firms to constantly align capital structure to a specific number. As a result, firm man-

agers usually set an optimal capital range: for example, 30%–50% debt rather than 40%. Target capital structure is often expressed using the book value of equity and debt. Market values can fluctuate substantially and seldom impact the appropriate level of borrowing. For management, the primary concern is the amount and types of capital invested by the company, not in the company (differs from perspective of investors who care about share prices). Capital structure policy is aligned to measures used by third parties like lenders, debt investors, and rating agencies.

5.10 Business models

The four major components of a business model framework are customers, products, channels, and pricing.

Global Industry Classification Standard. In the GICS each company is assigned to a sub-industry according to its principal business activity. Each sub-industry belongs to a particular industry; each industry belongs to an industry group; and each group belongs to a sector. In June 2009, the GICS classification structure comprised four levels of detail consisting of 154 sub-industries, 68 industries, 24 industry groups, and 10 sectors. Therefore, a sector is the broadest level of classification.

Cyclical businesses (e.g. mining) and capital-light businesses (e.g. software) usually do not use lots of debt because of business cycle volatility and no need respectively.

6 Alternative investments

6.1 Features, methods and structures

Alternative investments are investments other than ownership of public equity securities, fixed-income instruments, or cash. They are private capital, real assets (real estate, commodities, digital assets, IP, fine art, etc.), and hedge funds. Happens through investing in funds, co-investing, and direct investing. Greater diversification and higher expected returns but often involve longer-term, illiquid investments in less efficient markets. Investing in alternatives requires specialized knowledge.

Private capital - equity and debt. Private investors may have more information than public equity and debt investors who only have public-available information. Venture capital is for early growth companies; private equity (including LBO/MBO) typically for mature or declining companies.

Invest through funds (least sophisticated), co-investing (which is actually set up as a fund investment), then direct investing. Alternative funds usually in-

volve (1) the pre-commitment of funds prior to investment selection and an extended period during which the fund may not be sold, (2) higher management fees with more complex fee structures, and (3) less frequent transparency on periodic returns and fund positions versus equity or fixed-income funds.

Co-invest. Investors gain experience, reduce costs. Investment manager can accelerate investments if insufficient funds, can diversify better, and can expand universe of investments.

Limited partners - usually accredited investors in this context.

The illiquidity, complexity, and long-term nature of these investments require more complex structures to bridge potential gaps between manager and investor interests. Alternative investment structures may explicitly address both the roles and responsibilities of investors and managers to address these gaps. Examples of structures: limited partnerships, with the GP and many LPs (in a Limited Partnership Agreement - LPA). Master limited partnerships (MLPs) are like LPs but more liquid and traded. Use side-letters for fine-tuning adjustments.

Compensation. Alternative investment funds charge higher management fee + performance fee. Private equity funds might levy on committed capital instead of the management fee - this allows GP of private equity to deploy capital without rush. Hedge fund management fees based on assets under management (AUMs). Hurdle rate agreements sometimes distinguish between a hard hurdle rate, where the manager earns fees on annual returns in excess of the hurdle rate, or a soft hurdle rate, where the fee is calculated on the entire return when the hurdle is exceeded. Note there may be catch-up clauses, high-water mark (to protect LPs from paying the manager for the same returns multiple times), clawbacks (reduce by performance fee percentage if negative return; more common in private equity). There are two types of waterfalls: deal-by-deal (or American; more advantageous to GPs) waterfalls and whole-of-fund (or European; more advantageous to LPs) waterfalls.

Catch-up. E.g. if a GP+LP fund begins with 100 and then has a 8 percent soft hurdle for 2 years with catch-up and 80/20 GP/LP split on performance, and then at the end of 2 years earns 160, then LPs get 8+8, while GP catch-up for these 2 years is 2+2 because it is 80/20. Then the rest of the $160 - 100 - 16 - 4 = 40$ is based on the 80/20 split.

6.2 Performance and returns

Long-term, illiquid, less efficient markets, returns not normally distributed.

Private equity.

Long-term. J-curve with capital investment first, followed by capital returns. To evaluate, use IRR. Or MOIC (multiple of invested capital) as a shortcut. $MOIC = (\text{Realized value of investment} + \text{Unrealized value of investment}) / \text{Total amount of invested capital}$, where the total is paid-in capital minus management fees and expenses. For MOIC, no need to discount.

Because private equity and real estate investments involve longer holding periods, there is less emphasis on evaluating them in terms of shorter-term portfolio correlation benefits.

Performance appraisal for these investments is long-term so can be difficult to generalize, because results may vary significantly based on which investor has invested when in a particular vehicle.

Fair value hierarchy.

Level	Description	Sample Application and Method
Level 1	Quoted prices in active markets for identical asset/liability that may be accessed as of measurement date	Exchange-traded public equity securities (observed closing market price)
Level 2	Inputs other than quoted market prices in Level 1 that are directly or indirectly observable for an asset/liability	Over-the-counter interest rate derivatives (pricing model using quoted market prices)
Level 3	Unobservable inputs are used to measure fair value for asset/liability in which there is little, if any, market activity as of the measurement date	Private equity or real estate investments (cash flow projection models with reasonably available market participant assumptions)

For Level 3 asset pricing, regardless of the model used by a manager in such circumstances, that model should be independently tested, benchmarked, and calibrated to industry-accepted standards to ensure a consistency of approach. Because of the potential for conflicts of interest when applying estimates of value, hedge funds must develop procedures for in-house valuation, communicate these procedures to clients, and adhere to them consistently. Models tend to underestimate risk for e.g. private equity.

Hedge fund redemptions. HFs often charge a redemption fee and require a notice period before redemption. Moreover, there is a lockout period. Some impose a gate, a temporary restriction on redemptions to prevent fund damage.

Custom fee arrangements. Fees based on liquidity and asset size, founders class shares, either/or (either management or performance fees). High-water mark, to prevent investors paying for the same increase multiple times - high-water mark will also be different for different investors. Notes: in CFAI questions, (1) performance fee when net management fees is (the total increase in the fund size

- management fee) x performance fee percentage, rather than (total increase x performance fee) - management fee. (2) High-water mark is (end-of-year total minus end of high-water mark year) x performance fee, and then add management fee separately.

Benchmarking returns. Private equity and real estate indices might only make sense if same vintage ('since inception') of first year of investment because of illiquidity. Hedge funds fail a lot, so hedge fund indices have survivorship bias and backfill bias.

If a fund A invests in another fund B, there is a 'double' fee because have to pay management and performance fees of both A and B. However, some investors might still invest in funds of funds because A might provide diversification and fund B might be closed.

PE compensation. If there is a soft hurdle 8% preferred return with full catch-up, no management fee, 20% carried interest for the GP (general partner), American waterfall, and clawback:

Investment No.	Year		Amount (\$ m)		Profit			
	Invested	Sold	Invested	Sold	\$ m	%	LPs at 80%	GP at 20%
1	0	4	40	60	20	10.67%	16	4
2	0	4	40	100	60	25.74%	48	12
3	1	5	40	50	10	5.74%	10	0
4	1	5	40	120	80	31.61%	64	16
5	2	5	40	30	(10)	neg.	(8)	(2)
Total			200	360	160	12.47%	130	30

Note the preferred return soft hurdle is an ANNUAL rate. If an investment fails to make the soft hurdle, there is no carried interest. If the soft hurdle is cleared, the ENTIRE profit is eligible for the carried interest of 20% to the GP.

6.3 Private capital: equity and debt

Private equity strategies include leveraged buyout (LBO), venture capital (VC), and growth capital.

LBOs can be MBOs or MBIs (management buy-ins).

VC. Pre-seed or angel investing - idea stage. Seed financing - product development. Early-stage financing - moving towards operations. Later-stage financing - already in operations but before IPO. VC can be debt or equity; debt usually

before bankruptcy for greater control.

Mezzanine-stage financing - bridge financing to go from private to public. Note: mezzanine financing is different; it is hybrid equity-debt instruments, including convertible debt and convertible preferred shares.

Established companies in a transition phase may seek minority equity interest to expand, restructure, or acquire other companies. A private equity firm that engages in minority equity investing, also known as growth equity or growth capital.

A PIPE (private investment in public equity) transaction is a private offering to select investors with fewer disclosures and lower transaction costs that allows the issuer to raise capital more quickly and cost effectively than with other means that may be more regulated, expensive, and lengthy.

Exit strategies. There are two main exit strategies: trade sale and public listing. (1) Trade sale. In a trade sale, a portion or a division of the private company is sold either via direct sale or auction to a strategic buyer interested in increasing the scale and scope of the existing business. Often attracts regulatory scrutiny. PE can sell at higher price. Cheaper than IPO because it remains private. (2) Public listing. IPO (potentially highest price, more likely management will be retained so will have their support; but high cost, long time, onerous disclosures, stock market volatility, potential lockout periods where PE firm has to retain shares post-IPO; smaller companies, out-of-favor industries, unstable finances, limited operating history not suitable for IPO), direct listing (no underwriters; company just directly lists shares itself), SPAC (a “blank check” company exists solely for the purpose of acquiring an unspecified private company within a pre-determined period; otherwise, it must return capital to investors.) - more time for disclosures, flexibility in the transaction, and valuation is fixed. However, SPAC shareholders might exercise warrants (often sold in SPAC listings) which can increase cost of capital, SPAC shareholders might disagree with transaction (leading to stockholder overhang).

Other exit strategies include recapitalization, secondary sale, and write-off/liquidation. Recapitalization - increase or introduce leverage to its portfolio company and pay itself a dividend out of the new capital structure. Secondary sale - sale to other PE buyers.

PE returns. High returns for illiquidity, high leverage, specialized knowledge.

Private equity indices. Typically rely on self-reporting and are subject to survivorship, backfill, and other biases. This typically leads to an overstatement of returns. Failure to mark to market combined with the lag to mark investments due to inherent illiquidity will understate measures of volatility and correlations with other investments.

Private debt: venture debt, direct lending, mezzanine loans, levered loans and distressed debt. Requires specialized knowledge, life-cycle of the companies involved, fields (e.g. real estate) - modeling private equity or debt returns is not straightforward, due to a lack of good-quality data, more security-specific risk between assets, and artificially smooth returns.

Note: direct lending can involve a large capital commitment to a single borrower, with increased concentration risk and reduced diversification (increases risk, which reduces diversification because of higher risk correlation).

Venture debt for early-stage companies who have increased default risk + no collateral, often has features like for mezzanine loans. Direct lending requires due diligence and often require secured transactions. Can have levered loans - loans that are themselves levered. Mezzanine debt refers to private credit subordinated to senior secured debt but senior to equity in the borrower's capital structure. Distressed companies - high interest loans if good business plans. Other private debt: unitranche loans (blend of and structured between senior and subordinated debt) and specialty loans (e.g. litigation finance).

Diversification and private capital. Vintage year. The year the fund makes its first investment. Compare funds with same vintages - different vintages are subject to different economic (business cycle) environments. So, invest in multiple vintages. Funds seeded during the expanding phase of the business cycle tend to earn excess returns if they fund early-stage companies. Funds seeded during the contracting phase of the business cycle tend to earn excess returns if they fund distressed companies. Within a vintage, the risk-return tradeoffs look like this:



Private capital funds have a correlation of 0.65-0.85 with public market indices.

Private versus public capital. Private capital - early-stage financing has nega-

tive cashflows and little collateral; declining industries have declining prospects; cannot hedge in private capital.

6.4 Real estate and infrastructure

Current yield = NI / price .

Real estate has high initial costs, no two properties are the same, fragmented and local, illiquid (for some real estate), small investors can't diversify, hard to find a price (because of high transaction costs) and other differences.

Approximately 75 percent of global real estate values are residential instead of commercial. Real estate can be held privately or traded publicly through real estate investment trusts (REITs). Can own real estate directly, through pooled funds (e.g. mutual funds, REITs, etc.).

Direct private real estate owners can reduce their taxable income using non-cash property depreciation expenses and tax-deductible interest expenses (interest on their mortgages when investing in land for instance).

REITs (equity) and CMOs (debt) (collateral mortgage obligations) are PUBLIC real estate investment vehicles; real estate limited partnerships are PRIVATE. However, REITs have higher correlation with public equity markets than private real estate.

REITs. There are three main forms of REIT: equity REITs, which invest in properties outright or through partnerships and joint ventures (maximize occupancy and minimize expenses); mortgage REITs, which underwrite loans to real estate (mortgages) or invest in MBS; and hybrid REITs, which invest in both these types. The main appeal of the REIT structure is the elimination of double corporate taxation. Corporations pay taxes on their income, from which they make dividend distribution to their owners from after-tax earnings. The shareholders, in turn, are taxed at their personal tax rate. REITs can avoid corporate income taxation by distributing dividends equal to 90%–100% of taxable net rental income. Report in GAAP or IFRS but many report non-traditional measures, such as net asset value or variations of gross cash flow, such as funds from operations (FFO), which makes adjustments for depreciation, distributions, and preferred dividends, to better estimate future dividends, because non-cash depreciation expenses can be high for asset-intensive businesses. REIT advantages: tax structure, can manage property, can keep assets even if investors sell, BUT high correlation with public equity markets.

REITs and other private real estate funds. Core real estate strategies - often open-ended funds with high-quality real estate. Often closed-ended funds: core-plus, value-added, opportunistic - requiring various levels of redevelopment.

Investment characteristics. Predictable and stable income, price appreciation, inflation protection, low correlation to other assets, competitive returns.

Low-risk real estate. Senior debt, investment-grade commercial MBSs, etc. Bond-like properties.

Moderate-risk real estate. REITs. Stable cashflows, many lessors, sale-leasebacks. Bond-like properties.

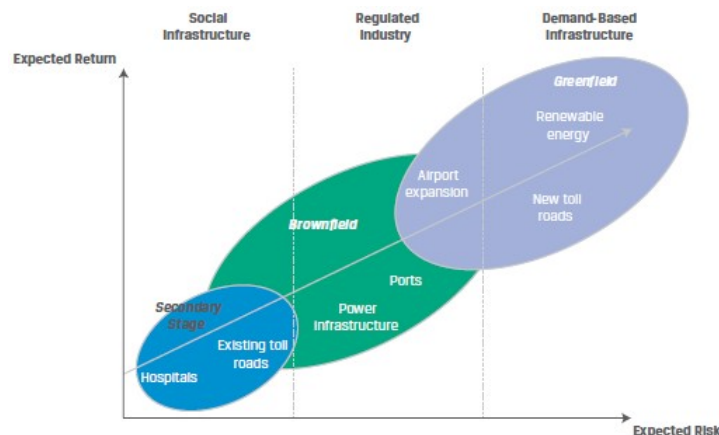
Higher-risk real estate. Core-plus and above. Requires more redevelopment and is more equity-like.

Real estate returns. About half of returns are income and other half price appreciation. But income is more steady than stock, so is a source of diversification - more income versus price appreciation, compared to stocks.

The market prices listed REITs continuously, whereas private real estate is appraised perhaps once a year. This mismatch in appraisal timing leads to correlation numbers that are artificially low. Additionally, equity investors in public real estate discount future cash flows, while appraisers of private real estate place heavy emphasis on current market conditions and recent trends.

During certain market conditions, equity REIT correlations with market benchmarks increase, particularly during steep market downturns, such as the 2007–08 financial crisis.

Infrastructure investments. In contrast to real estate, returns generated through contractual payments: availability payments, usage payments, and 'take or pay' arrangements (like offtakes). Greenfield projects - build-operate-transfer (BOT) lifecycle. Brownfield projects - shorter investment period and faster cash inflows. Secondary stage projects - no changes needed; just operate; cashflows immediate. Can be direct or indirect investments; but direct investing can introduce concentration risks because projects are expensive.



Social infrastructure = education, hospitals, prisons, etc.

Infrastructure investments typically support services that face inelastic demand and/or benefit from high barriers to entry, generate steady cash returns, and have a longer life cycle, equity investments in infrastructure offer lower correlation with public market equities and the broader economy. Given the stable underlying nature of cash flows, infrastructure debt tends to experience lower default rates and higher recovery than similar fixed-income instruments, and it is less subject to fluctuation over the economic cycle. Often used for pension funds and sovereign wealth funds - which have long investment horizons.

Economic infrastructure. Transport (includes bridges, roads, ports), communication, internet, utility, and energy.

6.5 Natural resources

Oil, grains, etc. Farmland, timberland, and raw land are similar to real estate investments in that they are unique, illiquid assets with distinct geographic location and features. But, not about buildings and tenancies; it's about geography. Farmland can be small, but timberland is large and has long lifecycles as trees grow. Direct or indirect investing. Smaller investors usually invest indirectly. Direct investment can be just owning (fixed income) or owning and operating (more variable income).

Timber. Timberland investment management organizations (TIMOs) manage timberland like REITs for property. TIMOs use their forest investment expertise to analyze and acquire suitable timberland holdings on behalf of institutional investors. Timberland provides flexibility in the timing of harvesting trees. Unlike timberland, farm products must be harvested when ripe, with little flexibility in production.

Commodities. Physical commodity markets lack price transparency. As such, commodity investments are usually made through financial derivative instruments, most frequently commodity futures and forwards and occasionally options on futures. Using derivatives to establish exposures to natural resources has several benefits: Because these instruments are traded on organized exchanges, they are very liquid and provide opportunities for price discovery.

Gain exposure through the underlying (forwards), futures and derivatives, exchange-traded products, etc.

Exchange-traded products. ETPs, either funds (ETFs) or notes (ETNs) may invest in commodities or commodity futures. For example, the SPDR Gold Shares ETF seeks to track the price of physical gold by holding bullion in vaults. It owned just under USD53 billion in gold bullion as of December 2022. ETPs may use leverage and may replicate the pay-offs from a long or short position.

Commodity-trading advisors. CTAs are managed futures funds that make directional investments primarily in futures markets based on technical and fundamental strategies. A commodity-focused CTA might concentrate on a specific commodity (such as grains) or be broadly diversified across commodities.

Specialized funds. An example of specialized funds is private energy partnerships, which are similar in structure to private equity funds.

Pricing. Commodities trade in both physical and financial markets. So, no-arbitrage means the difference in prices between the cash or spot markets and derivative markets is equal to the cost of carry, which is the opportunity cost of holding these assets, and mirrors the risk-free rate and the cost of storing, transporting, and insuring the commodity. The holder of the physical commodity should expect to be compensated via a higher forward price, $F_0(T)$, than the prevailing cash price. However, holding the commodity has a convenience yield. Because convenience yield is a benefit and accrues to the owner, it reduces the forward price.

Backwardation - when the spot price is above the forward prices. Inverted forward curve, e.g. convenience yield is positive and exceeds the cost of carry. Contango - when the spot price is below the forward prices, e.g. cost of ownership exceeds the convenience yield (e.g. low inventories and healthy demand). Contango generally lowers the return of the long-only investor, and backwardation enhances it. In a contango market, the long investor has to roll over by selling the near-dated contracts (at a lower price) and buying far-dated contracts (at a higher price). In a contango market, the long position price in the short term has declined because the forward price is fixed. $\text{Forward price} = \text{Spot price} + \text{Costs of carry} - \text{Benefits of ownership}$.

Supplies of physical commodities are determined by production and inventory

levels and secondarily by the actions of non-hedging investors. Demand for commodities is determined by the needs of end users and secondarily by the actions of non-hedging investors. Investor actions can both dampen and stimulate commodity price movements, at least in the short term.

Commodities provide greater returns in rising inflation environments at a cost of negative returns as inflation rates decline. In stable inflation environments, commodity returns have tended to be low but positive. This is where the inflation hedge argument comes from.

Note: very few commodity derivatives on crops, so buying farmland allows greater exposure to more crops.

6.6 Hedge funds

Hedge funds combine traditional debt and equity instruments with leverage, derivatives, short selling, and other strategies. Hedge funds are attractive for their diversification effects because their returns typically demonstrate low correlation with traditional asset investing. the investment strategy splits a portfolio such that each component helps hedge the risks from the other. Thus, by internally neutralizing market risk and by managing the portfolio components, the hedge fund manager can obtain enhanced risk-adjusted returns. Hedge fund managers have a great deal of freedom to make trading decisions and to decide how to allocate client funds. Mutual funds are highly regulated since they are available to public investors. Hedge funds are available only to institutional and accredited investors.

Equity	Event Driven	Relative Value	Opportunistic	Multi-Manager
Long/Short Equity	Merger Arbitrage	Convertible Bond Arbitrage	Global Macro	Fund of Funds
Short Biased	Distressed	Fixed-Income Arbitrage	Managed Futures	
Market Neutral	Special Situations Activist	Multi-Strategy		

Long/short - long promising stocks and short unpromising ones. Often long bias, but there can be a short bias. Short bias usually hasn't done well historically. Can be market-neutral, aiming for 0 beta.

Short-biased HFs. The beta in these types of funds is most often negative given the short bias. These strategies use quantitative (technical) and fundamental analysis to focus the bulk of their portfolio on shorting overvalued equity securities (against limited or no long-side exposures).

Market-neutral HFs. The hedge fund takes long positions in securities iden-

tified as undervalued and short positions in overvalued securities. The hedge fund tries to maintain a net position that is neutral with respect to market risk and other risk factors (size, industry, momentum, value, etc.). Ideally, the portfolio has an overall beta of approximately zero.

Event-driven. Tend to be long-biased, with merger-arbitrage having the least bias. Merger-arbitrage - short acquirer and long acquiree after possible merger announcement, hoping share price spreads between acquirer and acquiree close. Spread is often modest, so amplify with leverage. Distressed-restructuring - buy before bankruptcy; sometimes buy fulcrum securities - most senior security that, after undergoing restructuring, has the greatest likelihood of conversion into equity ownership; or might buy distressed debt which is still sufficiently secured. Special situations - e.g. spinoffs. Activist - buy some shares and get board seat and influence (different from private equity because activist HFs are buying into public equities).

Event-driven HFs are long-biased. Relative value HFs go for short-term pricing discrepancies.

Relative value. Tend to be based on short-term. Convertible bond arbitrage, fixed income (general), fixed-income (asset-backed securities), multi-strategy. Convertible bond arbitrage - market-neutral investment strategy seeks to exploit a perceived mispricing between a convertible bond and its component parts: the underlying bond and the embedded call option; buy issuer's debt and simultaneously sell stock based on delta of embedded call. This strategy can be sensitive to bankruptcy risks; however, they may be hedged away using either equity put options or credit default swap derivatives on the issuer. FI arbitrage - buy underpriced FI, short overpriced FI.

Opportunistic. Macro, managed futures. Managed futures - started off with commodity trading advisors (CTAs), but moved to financials using fundamental and technical analyses but trading financials means diversification risk fades. Managed futures leverage and borrow less than other types of HFs because they buy lots of futures and other derivatives.

Hedge fund investment forms. Direct forms: master feeder structures (offshore gets low tax, onshore can be structured to fit local tax and regulatory requirements, and efficient to feed into one fund), fund of one or separately-managed account (SMA). Indirect forms: fund-of-HFs, ETFs. By relying on quantitative tools, they imitate a broad spectrum of hedge fund returns or a specific style return. However, the returns from these strategies often fall short compared to pure hedge fund strategies because these instruments are publicly traded, are subject to a much heavier regulatory burden, do not impose restrictions on redemptions, and cannot use leverage to the same level.

Risk, return, diversification. Compared to traditional portfolios, hedge funds

seek to limit market exposure and returns from beta and primarily focus on generating idiosyncratic returns, or alpha. Excess returns from market inefficiencies (which may be short lived) and the skills of the manager in leveraging them. HFs get value from market beta (same as market), strategy beta (from hedge fund as a firm), and alpha (individual manager).

6.7 Digital assets

Distributed ledger technology. Uses cryptography. Capable of smart contracts - e.g. executing options once expired. Blockchain is a type of DLT using block of transactions glued together by cryptography. Blocks are glued together via consensus protocols - proof of work (longest chain is valid, glued by miners solving puzzles) and proof of stake (network participants pledge capital to glue blocks). Permissionless (e.g. BTC) and permissioned (where there are a few players who regulate).

Types of digital assets: cryptocurrencies, tokens. Tokenization. Instead of currencies, some kind of ownership rights as the currency. NFTs link digital assets to certificates of authenticity. Security tokens digitize ownership rights in public securities - can pay dividends. But utility tokens are just for paying for services and network fees. Governance tokens allow holders to decide what happens in permissionless networks if there is a problem.

Altcoins - cryptocurrencies other than BTC. Stablecoins - collateralized by another currency or asset - so can redeem assets from issuers. Asset-backed tokens - pegged. Financial regulators often class asset-backed tokens as securities.

Asset-backed tokens can allow fractional ownership of underlying assets, increasing liquidity of assets. However, transaction validations require significant computing power, so are expensive.

Many direct and indirect investing forms.

Exchanges. Central (privately held) and decentralized. Hard to hack decentralized exchanges.

dApps - decentralized applications for DeFi which seeks to develop open-source financial applications allowing transactions to take place (e.g. smart contracts) without central coordination.

Although purely speculative in nature, considered by some to have value drivers distinct from common equity and debt markets, cryptocurrencies have exhibited low correlations with traditional asset class returns. This may provide a hint that the long-term determinants of cryptocurrency prices differ from those of traditional investment assets. In practice, prices (or returns) of cryptocurrencies are driven more by market adoption, network effects, technological advance-

ment, regulatory development, speculation, and general market risk appetite. Some of these factors are unique to this asset class.

7 Derivatives

7.1 Derivative instruments and market features

Firm commitments (pre-determined amount): Forwards, futures, swaps (equity, interest-rate, etc.). Other types: contingent claim (i.e. options). One counterparty is usually a financial intermediary.

In a derivative contract, the security buyer is taking a long position and the security seller is taking a short position. Standalone versus embedded (e.g. callable bond) derivative.

Other than gaining and reducing exposures (hedging) and offsetting exposures, derivatives allow parties to gain very large exposures with little capital, and usually have much lower transaction costs and usually are more liquid than spot transactions of the underlying.

Derivative underlyings. Equities, indexes, equity or index swaps, options, warrants (when company issues options to buy the company's shares - sometimes issuing new shares). Bonds, interest-rate swaps. Currencies, commodities, credit (e.g. CDSs), other (crypto, weather, etc.). Note: underlying in a swap is the interest rate or market return rate, not the notional bond amount or bond or index or equity.

Swaps. Index swaps - pay return on one stock index and receive returns on another index. Interest-rate swaps - pay variable interest rate in exchange for a high fixed interest rate, allowing more or less exposure to underlying without trading the underlying (and can be used to change the portfolio duration without trading bonds). Credit default swaps - CDS seller agrees to pay in case of default of underlying in exchange for regular payments.

Derivative markets. ETD (exchange-traded derivative) markets have standardized derivative contracts with an exchange or central counterparty. OTC derivative markets are mostly amongst financial intermediaries acting as market makers without an exchange or central counterparty, offsetting risks to other parties or against each other. However, with the GFC, in the US there is a central clearing mandate: bilateral OTC trade first on a platform and then novated (replaced) by a central clearing party (CCP) which is the counterparty and clears and settles.

The futures exchange facilitates the trades but it's the clearinghouses which settle the trades.

7.2 Forward Commitment and Contingent Claim Features and Instruments

Forwards are OTCs, which are more customizable but have greater counterparty risk and less liquid.

Firm commitments are linear derivatives. S_0 is the current spot price, $F_0(T)$ is the forward price, S_T is the future spot price. So the buyer's payoff is linearly $S_T - F_0(T)$ and the seller's payoff is the inverse.

Futures. Counterparties deposit initial margins and keep maintenance margins (minimum margins). At the end of each day, there is a daily settlement or mark-to-market (MTM) at the settlement price. If the futures margins account falls below the maintenance margin, there is a margin call to replenish it (or the exchange pulls it from insurance) to the initial margin, which is the variation margin. Exchanges may limit by price limits and circuit breakers. Note margin accounts don't collect interest.

Swaps. Usually a floating (e.g. MRR or related to MRR) cashflow in exchange for a higher fixed cashflow. So the floating-rate payer is the fixed-rate receiver (the interest-rate swap seller and the short position) and vice versa. The floating rate resets, while the fixed rate (swap rate) is constant. It is equivalent to a series of forwards. Can be used to change portfolio duration without trading bonds. Can be OTC or ETD.

Options. Non-linear derivatives. Payoff to option buyer is always 0 or positive, can never be negative. The price is option 'premium'. OTC or ETD. European (exercised only at maturity) or American (exercised any time before maturity) options. A European option value before maturity is the intrinsic value (in/out/at the money if exercised at that point in time). Time value of option (even European option), also called the extrinsic value, is the market price of the option. Extrinsic value is made up of time until expiration, implied volatility, dividends and interest rate risks. Option premium = intrinsic + extrinsic value. Once the option seller gets the premium, no credit risk but the option buyer bears all credit risk.

Credit derivatives. Most common is a CDS. It is a contingent claim. CDSs are traded based on a CDS credit spread; the CDS spread depends on the probability of default (POD) and loss given default (LGD). Par spread is based on setting $PV = 0$. Credit protection seller is the CDS seller, long on credit risk and receives regular payments; credit protection buyer is shorting credit risk. Credit swap parties might not have anything to do with the company connected to the underlying; this is a derivative. Upon a default, the seller has to pay LGD times the notional CDS amount.

7.3 Derivative Benefits, Risks, and Issuer and Investor Uses

Covered call = long forward + short call. Payoff rises until a maximum at the call strike price.



Options reduce transaction costs (no need for underlying, e.g. transporting and storing oil), increase liquidity, low upfront costs (only need to fund the initial margin, don't need to fund the much more expensive temporary purchase of the underlying), and short positions are easier (no need to borrow the underlying). So, derivative prices often reflect fundamentals first.

Structured notes - add embedded options to debt for a bespoke security.

Derivative risks. Much greater leverage and risk. Leverage ratios = underlying asset price divided by derivative cost. Basis risk - occurs if the referenced index doesn't exactly match the underlying exposure because of timing issues or because the underlying is slightly different in nature from what is to be hedged. Counterparty risk - if OTC. Liquidity risk - if a party can't meet a margin call because of cashflow timing issues. MTM reduces credit risk. MTM must be on balance sheet unless it qualifies for hedge accounting (i.e. the derivative is embedded as part of a larger contract with that underlying asset or liability).

Hedge Type	Description	Examples
Cash Flow	Absorbs variable cash flow of floating-rate asset or liability (forecasted transaction)	Interest rate swap to a fixed rate for floating-rate debt FX forward to hedge forecasted sales
Fair Value	Offsets fluctuation in fair value of an asset or liability	Interest rate swap to a floating rate for fixed-rate debt Commodity future to hedge inventory
Net Investment	Designated as offsetting the FX risk of the equity of a foreign operation	Currency swap Currency forward

7.4 Arbitrage, Replication, and the Cost of Carry in Pricing Derivatives

Assets with a known future price (e.g. forward) must have a current spot price which is the future price discounted by the riskless rate. Otherwise, arbitrage.

Long forward = borrow at riskless rate S_0 (so the expected asset price at $t = T$ is $F_0(T) = S_0(1 + r)^T$) to buy asset, hold the asset and sell at whatever the market price S_T is, return the borrowed funds with interest at $F_0(T)$, giving a return of $S_T - F_0(T)$. So, if the forward contract price $F_0(T)$ is equal to the current spot S_0 with a riskless rate, there's no arbitrage opportunity between being long on the forward and borrowing at the riskless rate, buying at the spot price, selling later at the forward price and then repaying the principal + interest.

There are three long-forward cases. (1) Investor has no capital and so just goes long on the forward contract - they get $S_T - F_0(T)$. *No capital required, no capital costs.* (2) Investor has capital S_0 to replicate the long forward so invests it and gets $S_T - F_0(T)$ MINUS S_0r (opportunity cost of capital) if the asset price rises more than the riskless rate. *Investor gets the increase in price + interest on capital, but this requires capital.* (3) Investor borrows from S_0 from the bank and buys at the asset at S_0 so the bank gets S_0r and the investor just gets $S_T - S_0 - S_0r = S_T - S_0(1 + r) = S_T - F_0(T)$. *Investor gets the increase in price minus cost of capital from bank.*

If the current spot price of the underlying is too high. (1) Short a forward contract on it - since the price $S_{high} > S_0$ where S_0 is the 'real value' price, the forward contract price would be $S_{high}(1 + r)^T = F_{high}T$ but since the price is too high, the underlying asset price will be $F_{high}T > S_{high,1}$, e.g. $S_{high,1} = S_T$. (2) If the investor has capital, short the asset by selling it at S_{high} , invest the proceeds at the riskless rate and buy the asset back at S_T . (3) If the investor doesn't have capital, borrow funds to short the asset as in (2) but this will incur

at an extra cost of $S_{high}r$.

If an asset is underpriced relative to its forward price based on the riskless rate, buy it using own or borrowed funds.

Short forward = sell asset at S_0 (so the expected asset price at $t = T$ is $F_0(T) = S_0(1 + r)^T$) at $t = 0$ and invest the funds at the riskless rate to get at $t = T$ $F_0(T)$ and then buy back the asset at whatever the market price S_T is to get a payoff of $F_0(T) - S_T$.

There are three short-forward cases. (1) Investor has no capital and so just goes short on the forward contract - they get $F_0(T) - S_T$, but *no capital was required; no capital costs*. (2) Investor has the underlying asset valued at S_0 (or equivalent funds and buys it) to replicate the short forward so sells it and invests the proceeds at the riskless rate to get at $t = T$ $S_0 + S_0r - S_T = F_0(T) - S_T$ if the asset price declines. *Investor gets decrease in price, but capital required*. (3) Investor borrows the underlying asset or equivalent funds S_0 from the bank to replicate the short forward so sells the underlying asset at S_0 , invests the proceeds to get $F_0(T)$ at $t = T$ and buys it back at S_T ; the bank gets S_0r , so the investor just gets $F_0(T) - S_T - S_0r$. *Investor gets decrease in price minus borrowing cost*.

Short forward = short sell and buy back later (using case (2) - investor's own capital). Both give payoffs of $F_0(T) - S_T = S_0(1 + r)^T - S_T$.

When costs of carry (r , the riskless rate), income benefits (e.g. dividends on an underlying stock, I) and costs (e.g. storage costs, C) are included:

$$F_0(T) = (S_0 - PV_0(I) + PV_0(C))(1 + r)^T$$

FX. FXs expressed as price currency divided by base (domestic) currency. For e.g. a long USD/EUR FX forward position is the sale of USD and purchase of EUR. To solve for the future FX rate $F_0(T)$ based on current FX rate S_0 and the interest rates in both the base (domestic) r_d and price (foreign) r_f currencies

$$F_0(T) = S_0 e^{(r_f - r_d)T}$$

If $r_f > r_d \Rightarrow F_0(T) > S_0$, i.e. there is a forward premium, the domestic currency is expected to appreciate, and vice versa. The side with the higher interest rate experiences 'more inflation' so its currency weakens in the future.

Storage and insurance costs increases the difference between the spot and forward price. The forward price significantly below the no-arbitrage price may indicate the presence of a convenience yield (non-cash benefit to holding the asset, e.g. arising from tight supply).

Asset Class	Examples	Benefits (<i>i</i>)	Costs (<i>r, c</i>)
Asset without Cash Flows	Non-dividend-paying stock	None	Risk-free rate
Equities	Dividend-paying stocks	Dividend	Risk-free rate
	Equity indexes	Dividend yield	
Foreign Exchange	Sovereign bonds (foreign)	None	Difference between foreign and domestic risk-free rates ($r_f - r_d$)
Commodities	Market exchange rates		
	Soft and hard commodities	Convenience yield	Risk-free rate
	Commodity indexes		Storage cost
Interest Rates	Sovereign bonds (domestic)	Interest income	Risk-free rate
	Market reference rates		
Credit	Single reference entity	Credit spread	Risk-free rate
	Credit indexes		

Initial value with convenience yield being positive:

$$V_t(T) = S - t - (\gamma - \theta)(1 + r)^t - F_0(T)(1 + r)^{-(T-t)}$$

7.5 Pricing Forward Contracts

The value, including MTM value, of a long forward at time t between inception and maturity:

$$V_t = S_t - F_0(T)(1 + r)^{-(T-t)}$$

where r is the riskless rate. Incorporating income (e.g. dividends) and costs,

$$V_t = (S_t - PV_t(I) + PV_t(C)) - F_0(T)(1 + r)^{-(T-t)}$$

where

$$F_0(T) = (S_T - PV_T(I) + PV_T(C))(1 + r)^T$$

Subtract dividends which won't be exchanged in the forward, even if the last dividend occurs at the forward date.

FX:

$$S_t = F_0(T)e^{-(r_f - r_d)(T-t)}$$

For a long FX (i.e. investor believes base/domestic currency will appreciate versus the foreign currency), USD/EUR contract (buy EUR, sell USD), if the US interest rate rises, there is an MTM gain because the USD will depreciate more - so the spot USD/EUR will decline for a given forward FX. It's like the long position needs to give 'less' USD now, so gets an MTM gain relative to what it was before the rate rise.

Zero (i.e. spot) rates. The term structure of interest rates provides default-free (theoretically) zero-coupon rates. However, sometimes there is no term structure to use, so need to find the zero rates from a series of coupon bonds and their prices by bootstrapping - in the first year, use the YTM of year 1 z_1 , then use the coupon rate of the 2-year bond with the YTM of year 1 and calculate the YTM of year 2 z_2 , etc. From these, calculate discount factors, e.g.

$$DF_1 = \frac{1}{(1 + z_1)}$$

$$DF_2 = \frac{1}{(1 + z_2)^2}$$

IFRs or implied forward rates are obtained from the zero rates. Example of calculating the zero rate from the implied forward rate:

$$(1 + r_{z_2})^2 = (1 + r_1)(1 + f_{1,1})$$

because the left side is the discount rate. Beyond one year, YTM's can't be used for discounting zero-coupon bonds (although it is a close approximation); must use zero discount rates. YTM's aren't zero rates after 1 year because of prior years, e.g. 1-year rates, 'distort' future rates.

Cannot use implied forward rates as discount rates, as they are FUTURE rates. E.g. can't do this:

$$\frac{2.5}{1.0296} + \frac{102.5}{(1 + IFR)^2}$$

However, can do this:

$$\frac{2.5}{1.0296} + \frac{102.5}{(\sqrt{(1.02396)(1 + IFR)})^2}$$

Note: Interest rates given as APRs. A 3-month 1.25% MRR is 1.25%/4 for each quarter. An $IFR_{3m,3m} = 2.25\%$ means the quarterly IFR is 2.24%/4. Note 2: a 2-year bond yielding 5% means it has a yield per year of 5%.

Forward rate agreement (FRA). OTC derivatives contract in which counterparties agree to apply a specific interest rate to ONE (not multiple in contrast to swaps) future period. The FRA buyer, the long position, agrees to pay the agreed fixed interest rate (gains if the MRR rises) on a notional amount at the END of the period in exchange for MRR settled (i.e. calculated) at the BEGINNING of the period (in contrast to swaps, which are at the end of the end of the period):

$$MTM = PV((MRR_{B-A} - IFR_{A,B-A}) \times Period \times \text{Notional amount})$$

where the period is e.g. 90 days, and the PV is from the beginning of that 90-day period discounted by the MRR_{B-A} rate.

7.6 Pricing Futures Contracts

Forward contract 'notional' MTM changes are based on differences between spot and the present value of the forward price - there is a lot more counterparty credit risk than in futures because there is no actual MTM. In contrast, the MTM of a futures price is based on FUTURES PRICE CHANGE, and following MTM settlement, the futures contract price resets to 0 at the new closing price. Both parties to the futures price contract deposit initial margins, settle variation margins, and get margin calls if they fall below maintenance margins. The sum of futures MTMs are about the same as forward MTMs.

Futures on short-term interest rates are available as an alternative to FRAs, based on price:

$$f_{A,B-A} = 100 - (100 \times MRR_{A,B-A})$$

If the MRR is expected to be negative in the future, then $f > 100$. The subtraction in the formula reverses the price/yield relationship from the FRA agreement. The buyer (lender) is bettering the price will rise, i.e. the interest rate will fall. Interest rate futures daily settlement is based on futures contract basis point value (BPV):

$$BPV = \text{Notional Principal} \times 0.01\% \times \text{Period}$$

The BPV change is linear, so a one basis point (0.01%) change in MRR affects price linearly.

The short position (borrower) gains if the MRR rises in contrast to FRAs:

Contract Type	Gains from Rising MRR	Gains from Falling MRR
Interest rate futures	Short futures contract	Long futures contract
Forward rate agreement	Long FRA: FRA fixed-rate payer (FRA floating-rate receiver)	Short FRA: FRA floating-rate payer (FRA fixed-rate receiver)

Forward and futures price differences. (1) If interest rates are constant AND/OR interest rates are not correlated with futures prices, then there is no difference. However, if futures prices are positively correlated with interest rates, LONG futures contracts are more attractive than LONG forward positions for the same underlying and maturity. LONG futures preferred over LONG forwards where futures prices are positively correlated with interest rates - if the underlying price is rising, then long position can reinvest MTM profits at high interest rate; if the underlying price is falling, then long position can borrow MTM payments at lower interest rates. If futures prices are negatively correlated with interest rates, long forwards will be favored. (2) The short maturity of most futures contracts and the ability of most market participants to borrow near

risk-free rates for these maturities typically results in little to no distinction between futures and forward prices. An exception to this is the so-called convexity bias. While futures BPVs are linear, because FRA MTM values are discounted, there is convexity - so a reduction in MRR leads to a bigger change than an increase for FRAs, whereas it is linear for STIR futures.

There are margining requirements for exchange-traded derivatives; traditionally OTC derivatives didn't but they are increasingly requiring them.

7.7 Pricing and Valuation of Interest Rates and Other Swaps

Swaps versus forwards (particularly for interest rate swaps). While forwards and futures can be used, swaps are more customizable (which reduces administrative burden of combining multiple forwards). FRAs exchange interest rates once; standard IR swaps are each period (at the END of each period). Moreover, swaps have CONSTANT fixed payments.

IR swaps as a combination of FRAs. From the term structure, calculate the zero (spot) rates using government coupon bonds. Then calculate IFRs. Use the IFRs (not the spot rates) in FRAs because we are getting the interest a period ahead. But these fixed payments are different for each year, so combining FRAs like this doesn't result in a IR swap. Instead, solve for the par rate (i.e. constant PMTs), which are fixed! So, the fixed swap rate as an IRR on the IFRs to maturity.

IR swaps are like cash bond investments: fixed-rate payer is like shorting a bond; MRR payer is like taking a long position on a bond. Investors often use swaps instead of bonds to address rate changes. If interest rates are expected to fall, a portfolio manager may choose to receive fixed on a swap rather than purchase underlying bonds to realize a gain in a lower rate environment.

For the fixed-rate payer of an IR swap, if there are periodic settlements, they could be:

$$\text{Periodic settlement} = (MRR - s_N) \times \text{Notional amount} \times \text{Period}$$

where s_N is the fixed swap (e.g. par) rate, while the MRR is the IFR zero-rate. The value of a swap (to determine MTM) on any settlement date equals the current settlement value above plus the present value of all remaining future swap settlements. Although swap market conventions vary, here assume the MRR sets at the beginning of each interest period, with the same periodicity and day count convention as the swap rate. The fixed versus floating difference is exchanged at the end of each period. The par swap rate s_i comes from:

$$\sum PV(\text{Floating payments received}) = \sum PV(\text{Fixed payments paid})$$

$$\sum \frac{IFR_i}{(1+z_i)^i} = \sum \frac{s_i}{(1+z_i)^i}$$

An IR swap is also basically the combination of a long FRN (floating rate note) + shorting a fixed-coupon bond, where the fixed rate is the fixed-rate of the IR swap.

7.8 Pricing and Valuation of Options

Option pricing. Exercise value of call or put option. Take the (exercise) strike price and discount, compare it to the current spot price. The call option will resemble a forward. Call or put option price is the sum of the exercise value + time value (TV):

$$c_t = \max(0, S_t - X(1+r)^{-(T-t)}) + TV_t$$

$$p_t = \max(0, X(1+r)^{-(T-t)} - S_t) + TV_t$$

where TV is a function of time and volatility. TV approaches zero near maturity.

Moneyness. In/out/at the money. If deeply in the money, more likely to be exercised; if deeply out then not likely.

Price bounds. Call option:

$$\max(0, S_t - X(1+r)^{-(T-t)}) < c_t < S_t \text{ (share price)}$$

Put option:

$$\max(0, X(1+r)^{-(T-t)} - S_t) < p_t < X \text{ (strike price)}$$

The upper bound of a call option is the underlying, even if there is a positive exercise price. The upper bound is not the PV of the underlying asset minus the exercise price. If the call option price were higher than the (current) underlying asset price, if the underlying were to rise in value, you get less from exercising the call option than from holding the underlying asset itself. E.g. if the underlying is currently 10, and the call option is 15, then even if the call option had the minimum exercise price of 0, the underlying gets a higher payoff. Can't have a negative exercise price; otherwise, you always exercise it, and there's no uncertainty!

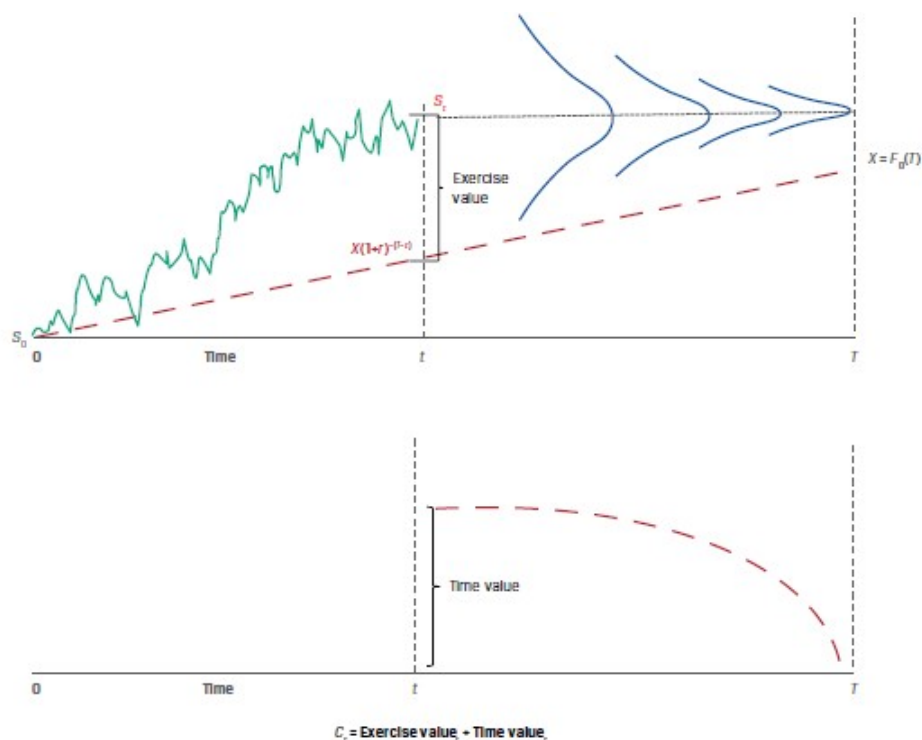
Replication. Call option is like an uncertain long forward. To replicate the long call option, borrow at r and buy a *proportion* of the underlying (where the proportion depends on the likelihood of exercise at T), i.e a proportion of $X(1+r)^{-T}$. Constantly adjust proportion during call option life. At maturity, if in the money sell it and if out of the money do nothing if out-of-the-money.

Short put option is like an uncertain short forward. To replicate the short

put option, short a proportion of the underlying, i.e. $X(1+r)^{-T}$ - sell the underlying and lend at r . Constantly adjust during put option life. At maturity, if put option is in money, buy back the underlying and do nothing otherwise.

Factors affecting options prices. Higher exercise price reduces call option value and increases put option value. Longer time to expiration increases call option value and put option value BUT since put options have limited downside values (in contrast to call options which have unlimited upsides), more time can actually begin to reduce (because of present value effects) and eventually become a net negative for put option value. A higher riskless rate reduces the present value of strike prices and therefore increases call option values and reduces put option values. Volatility increases call and put values. Higher dividends reduce call option values and increase put option values; higher carry costs (e.g. storage) reduce call values and increase put values.

Note. A purchased option declines in value, all else equal, because of the time value decline. A sold option increases in value, all else equal, because of the time value decline.



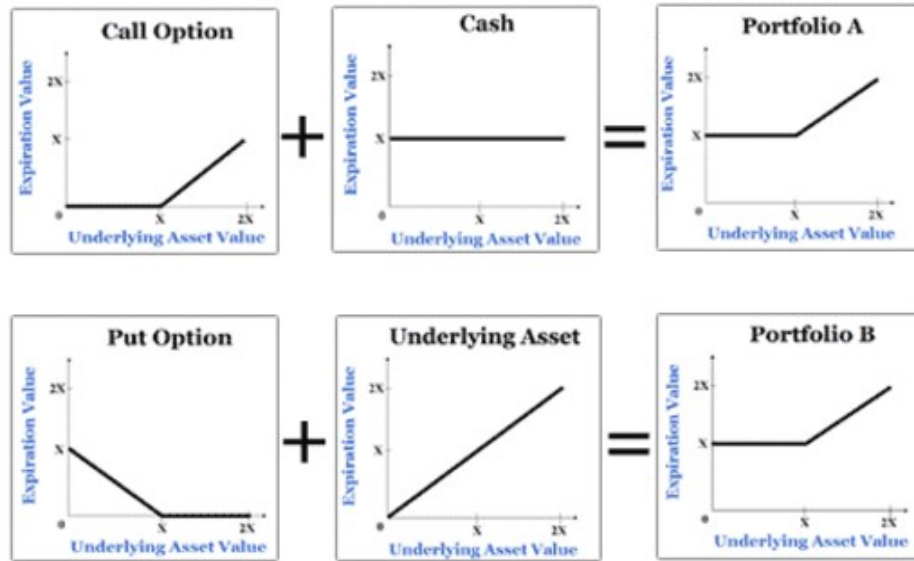
Note. Whether an option is in-the-money or out-of-the-money depends only on the CURRENT price (NOT discounted) compared to the exercise price (NOT

discounted).

7.9 Option Replication Using Put–Call Parity

Fiduciary call. Buy (long) call option at $-c_0$ on an asset Z with strike price X and buy (long) riskfree bond at $-X(1+r)^{-T}$ where X is the par value of a couponless bond. At T, investor gets X and any proceeds of exercising call option ($= X$ from bond $+ S_T - X$ from exercising call option).

Protective put. Buy (long) put option on an asset Z with strike price X at $-p_0$ and buy (long) underlying asset Z at $-S_0$. At T, investor gets X if Z's price is below X and more if Z's price is above X ($= S_T$).



Put-call parity. Buying a fiduciary call and buying a protective put are equivalent, for a riskfree bond and asset Z:

$$-p_0 - S_0 = -c_0 - X(1+r)^{-T}$$

$$p_0 + S_0 = c_0 + X(1+r)^{-T}$$

If there is an inequality, then buy the lower side and sell the higher side. Note: covered call (short call + long asset) and short put + long riskfree bond is just a re-arrangement of the put-call parity:

$$-c_0 + S_0 = -p_0 + X(1+r)^{-T}$$

Replicating forwards. Synthetic assets. A long underlying asset position can be replicated by entering a long FORWARD contract and purchasing the risk-free

asset.

$$S_0 = F_0(T)(1+r)^{-T}$$

Note however that the PAYOFF of the long forward + risk-free asset is:

$$Payoff = S_T - F_0(T) + Bond = S_T - F_0(T) + F_0(T)$$

Substitute this into put-call parity to get the put-call FORWARD parity (where the LHS is the synthetic protective put):

$$F_0(T)(1+r)^{-T} + p_0 = c_0 + X(1+r)^{-T}$$

Rearranging the above,

$$p_0 - c_0 = (X - F_0(T))(1+r)^{-T}$$

Firm value as a combination of options. A firm's value at $t = 0, T$ is V_0, V_T , and it has debt with value D at maturity ($t = T$). With equity E :

$$V_0 = E_0 + PV(D)$$

At $t = T$, the firm is solvent if $V_T > D$ and insolvent if $V_T < D$. If solvent, debtholders get D and shareholders get $E_T = V_T - D$; if insolvent, debtholders get V_T and shareholders get $E_T = 0$. So, debtholders are limited to a maximum of D , i.e. $\min(V_T, D)$, while shareholders get the residual $\max(0, V_T - D)$. Debtholders have a covered call, i.e. a short put + debt value = $PV(D) - p_0$ while shareholders have a long call = c_0 - where the strike price of the call and put options are D . Take the put-call parity and replace underlying with the firm value V_0 and the strike price X with D :

$$V_0 = (PV(D) - p_0) + c_0$$

So, the put premium p_0 is the price the firm (shareholders) pays for debtholders to assume the risk so that shareholders don't have to pay extra if the firm doesn't have enough money to pay debt. The higher the p_0 , the higher the credit risk. This put option increases in value to shareholders as the likelihood of insolvency increases.

7.10 Valuing a Derivative Using a One-Period Binomial Model

We want to price call (and put) options.

Binominal model. Let S_0, S_1^u, S_1^d be the prices of the underlying at $t = 0, 1$ with $S_1^d < S_0 < S_1^u$. Let X be the strike price of the call and put options, and $S_1^d < X < S_1^u$. At $t = 0$, the call option is worth c_0 , and at $t = 1$,

$$c_1^u = \max(0, S_1^u - X) = S_1^u - X$$

$$c_1^d = \max(0, S_1^d - X) = 0$$

The only unknown value is c_0 , which may be determined using replication and no-arbitrage pricing. At $t = 0$, SELL the call option at c_0 and purchase h (hedge ratio) units of the underlying. Then the value of the call option at $t = 0$ is V_0 and at $t = 1$ it is:

$$V_0 = hS_0 - c_0$$

Defining $R^u = \frac{S_1^u}{S_0}$ and $R^d = \frac{S_1^d}{S_0}$,

$$V_1^u = hS_1^u - c_1^u = hR^u S_0 - \max(0, S_1^u - X)$$

$$V_1^d = hS_1^d - c_1^d = hR^d S_0 - \max(0, S_1^d - X)$$

We assume $V_1^u = V_1^d$ and solve for h :

$$h = \frac{c_1^u - c_1^d}{S_1^u - S_1^d}$$

Then discount V_1 to solve for c_0 :

$$V_0 = hS_0 - c_0 = \frac{V_1}{1+r}$$

$$c_0 = hS_0 - \frac{V_1}{1+r} = hS_0 - \frac{hS_1^u - (S_1^u - X)}{1+r} = hS_0 - \frac{S_1^u(h-1) + X}{1+r}$$

Alternatively, though less useful,

$$c_0 = hS_0 - \frac{V_1}{1+r} = hS_0 - \frac{hS_1^d}{1+r}$$

To calculate the equivalent put, use the put-call parity to calculate p_0 . Or just replace the c with a p above and re-derive analogous expressions using p_0 , though note the hedge ratio will be negative:

$$h = \frac{p_1^u - p_1^d}{S_1^u - S_1^d} < 0$$

because

$$\begin{aligned} p_1^u &= 0 \\ p_1^d &= X - S_1^d \end{aligned}$$

which can be rectified by saying the put must be bought rather than sold. In the binominal pricing model, the call option is SOLD. The put option is BOUGHT.

The binomial pricing model doesn't need the probabilities of the binomial branches! But there is a way to specify them - see below.

Another way of calculating c_0 is to first calculate the probabilities of high and

low ASSUMING RISK-NEUTRALITY, i.e. no risk-aversion or risk-seeking. We then want:

$$S_0 = \frac{S_1^u \pi + S_1^d (1 - \pi)}{1 + r}$$

Defining $R^u = \frac{S_1^u}{S_0}$ and $R^d = \frac{S_1^d}{S_0}$,

$$\pi = \frac{(1 + r) - R^d}{R^u - R^d}$$

Notes. (1) This 'risk-neutral probability' π (RNP) will differ from 'real-world' probabilities because the real world is not risk-neutral. The RNPs are 'virtual probabilities'. (2) If the risk-free rate changes, the RNP changes.

Using risk-neutral probabilities,

$$c_0 = \frac{c_1^u \pi + c_1^d (1 - \pi)}{1 + r}$$

Use put-call parity to derive p_0 .

Use the risk-neutral probabilities even if real probabilities are given; the RNPs are borne out of arbitrage. It is a change of probability measure to one that is risk-neutral, i.e. without arbitrage. Real probabilities mean that something is not risk-neutral or there is an arbitrage opportunity.

8 Financial Statement Analysis

8.1 Introduction

International Organization of Securities Commissions (IOSCO), most of the world's financial regulators are members. Objectives and Principles of Securities Regulation is benchmark - to protect investors, efficient/transparent markets, reduce system risk.

US SEC. Created after 1929 stock market crash. Most significant laws: Securities Act 1933, Securities Exchange Act 1934, Sarbanes-Oxley 2002. Securities Act 1933: specifies the financial and other significant information that investors must receive when securities are sold, prohibits misrepresentations, and requires initial registration of all public issuances of securities. Securities Exchange Act 1934: This law created the SEC, gave the SEC authority over all aspects of the securities industry, and empowered the SEC to require periodic reporting by companies with publicly traded securities. Sarbanes-Oxley 2002: Public Company Accounting Oversight Board (PCAOB) to oversee auditors, overseen by the SEC; auditor independence, management certification of financial reports, management must report on internal controls. Usually fulfill SEC requirements by regulatory filings and reply to SEC questions.

US SEC forms. Securities Offerings Registration Statement: registration and information disclosure about securities offering. Forms 10-K (US registrants), 20-F (Canadian), and 40-F (all other) - annual comprehensive overview with audited financial statements. Annual reports - for shareholders so much more polished and not required by SEC. Proxy Statement/Form DEF-14A - before annual and special meetings shareholders get a Proxy Statement about proposals, shareholders composition, directors; DEF-14A is for shareholders to assign proxy and filed with SEC. 10-Q (US registrants) and 6-K (overseas) for interim periods with unaudited financial statements, significant changes; 10-Q provided to shareholders or an abbreviated version. 8-K: 'current reports' if a significant change to the corporation occurs (e.g. changes in securities). Forms 3 (initial statement), 4 (changes), 5 (annual report), and 144 (proposed securities sale) by beneficial owners of corporate insiders (directors, officers, more than 10 percent of an equity class) to examine their trades. 11-K (annual report): records all insider or employee activity involving the buying and selling of a company's stock. The form is used to report employee transactions as well as transactions involving employee stock purchase savings or retirement plans.

EU. Each state regulates, but from 2005 for consolidated accounts of EU-listed companies use IFRS. IASB issues new IFRS standard, European Commission looks at it with the help of (1) European Financial Reporting Advisory Group, (2) Standards Advice Review Group, (3) Accounting Regulatory Committee. It then goes to the EU Parliament and Council of the EU for a vote. European Securities Committee (ESC) advises the EU on securities regulation; European Securities and Market Authority (ESMA) is a cross-border supervisor.

Financial report notes include information about operating segments. An operating segment (1) may generate revenue and expenses; (2) results regularly reviewed by management; (3) discrete financial information available. An operating segment is 10 percent or more of the combined operating segments' revenue, profit, or loss (each separately; revenue with revenue, profit with profit only, loss with losses only). The REPORTABLE operating segments must be more than 75 percent of the combined; the other is 'all other segments'. Companies must identify how the factors used to divide into segments and describe their products and services. Each segment's revenue, profits, assets, liabilities, etc. need to be reported in the notes.

Large customer. Must report if any single customer accounts for more than 10 percent of total revenues.

MD&A. Typically unaudited, though Germany is an exception (required and audited). IASB provides a 'Management Commentary' IFRS practice guidance on content: (1) the nature of the business; (2) management's objectives and strategies; (3) the company's significant resources, risks, and relationships; (4) results of operations; and (5) critical performance measures. US SEC requires

listed companies to provide MD&A and requires: (1) favorable/unfavorable factors impacting business; (2) economic and market uncertainties that could materially impact the business; (3) off-balance sheet and contractual commitments; (4) accounting policies and judgments.

Auditor reports. Annual reports usually have to be audited. Auditor's opinion is the 'audit report'. In US, PCAOB under Sarbanes-Oxley has audit standards; in most other places, International standards on auditing (ISA). ISA: auditor is to (1) get reasonable assurance (because it is sampled) that financial statements are on the whole free from material misstatement; (2) report on findings. Opinions: (1) unqualified (clean, unmodified) - e.g. 'true and fair', 'fairly presented'; (2) qualified - see report; (3) adverse - materially departs from standard; (4) disclaimer of opinion - scope limitation. Key or critical audit matters - matters which have a high probability of being misrepresented or are very complex. Under Sarbanes-Oxley, US auditors must express an opinion on the internal control systems (process for generating financial reports).

IFRS versus other systems. IFRS is principle-based; US GAAP is rules-based.

Trailing earnings.

An analyst gathers the following information (in € thousands) about a company whose fiscal year ends on 31 December:

Earnings for the six months ended 30 June of Year 1	2,000
Earnings for the year ended 31 December of Year 1	1,500
Earnings for the six months ended 30 June of Year 2	2,200

Q. The company's trailing 12 month earnings (in € thousands) for the period ended 30 June of Year 2 is:

- A. 1,700.
- B. 3,700.
- C. 4,200.

Trailing earnings or income calculation - answer is A because Y1 total minus Y1 half + Y2 half

8.2 Financial statement changes

Correction of an error for a prior period is handled by restating the financial statements (including the balance sheet, statement of owners' equity, and cash flow statement) for the prior periods presented in the current financial statements. Thus, it uses the "full retrospective method," which requires companies to restate prior periods' results.

New standards. Depending on the standard, companies may be permitted to

adopt the standards prospectively (in the future) or retrospectively (restate financial statements as though the standard existed in the past) while the new revenue recognition standard also offered companies the option of using a “modified retrospective” method of adoption. Under the modified retrospective approach, companies were not required to revise previously reported financial statements. Instead, they adjusted opening balances of retained earnings (and other applicable accounts) for the cumulative impact of the new standard.

8.3 Analyzing income statements

Income statement information. Revenue is both operating and non-operating revenue. Net revenue is revenue less returns and allowances. Gross profit is total revenue minus the cost of goods sold (COGS). EBITDA = operating income (pre-interest and pre-tax) + depreciation + amortization. Operating income = gross OPERATING profit (operating revenue minus COGS) minus operating expenses (wages, depreciation); sometimes, analysts say operating income = EBIT but EBIT includes non-operating revenue such as dividends and gains/losses on investment securities. Earnings = net income. Net income excludes OCI but net income includes non-recurring income (e.g. gains on PPE sales).

Accrual accounting is when revenues and expenses are recorded when transaction occurs rather than when money is exchanged (cash-based accounting). Accrual accounting - revenue recognized when it is EARNED, when risk and reward are transferred. If a customer pays on credit in exchange for the company's goods/services, then AR is recorded on the BS and revenue is recorded in the IS even if payment not received - because the revenue has been 'earned'. If payment received first, then record unearned (deferred) revenue as assets and need to deliver goods/services as liability in the BS and then later transfer to IS.

Service revenues are recognized by reference to the stage of completion of the service contract, e.g. if historical patterns provide evidence that 25% of the services performed under the contract are incurred during the contract's first year, only 25% of the contract revenue would be recognized in the first year.

Revenue recognition similar but slightly different under IASB, FASB - however, they converged in 2014.

Under the converged accounting standards, the incremental costs of obtaining a contract and certain costs incurred to fulfill a contract must be capitalized. They are capitalized as assets on the BS if the company expects to recover them through a contract; if not, then expense immediately. They can be depreciated or amortized over the life of the contract. If a company expensed these incremental costs in the years prior to adopting the converged standards, all else being equal, its profitability will appear higher under the converged standards.

Revenue recognition. Each requires probable collectivity under the contract,

though IFRS says more likely than not and GAAP is likely. Each identified performance is identified separately. The transaction price is allocated to each performance obligation. Revenue is recognized in the income statement when the performance obligation is satisfied (and as deferred revenue in the balance sheet if performance not satisfied yet). Revenue recognized only if 'highly probable' it will not be reversed; otherwise, recognize small amount and record 'right to refund' liability. If revenue recognized but payment is contingent on a future action, a contract asset is recorded. If consideration received before performance, a contract liability is recorded.

Examples. If acting as agent for another seller, only recognize fees as revenues. For upfront new stores of a franchisee, recognize deferred revenue and amortize straight-line over life of the franchise agreement.

Expense recognition. This is subtracted from total revenue to get operating revenue, i.e. expenses are an operating EXPENSE - whether immediately expensed, capitalized, amortized. Generally recognize when incurred or economic benefit consumed. Three expense-recognition models: matching, expensing as incurred, capitalization with subsequent amortization or depreciation.

Matching. Match costs with revenues at the same time or period. E.g. if inventory purchased and some sold, inventory is added to asset-side of BS, revenue is matched with sold inventory but leftover inventory remains an asset.

Expensing as incurred. Usually period costs, like overhead and employee compensation, aren't matched with anything so just expensed as-is in the BS. In the CFS, they are operating cashflows.

Capitalization with amortization or depreciation. Doesn't apply to land or IP. Initially recognize expense as asset and then depreciate or amortize as benefits used up.

Expense (e.g. equipment) can be expensed immediately as cash (and in the CFS affect operating cashflows) or recognized as investment and then capitalized and depreciated (and in the CFS affecting investing cashflows). Income statement: Expensing immediately - if it exceeds the amortized or depreciated expenses of that year - increases operating expenses, reduces operating income, and reduces taxes. Balance sheet: Expensing immediately reduces assets because it exhausts the asset immediately; capitalizing and depreciating means the asset is exhausted over a longer time - note this affects shareholder equity.

Capitalizing interest - if the interest is incurred for a company's asset (e.g. PPE), it is 'expensed', i.e. the interest capitalized and depreciated over a long time as part of the PPE (so it's basically an ASSET and appears in the balance sheet); if the interest is incurred for something a company will sell, it is capitalized and depreciated as inventory in the balance sheet, and then expensed

in the income statement when the asset is sold. Total interest cost might be divided between immediately expensed and capitalized (as part of an asset for itself or for inventory; the entire capitalized amount will show up, but it will be depreciated or amortized over several years).

The ICR interest coverage ratio (= times interested earned (TIE)):

$$ICR = \frac{\text{EBIT} + \text{amortization of deferred financing for that same year}}{\text{expensed interest} + \text{entire capitalized interest for that year}}$$

The amortization is added because it's not 'fair' to add the entire capitalized lease - which is depreciated or amortized over several years without 'getting a refund' on the interest amortized that year.

To adjust capitalized costs to expensed costs (i.e. as if the expensed asset is exhausted immediately), (1) income statement - add expenses for each year, remove present amortizations originating in prior years, (2) balance sheet - remove capitalized costs from assets (don't add anything back to assets because the asset is being exhausted immediately), (3) cashflows - reduce operating cashflows and increase investing cashflows.

Balance sheet information. Ending shareholders' equity = Beginning shareholders' equity + Net income + Other comprehensive income - Dividends + Net capital contributions from shareholders (=paid-in capital). Retained earnings = net income + OCI - dividends; funds reinvested back into the business. So, ending shareholder's equity = beginning shareholder's equity + retained earnings + paid-in capital.

IFRS requires that items of income or expense that are material or relevant to the understanding of the entity's financial performance should be disclosed separately, e.g. unusual or infrequent items - litigation, restructuring, goodwill impairment, buying/selling and integrating assets. Similar to GAAP. They might be shown as 'other operating income or expenses, and separate 'recurring income' from 'operating income'. Discontinued operations are separated.

Under US GAAP, material items that are unusual or infrequent and that are both as of reporting periods beginning after 15 December 2015 are shown as part of a company's continuing operations but are presented separately.

Other items. Consolidation changes scope of financial information, exchange rates also have an effect. The specific effects of these aren't required to be disclosed.

Earnings per share (EPS). IFRS and GAAP require basic and diluted EPS and continuing operations EPS in the income statement using IFRS: 'ordinary shares' (subordinate to all other equity); GAAP: 'common stock' or 'common shares'. This is 'basic EPS'. Common stock or ordinary shares of OUTSTAND-

ING (i.e. not part of company treasury).

$$\text{Basic EPS} = \frac{\text{Net income} - \text{preferred dividends}}{\text{Weighted average no. of shares}}$$

The weighted average number of common shares outstanding would be the sum of 2,000,000 shares $\times$ 1/2 year + 1,900,000 shares $\times$ 1/2 year, or 1,950,000 shares. If a stock split or stock dividend increases the number of common stock, then find the weighted average and then apply this change as if it occurred on Jan 1 of that year.

If a company has financial instruments convertible into common stock (e.g. convertible bonds, warrants, employee stock options, convertible preferred stock), then it has a 'complex capital structure'; otherwise, 'simple capital structure'. If all instruments converted to common stock, then 'diluted EPS'.

Convertible preferred stock. Use 'if-converted' method - as if it was converted at the beginning of the fiscal year, and this also means NO PREFERRED DIVIDENDS. So,

$$\text{Diluted EPS} = \frac{\text{Net income (i.e. before preferred dividends)}}{\text{Outstanding common stock} + \text{NET converted preferred stock}}$$

Convertible debt. Similar to convertible preferred stock, use 'if-converted' method.

$$\text{Diluted EPS} = \frac{\text{Net income} + \text{after-tax convertible debt interest} - \text{preferred dividends}}{\text{Outstanding common stock} + \text{converted convertible debt}}$$

Stock options and warrants. Use GAAP 'treasury-stock' method where all options are exercised at the beginning of the fiscal year to buy shares at the price of that time and then treasury uses the proceeds to buy back shares at the fiscal year average price. The number of shares increases and prorated by date. No change to net income. Use outstanding common stock again (i.e. remove company treasury shares). Calculate the NET increase in shares - i.e. shares are issued AND THEN BOUGHT BACK, so calculate the net!!! The IFRS uses a method which is equivalent, it just calls the stock the company treasury buys as 'inferred shares'.

Stock dividends are treated as having been issued retroactively to the beginning of the period, e.g. if common shares are issued as stock dividends, it's as if they were issued at the beginning of the year.

Antidilutive securities are not counted in diluted EPS. Examine each security in isolation from others. Diluted EPS is minimum possible EPS. E.g. convertible preferred stock which has a high dividend rate and low share dilution. In this case, the final diluted EPS is simply reported as basic EPS.

Whether a warrant will be exercised for diluted EPS calculations depends on

its exercise price compared to the average annual price.

Income statement ratios: common size analysis, ratios. Common size analysis - each line item in the income statement is a percentage of REVENUE, allowing one some analysis of strategies. Relevant ratios:

$$\text{Gross profit margin} = \frac{\text{Gross profit}}{\text{Total revenue}}$$

$$\text{Net profit margin} = \frac{\text{Net income}}{\text{Total revenue}}$$

The latter is also called 'profit margin' or 'return on sales'. Other ratios: operating profit margin, pretax margin (profit before tax but after interest divided by total revenue).

Non-recurring items (non-operating income). Part of net income but not operating income. Generally taxed. (1) Realized gains or losses on PPE upon sale. (2) Litigation expenses. (3) Changes in accounting. (4) Other infrequent or unusual changes. (5) Asset impairments (GAAP but not IFRS).

Other comprehensive income (OCI). Not on income statement, but included as part of shareholder equity. Generally taxed but is complicated. (1) Unrealized gains/losses on certain securities - debt securities held for interest AND principal (IFRS, GAAP) and elected equities (IFRS) and derivative contracts used as hedges (IFRS). (2) Actuarial gains/losses on defined benefit pension assets and liabilities (IFRS and GAAP). (3) Asset revaluations (and reversals of revaluation surpluses but beyond this, they're asset impairments) (IFRS). (4) Foreign currency adjustments (in consolidating statements of foreign subsidiaries).

NOTE: If an accounting policy change is made, if practical, it should be retrospective for all SHOWN years in the report.

NOTE2: Under IFRS, income includes increases in economic benefits from increases in assets, enhancement of assets, and decreases in liabilities.

NOTE3: Issuing shares doesn't change the IS; it's 'common stock' and if higher than par value, then 'additional paid-in capital' in the BS.

NOTE4: Units of production depreciation mean assets are depreciated depending on how much output they produce.

8.4 Capitalizing interest example

Consider a company that builds a small production facility worth \$5 million with a useful life of 20 years. It borrows the amount to finance this project at an interest rate of 10%. The project will take a year to complete to put the

building to its intended use, and the company is allowed to capitalize its annual interest expense on this project, which amounts to \$500,000.

The company capitalizes interest by recording a debit entry of \$500,000 to a fixed asset account and an offsetting credit entry to cash. At the end of construction, the company's production facility has a book value of \$5.5 million, consisting of \$5 million in construction costs and \$500,000 in capitalized interest.

In the next year, when the production facility is used, the company books a straight-line depreciation expense of \$275,000 (\$5.5 million of the facility's book value divided by 20 years of useful life) of which \$25,000 (\$500,000 of capitalized interest divided by 20 years) is attributable to the capitalized interest.

8.5 Analyzing balance sheets

If interest is capitalized, it appears as part of the PPE as an ASSET in the balance sheet, and not a liability.

Intangible assets. Some are separable from the company (e.g. patents) while others not (goodwill). IFRS allows reporting on 'cost model' or 'revaluation model'. GAAP only allows 'cost model'. Intangible with finite life is amortized over best life estimate; intangible with indefinite life is reviewed each year and not amortized. Financial analyst might delete and add the amortization, depreciation, or impairment to income and recalculate 'tangible book value'. IFRS allows identification of intangibles if (1) it has economic benefits and (2) reliably costed.

R&D. IFRS requires research to be expensed and development to be capitalized. GAAP in contrast requires both be expensed. Under IFRS and GAAP, externally acquired intangibles are capitalized.

Capitalizing internal development costs. E.g. GAAP software research costs are expensed but development costs capitalized once software is 'feasible'. This is similar to IFRS development.

Goodwill. Economic goodwill versus accounting goodwill (only recorded for acquisitions). IFRS and GAAP: capitalize goodwill, test for impairment annually.

Financial instruments. A contract that gives rise to a financial asset of one entity, and a financial liability or equity instrument of another entity. After initial acquisition, a financial instrument is valued by 'amortized cost' or 'fair value'.

Amortized cost ('measured at cost') is initial value minus any principal repayments, plus or minus any amortization of discount or premium (at the time

of purchase, not changes thereafter), and minus any reduction for impairment. Amortized cost if asset is to be 'measured at cost' (IFRS) or 'held-to-maturity' (GAAP). Market value changes not reflected, no unrealized gains/losses in income statement nor balance sheet.

Fair value: unrealized (y/y) gains and losses are recorded as income OR comprehensive income; realized gains or losses are income, not comprehensive income.

Fair value OCI - 'measured at fair value through OCI' (IFRS) or 'available-for-sale securities' (GAAP) (security being held for cashflow AND sale). IFRS: debt (and equity if declared as such at the beginning); GAAP: debt only. Unrealized gains/losses on OCI (and therefore not on income statement) but shows up in balance sheet as asset change.

Fair value income (not OCI) - 'measured at fair value through PNL' (IFRS) or 'trading security' (GAAP). Unrealized gains/losses on income statement AND balance sheet. IFRS: all other securities, and can make an initial irrevocable election for any security; GAAP: all equities and trading (not for cashflows, rather for sale) debt.

Measured at Cost or Amortized Cost	Measured at Fair Value through Other Comprehensive Income	Measured at Fair Value through Profit and Loss
▪ Debt securities that are to be held to maturity. ▪ Loans and notes receivable ▪ Unquoted equity instruments (in limited circumstances in which the fair value is not reliably measurable, cost may serve as a proxy [estimate] for fair value)	▪ "Available-for-sale" debt securities (US GAAP); debt securities for which the business model involves both collecting interest and principal and selling the security (IFRS) ▪ Equity investments for which the company irrevocably elects this measurement at acquisition (IFRS only)	▪ All equity securities unless the investment gives the investor significant influence (US GAAP only) ▪ "Trading" debt securities (US GAAP) ▪ Securities not assigned to either of the other two categories, or investments for which the company irrevocably elects this measurement at acquisition (IFRS only)

Debt securities at fair value: interest income is recognized in profit and loss while unrealized gains and losses are recognized in other comprehensive income.

Non-current liabilities. All liabilities that are not classified as current are considered to be non-current or long-term. Non-current is beyond 12 months following reporting period. Long-term financial liabilities such as loans or bonds payable are amortized at cost so that at maturity they are par value. In some cases, financial liabilities are held at fair value, if for trading, e.g. shorting a security by borrowing it (financial liability is to return the security).

Deferred tax liabilities and assets. DT liability: when taxes are paid later, e.g. income is reported later or use accelerated depreciation over straight-line depreciation. DT asset: if income is reported earlier, e.g. loss carryover, prepaid expense ends up on balance sheet and will be used later so will be expensed later.

Balance sheet analysis for liquidity (ability to meet short-term financial commitments) and solvency (longer-term meeting obligations). Use ratios and common-size analysis.

Common-size analysis. Vertical common-size analysis - state each line item as a percentage of total assets. Compare cash and marketable securities from current assets with current liabilities to examine liquidity. High goodwill can mean recent acquisitions; low goodwill can mean organic growth.

Balance sheet ratios. Note current assets include inventories.

Liquidity Ratios	Calculation	Indicates
Current	$\text{Current assets} \div \text{Current liabilities}$	Ability to meet current liabilities
Quick (acid test)	$(\text{Cash} + \text{Marketable securities} + \text{Receivables}) \div \text{Current liabilities}$	Ability to meet current liabilities
Cash	$(\text{Cash} + \text{Marketable securities}) \div \text{Current liabilities}$	Ability to meet current liabilities
Solvency Ratios		
Long-term debt-to-equity	$\text{Total long-term debt} \div \text{Total equity}$	Financial risk and financial leverage
Debt-to-equity	$\text{Total debt} \div \text{Total equity}$	Financial risk and financial leverage
Total debt	$\text{Total debt} \div \text{Total assets}$	Financial risk and financial leverage
Financial leverage	$\text{Total assets} \div \text{Total equity}$	Financial risk and financial leverage

Note: accumulated depreciation of an asset is recorded as a CONTRA-ASSET in the balance sheet.

8.6 Analyzing Cashflows

Primary financial statements. Balance sheet is a stock. Income, cashflow, and statement of shareholder equity statements are flows between balance sheet reporting dates. Cashflows are cash accounting while other accounts are generally accrual-based. Changes in the balance sheet (i.e. assets $\neq$ liabilities in a transaction) originate/get reflected in the income statement, e.g. interest payment on loan is an expense in the income statement and a reduction in the asset (cash) but not change in liability (so, RE changes).

Accounts receivables. Beginning AR from balance sheet. Add revenues from income statement, minus cash collected, to get the final AR reported to the balance sheet. Similar for accounts payable.

Example. Buy inventory on credit. Balance sheet: inventory recorded as asset, AP recorded as liability. When paid, cash decline is recorded as asset decline, AP liability declines, and operating cash declines. When sold, inventory declines, cash increases, while in the income statement, revenue AND costs recorded.

Operating cashflows. Direct or indirect. The direct method uses the major categories of gross cash receipts and payments, and the indirect method reconciles net income to net cash flow.

Direct operating. Use revenue to find cash paid to company (by customers) and cash paid out - cash paid out decomposed into suppliers, employees, other operating expenses, interest, taxes. Cash received: revenue +/- AR change. Cash paid to suppliers: COGS +/- inventory +/- AP change. Cash paid to employees: Salaries and wages expense +/- salaries and wages payable. Cash for other operating expenses: other operating expenses +/- prepaid expenses +/- accrued liabilities. Cash for interest (GAAP: operating; IFRS: operating or financing): Interest expense +/- interest payable change. Cash for income taxes: income taxes +/- tax payable change.

Indirect operating. Adjusts net income for non-cash items (e.g., depreciation, amortization) and changes in current assets and liabilities. Net income is adjusted using current operating assets and liabilities. Adjust for non-operating activities (GAIN on sale of equipment is non-operating and belongs in investing cashflows, GAIN on sale means subtract from net income; asset value changes upon retiring debt early - e.g. loss on debt retirement means addition to net income), non-cash expenses (e.g. add back depreciation/amortizations), changes in operating working capital (add CHANGES in AR, AP, wages payable, deferred/prepaid expenses, and changes in inventory). Finally, adjust for deferred tax liabilities - an increase in deferred tax liability means an addition to net income.

Convert indirect to direct operating cashflows in three steps:

<i>Step 1</i>	Total revenues	USD23,803
Aggregate all revenue and all expenses	Total expenses	21,593
	Net income	USD2,210
<i>Step 2</i>	Total revenue less noncash item revenues:	
Remove all noncash items from aggregated revenues and expenses and break out remaining items into relevant cash flow items	(USD23,803 – USD205) =	USD23,598
	Revenue	USD23,598
	Total expenses less noncash item expenses:	
	(USD21,593 – USD1,052) =	USD20,541
	Cost of goods sold	USD11,456
	Salary and wage expenses	4,123
	Other operating expenses	3,577
	Interest expense	246
	Income tax expense	1,139
	Total	USD20,541
<i>Step 3</i>	Cash received from customers ^a	USD23,543
Convert accrual amounts to cash flow amounts by adjusting for working capital changes	Cash paid to suppliers ^b	(11,900)
	Cash paid to employees ^c	(4,113)
	Cash paid for other operating expenses ^d	(3,532)
	Cash paid for interest ^e	(258)
	Cash paid for income tax ^f	(1,134)
	Net cash provided by operating activities	USD2,606

Calculations for Step 3:

^aRevenue of \$23,598 less increase in accounts receivable of \$55.

^bCost of goods sold of \$11,456 plus increase in inventory of \$707 less increase in accounts payable of \$263.

^cSalary and wage expense of \$4,123 less increase in salary and wage payable of \$10.

^dOther operating expenses of \$3,577 less decrease in prepaid expenses of \$23 less increase in other accrued liabilities of \$22.

^eInterest expense of \$246 plus decrease in interest payable of \$12.

^fIncome tax expense of \$1,139 less increase in income tax payable of \$5.

To be added to NI to get operating cashflows:

- Non-cash items
 - Depreciation expense of tangible assets
 - Amortization expense of intangible assets
 - Depletion expense of natural resources
 - Amortization of bond discount
- Non-operating losses
 - Loss on sale or write-down of assets
 - Loss on retirement of debt
 - Loss on investments accounted for under the equity method
- Increase in deferred income tax liability
- Changes in working capital resulting from accruing higher amounts for expenses than the amounts of cash payments or lower amounts for revenues than the amounts of cash receipts
 - Decrease in current operating assets (e.g., accounts receivable, inventory, and prepaid expenses)
 - Increase in current operating liabilities (e.g., accounts payable and accrued expense liabilities)

Note: Don't subtract dividend payments from NI to get operating cashflows. They remain part of operating cash inflows, though the dividends also become part of financing cash outflows.

Investing cashflows. Direct and indirect methods are the same here. PPE purchases, including the capex or purchase price of the PPE. For PPE sales, e.g. take beginning equipment from balance sheet and ending equipment, get equipment purchased from notes. The difference must be the historical price of the equipment SOLD. Take beginning accumulated depreciation (a CONTRA-ASSET) from balance sheet, ending accumulated depreciation, and depreciation expense from income statement. The difference must be depreciation thrown away with the equipment sale. From this, calculate book value of equipment sale, then add this to GAIN ON SALE cash from the income statement (i.e. if equipment with book value 100 was sold for 120, then 20 is the gain on sale). This is the 'cash' received from the sale.

NOTE: the book value of an asset is just a change from one asset in the balance sheet (asset book value) to another asset (cash), so it's not part of revenue! Only the realized gain/loss ON TOP of the book value is recorded as revenue!

Financing cashflows. Indirect and direct methods same. If long-term debt has declined, it was probably retired - a financing cash outflow.

GAAP v IFRS. US GAAP classify interest and dividends received from investments as operating activities, whereas IFRS allow companies to classify those items as either operating or investing cash flows. Likewise, US GAAP classify interest expense as an operating activity, even though the principal amount of

the debt issued is classified as a financing activity. IFRS allow companies to classify interest expense as either an operating activity or a financing activity. US GAAP classify dividends paid to stockholders as a financing activity, whereas IFRS allow companies to classify dividends paid as either an operating activity or a financing activity.

Topic	IFRS	US GAAP
Classification of cash flows:		
• Interest received	Operating or investing	Operating
• Interest paid	Operating or financing	Operating
• Dividends received	Operating or investing	Operating
• Dividends paid	Operating or financing	Financing

US GAAP classify all income tax expenses as an operating activity. IFRS also classify income tax expense as an operating activity, unless the tax expense can be specifically identified with an investing or financing activity (e.g., the tax effect of the sale of a discontinued operation could be classified under investing activities).

8.7 Analyzing cashflows 2

Usually, if operating cashflow exceeds net income, this is usually a positive sign for company health. It usually means depreciation expenses from prior investments are reducing net income but leaving operating cashflows high.

Common-size analysis. For each line item in cashflows, use as the base cash inflows (for cash inflow line items) or outflows (for cash outflow line items) OR net revenue. Can do this for both direct and indirect operating cashflows.

CF. Cashflows. Usually refers to just operating cashflows, CFO.

FCF. Free cashflow. CFO + investing cashflows.

$$FCF = CFO - FCInv(capex)$$

Using free cash flow, investors can see if the company has enough cash on hand to repay creditors and equity investors after paying for its operating expenses and capital expenditures. NOTE: only add interest if GAAP or interest is classified as operating instead of financing cashflows under IFRS.

FCFE. FCF to equityholders. CFO + investing cashflows + financing cashflows.

$$FCFE = CFO - FCInv + \text{Net borrowing}$$

which is equivalently

$$FCFE = CFO - FCInv - \text{Debt repayments}$$

When net borrowing is negative, debt repayments exceed receipts of borrowed funds.

FCFF. FCF to the firm (debt and equityholders):

$$FCFF = NI + NCC + Interest(1 - \text{tax rate}) - FCInv - WCInv$$

where NCC is non-cash items such as depreciation; FCInv is fixed capital like equipment; and WCInv is working capital. The reason for adding back interest is that FCFF is the cash flow available to the suppliers of debt capital as well as equity capital. FCFF can be restated as:

$$FCFF = CFO - FCInv + Int(1 - \text{tax rate})$$

where CFO is cashflow from operating activities. CFO includes interest paid in operating activities - but if interest paid was included in financing, remove $Int(1 - \text{tax rate})$.

Cashflow ratios. Profitability or performance ratios versus interest coverage ratios - there are many kinds.

Performance Ratios	Calculation	What It Measures
Cash flow to revenue	$CFO \div \text{Net revenue}$	Operating cash generated per dollar of revenue
Cash return on assets	$CFO \div \text{Average total assets}$	Operating cash generated per dollar of asset investment
Cash return on equity	$CFO \div \text{Average shareholders' equity}$	Operating cash generated per dollar of owner investment
Cash to income	$CFO \div \text{Operating income}$	Cash generating ability of operations
Cash flow per share ^a	$(CFO - \text{Preferred dividends}) \div \text{Number of common shares outstanding}$	Operating cash flow on a per-share basis
Coverage Ratios	Calculation	What It Measures
Debt coverage	$CFO \div \text{Total debt}$	Financial risk and financial leverage
Interest coverage ^b	$(CFO + \text{Interest paid} + \text{Taxes paid}) \div \text{Interest paid}$	Ability to meet interest obligations
Reinvestment	$CFO \div \text{Cash paid for long-term assets}$	Ability to acquire assets with operating cash flows
Debt payment	$CFO \div \text{Cash paid for long-term debt repayment}$	Ability to pay debts with operating cash flows

Coverage Ratios	Calculation	What It Measures
Dividend payment	CFO ÷ Dividends paid	Ability to pay dividends with operating cash flows
Investing and financing	CFO ÷ Cash outflows for investing and financing activities	Ability to acquire assets, pay debts, and make distributions to owners

Notes:

^a If the company reports under IFRS and includes total dividends paid as a use of cash in the operating section, total dividends should be added back to CFO as reported and then preferred dividends should be subtracted. Recall that CFO reported under US GAAP and IFRS may differ depending on the treatment of interest and dividends, received and paid.

^b If the company reports under IFRS and included interest paid as a use of cash in the financing section, then interest paid should not be added back to the numerator.

8.8 Working capital

Working capital turnover ratio.

$$\text{WC turnover ratio} = \frac{\text{Net revenue}}{WC}$$

8.9 Inventories

Inventories are stock variables, COGS is a flow variable, so inventory write-downs lead to an INCREASE in COGS.

$$COGS = \text{beginning inventory} + \text{inventory purchases} - \text{ending inventory}$$

IFRS allows weighted average, retail, standard or FIFO - FIFO because it reflects reality, cannot use LIFO. Weighted average - the cost of each item is determined from the weighted average of the cost of similar items at the beginning of a period and the cost of similar items purchased or produced during the period. Retail - inventory is measured based on its selling price reduced by the relevant profit margin. Standard costing - inventory is measured at the standard cost of each unit reflecting predetermined rates for the material, labor and overhead expenses at normal level of output and efficiency.

IFRS states that inventories shall be measured (and carried on the balance sheet on an item-by-item basis) at the lower of historical cost and net realizable value NRV (i.e. net of costs of selling and preparing for sale). If the inventory value falls below the inventory carrying amount (cost, but not replacement cost), then write-down to net-realizable value and the loss is recognized as an expense. If inventory value rises above inventory carrying amount (limited to the original write-down; reversal must use inventory existing in the current, not previous year), then this is a 'reversal' and reduction in COGS (and therefore an increase in operating income). Producers/commodity brokers' ag/forestry/mineral/mineral product inventories simply always use net realizable value.

Both IFRS and US GAAP allow agricultural inventories to be valued at net realizable value.

GAAP. Weight cost, FIFO or LIFO. After 2016, inventories are measured at lower of cost or net realizable value - similar to IFRS. However, GAAP prohibits reversal of write-downs. Furthermore, for inventories measured using LIFO (cost of sales better reflects current replacement values) or retail inventory methods, it is lower in cost and market value. Market value is current replacement cost subject to min (NRV minus normal profit margin) and max (NRV) limits.

LIFO. Usually, inventories rise in prices so rise in carrying costs. LIFO means oldest inventories remain the longest, so the net realizable value is conservative - meaning fewer write-downs.

FIFO and LIFO (for expenses in the income statement and ending inventory on the balance sheet). Per unit, if inventory prices are falling, FIFO has higher cost of sales in the income statement and profit will be lower than LIFO and vice versa; moreover, it will be much higher than ending inventory which will have more later-purchased inventory. The carrying amount of inventories under FIFO will more closely reflect current replacement values because inventories are assumed to consist of the most recently purchased items. The cost of sales under LIFO will more closely reflect current replacement value.

$$\text{LIFO reserve} = \text{FIFO inventory value} - \text{LIFO inventory value}$$

In inflationary environments, LIFO COGS will be higher than FIFO COGS and FIFO inventories will be higher than LIFO inventories, leading to higher LIFO reserves. A falling LIFO reserve in an inflationary environment means more inventories are being sold than bought.

$$\text{FIFO COGS} = \text{LIFO COGS} - X$$

$$FIFO_0 + IP - FIFO_1 = LIFO_0 + IP - LIFO_1 - X$$

$$X = (FIFO_1 - LIFO_1) - (FIFO_0 - LIFO_0)$$

$$X = \text{end LIFO reserve} - \text{start LIFO reserve}$$

$$\text{FIFO COGS} = \text{LIFO COGS} - (\text{end LIFO reserve} - \text{start LIFO reserve})$$

This makes sense because the last term is (end FIFO - start FIFO) - (end LIFO - start LIFO), where (end FIFO - start FIFO) should rise in an inflationary environment. As inventory shrinks, it is usually expected that the difference between FIFO and LIFO inventories shrinks, so LIFO reserve to shrink.

Ratios.

$$\text{Inventory turnover} = \frac{COGS}{\text{average inventory}}$$

where average inventory is after any writedowns.

Notes:

- Note distinction between production and purchase costs.
- Note perpetual (inventory data updated real-time) versus periodic (inventory data updated e.g. once a year) inventory systems - the distinction is most important for LIFO but not much to FIFO.

Disclosures.

IFRS requires the following financial statement disclosures concerning inventory:

- a. the accounting policies adopted in measuring inventories, including the cost formula (inventory valuation method) used;
- b. the total carrying amount of inventories and the carrying amount in classifications (e.g., merchandise, raw materials, production supplies, work in progress, and finished goods) appropriate to the entity;
- c. the carrying amount of inventories carried at fair value less costs to sell;
- d. the amount of inventories recognized as an expense during the period (cost of sales);
- e. the amount of any write-down of inventories recognized as an expense in the period;
- f. the amount of any reversal of any write-down that is recognized as a reduction in cost of sales in the period;
- g. the circumstances or events that led to the reversal of a write-down of inventories; and
- h. the carrying amount of inventories pledged as security for liabilities.

Inventory-related disclosures under US GAAP are similar to these disclosures, except that requirements (f) and (g) are not relevant because US GAAP does not permit the reversal of prior-year inventory write-downs. US GAAP also requires the disclosure of significant estimates applicable to inventories and of any material amount of income resulting from the liquidation of LIFO inventory.

Note: Inventory valuation allowance accounts are contra accounts in the BS for reduced inventory values. E.g. if carrying value is 100 and NRV is 90, then inventory valuation allowance is 10.

IFRS. The reversal of any write-down of inventories is recognized as a reduction in cost of sales (reduction in the amount of inventories recognized as an expense)

8.10 Long-term assets

On the statement of cash flows, costs of internally developing intangible assets are classified as operating cash outflows (and expensed immediately) whereas costs of acquiring intangible assets are classified as investing cash outflows (and

capitalized).

Asset valuation models. Revaluation model - revaluation. Cost model - historical cost minus accumulated depreciation.

Revaluation. A reverse of an impairment. Increases asset in balance sheet. Doesn't affect net income, but it affects other comprehensive income. Doesn't affect cashflows.

An asset is impaired when the carrying amount exceeds its recoverable amount. Expensed in the income statement. Reduces asset value in the balance sheet. Impairment doesn't affect any cashflows.

IFRS: The recoverable amount is the higher of its fair value less costs to sell and its value in use. Value in use is based on the present value of expected future cash flows.

GAAP: assessing recoverability is separate from measuring impairment loss. The carrying amount of an asset is not recoverable when it exceeds the UNDISCOUNTED expected future cash flows of the group. If the asset's carrying amount is not recoverable, the impairment loss is the difference between the asset's FAIR VALUE and carrying amount.

PPE (long-lived assets held for USE), intangible assets with definite lives - test for impairment in a year only if there are INDICATIONS of impairment. Intangibles with indefinite lives: tested annually for impairment; compare carrying amount versus fair value. Land - test for impairment if indications.

PPE (long-lived assets reclassified for SALE) - reclassify if management now intends to sell and sale is highly probable; upon reclassification test for impairment, comparing to fair value less costs to sell. Assets held for sale are no longer depreciated.

Reversals. IFRS allows impairment reversals on assets held for use or sale. GAAP allows impairment reversals on assets for held for sale ONLY.

The gain or loss on the sale of long-lived assets is computed as the sales proceeds minus the carrying amount of the asset at the time of sale. For exchanges, assume acquired asset has the fair value of the asset given up, but if that's not clear then simply let fair value of new asset = carrying amount of old asset (i.e. no gain/loss).

The fixed asset turnover ratio (total revenue divided by average net fixed assets or net PPE) reflects the relationship between total revenues and investment in PPE. Other ratios such as useful life and age can be obtained in straight-line depreciation by dividing historical cost or accumulated depreciation by the an-

nual depreciation expense.

Notes: IFRS requires useful lives, amounts of disposals, impairment losses, amortization rates to be disclosed, but not fair values or acquisition dates. For long-lived intangible assets, under US GAAP, companies are required to disclose the estimated amortization expense for the next five fiscal years; GAAP doesn't require disclosure of useful life.

Notes2: Internally developed intangible assets are generally expensed, but acquired intangible assets - e.g. in a business combination - are capitalized.

Disclosures. PPE IFRS: for each class of PPE, must disclose:

- measurement basis
- depreciation method
- the useful life (or, equivalently, the depreciation rate) used
- gross carrying amount
- accumulated depreciation at the beginning and end of the period
- reconciliation of the carrying amount at the beginning and end of the period.
- restrictions on title and pledges as security of PPE and contractual agreements to acquire PPE are required
- if the revaluation model is used, the date of revaluation, details of how the fair value was obtained, the carrying amount under the cost model, and the revaluation surplus must be disclosed
- depreciation expense for the period, the balances of major classes of depreciable assets, accumulated depreciation by major classes or in total, and a general description of the depreciation method(s) used in computing depreciation expense with respect to the major classes of depreciable assets.

Intangible assets IFRS. For each class of intangible assets, a company must disclose:

- whether the useful lives are indefinite or finite
- if finite, the useful lives (or, equivalently, the amortization rate) used, the amortization methods used, the gross carrying amount and the accumulated amortization at the beginning and end of the period, where amortization is included on the income statement, and a reconciliation of the carrying amount at the beginning and end of the period

- if an asset has an indefinite life, the company must disclose the carrying amount of the asset and why it is considered to have an indefinite life
- restrictions on title and pledges as security, contractual agreements to acquire intangible assets.
- if the revaluation model is used, the date of revaluation, details of how the fair value was obtained, the carrying amount under the cost model, and the revaluation surplus must be disclosed.

Intangible assets GAAP:

- gross carrying amounts and accumulated amortization in total and by major class of intangible assets
- the aggregate amortization expense for the period
- the estimated amortization expense for the next five fiscal years

The disclosures related to impairment losses also differ under IFRS and US GAAP. Under IFRS, a company must disclose for each class of assets the amounts of impairment losses and reversals of impairment losses. The company must also disclose in aggregate the main classes of assets affected by impairment losses and reversals of impairment losses and the main events and circumstances leading to recognition of these impairment losses and reversals of impairment losses. Under US GAAP, the company must disclose a description of the impaired asset, what led to the impairment, the method of determining fair value, the amount of the impairment loss, and where the loss is recognized on the financial statements.

Revaluation surplus is in the balance sheet. But overflow goes to income. See example below.

An analyst gathers the following information about a company's non-depreciable asset reported under the revaluation model:

Original cost	€25,000
Reported value after initial revaluation	€27,500
Reported value after second revaluation	€22,500

Q. The revaluation surplus after the second revaluation is:

- A. -€2,500.
- B. €0.
- C. €2,500.

Solution

- A. Incorrect because as a result of the revaluations, the €2,500 (= €27,500 - €25,000) increase in the value of the asset during the first year is recorded in the revaluation surplus account in equity, and the -€5,000 (= €22,500 - €27,500) decrease in fair value recorded in the second year is assumed to be applied only to the revaluation surplus account leading to a balance of -€2,500 (= €2,500 - €5,000).
- B. Correct because if a revaluation initially increases the carrying amount of the asset class, the increase in the carrying amount of the asset class bypasses the income statement and goes directly to equity under the heading of revaluation surplus. Any subsequent decrease in the asset's value first decreases the revaluation surplus and then goes to income. As a result, the €2,500 (= €27,500 - €25,000) increase in the value of the asset during the first year is recorded in the revaluation surplus account in equity, and out of the -€5,000 (= €22,500 - €27,500) decrease in fair value recorded in the second year, -€2,500 is used to decrease the revaluation account and the remaining -€2,500 flows to the income statement. Therefore, the balance in the revaluation surplus account is zero.

8.11 Long-term liabilities and equity

Leases. A lease that resembles a purchase is classified as a finance lease. All other leases are operating leases. If at least one criteria satisfied of below, it's a financing lease: (1) The lease transfers ownership of the underlying asset to the lessee. (2) The lessee has an option to purchase the underlying asset and is reasonably certain it will do so. (3) The lease term is for a major part of the asset's useful life. (4) The present value of the sum of the lease payments equals or exceeds substantially all of the fair value of the asset. (5) The underlying asset has no alternative use to the lessor.

Lessee. IFRS - same for finance and operating leases. At lease inception, the lessee records a lease payable (LP) in liabilities and a right-of-use (ROU) in assets in the BS, both equal to the present value of future lease payments (they can slightly differ if there are lease incentive payments, etc.). This is the PV of BOTH principal + interest payments. LP declines (using effective interest method) by ONLY the value of the principal each period. ROU is amortized, usually straight-line. LP and ROU will diverge. Lease interest paid + ROU depreciation are expenses in the IS; lease principal payment is a financing cash outflow, lease interest payment is an operating or financing cashflow.

Lessee. GAAP. Different for finance and operating leases. Finance leases: same

as IFRS. Operating leases (to 31 Dec 2018; afterwards, same as finance lease): both ROU and LP decline by lease repayments, so they're equal on the BS; their sum equals the lease payment and is expensed as a single combined line in the IS (it's a 'lease expense', not interest nor amortization nor principal - be careful in EBITDA calculations); they are recorded combined in operating cashflows.

Lease liability = ROU at the beginning. Record interest payment from lease liability + amortization from ROU in the IS; record lease interest payment as operating CF and principal repayments under financing cashflows.

	Lease Payment	Interest Expense (10% × Lease Liability)	Principal Repayment (Payment – Interest)	Lease Liability
	FO.1	FO.2	FO.3	FO.4
Year 0				379,079
Year 1	100,000	37,908	62,092	316,987
Year 2	100,000	31,699	68,301	248,685
Year 3	100,000	24,869	75,131	173,554
Year 4	100,000	17,355	82,645	90,909
Year 5	100,000	9,091	90,909	0
Total	500,000	120,921	379,079	

	Amortization Expense	ROU Asset
	Straight-Line F.1	F.2
Year 0		379,079
Year 1	75,816	303,263
Year 2	75,816	227,447
Year 3	75,816	151,631

(last several lines are cutoff).

Lessees. Lease accounting exemptions for certain lease contracts: If its term is 12 months or less (IFRS and US GAAP) or it is for a low-value asset (up to USD 5000) in sales price (IFRS only), then lessee can elect to expense the lease payments on a straight-line basis.

Lessor. Same under IFRS and GAAP. Financing lease: in the BS, enter a lease receivable (PV of future payments) and remove leased asset - and recognize gain/losses. This is like the sale of an asset, as it's a sales-type lease. Lease interest is recorded as revenue in the IS; entire lease payment (interest + principal repayment) is operating cashflow and recorded as cash assets in the BS (so the lessor is getting cash for both the gain/loss over the book value + book value = principal, as well as interest); lease receivable net of principal

payments is asset in the BS. Operating leases: they're like rentals, so the leased asset remains in the BS and entire lease payment recognized as revenue (a type of deferred revenue) in the IS. Depreciation of the asset is recognized as expense in the IS. Entire lease payments are recognized as cash/receivable assets in the BS; lessors get principal payments to approximately offset asset depreciation. Entire lease payment (interest + principal repayment) in operating cashflows, like financing leases.

Disclosures. Sometimes leases are specified but other times in 'Other assets', 'Other liabilities' in BS.

Employment wages and salaries. Short-term expenses, matched with employee work. Operating CF.

Postemployment and share-based compensation. Salaries vest in the period in which the employee works, but pensions vest with a lag.

In defined-contribution pensions, the company contribution is the pension expense and is reported as an operating cash outflow. Accounting is simple.

Defined-benefit plans (e.g. retiree health plans) are expensed when incurred because it's not known if/when they will be incurred, and are still operating CFs. Defined-benefit pensions. Have to estimate PV of contribution. IFRS. Pension present value of liabilities (estimated by actuaries) minus fair value of pension plan assets = retirement benefit obligations. Under IFRS and GAAP, net pension assets or liabilities (also like retirement benefit obligations) in the BS comprise service costs (more pensions to pay in the future as employee works more in the company) and past service costs, net interest or net income (in IS) = change in PV of net defined benefit asset or liability = net defined benefit asset or liability times the discount rate. The net pension assets or liabilities also implicitly include remeasurements = actuarial gains or losses (based on changes in numerical assumptions) + realized/unrealized gains on assets (e.g. dividends) which are in OCI.

Defined-benefit plan reporting. Report a net pension asset or liability (obligation). Compare fair value of pension assets to PV of pension obligations.

Pension expenses. For defined-contribution plans, easy. For defined-benefit plans, it's the service cost + interest expense - expected return on plan assets; ignore actuarial gains/losses (part of OCI).

An analyst gathers the following information (in \$ thousands) about a company reporting under US GAAP:

Defined Contribution Plan:	
Contributions to defined contribution plan	1,000
Defined Benefit Plan:	
Contributions to defined benefit plan	1,500
Employees' service cost for the period	1,400
Interest expense accrued on the beginning pension obligation	200
Expected return on plan assets	400
Actuarial gains for the period	100

Q. The pension expense (in \$ thousands) reported in the current year is:

- A. 2,200.
- B. 2,400.
- C. 2,500.

Solution

A. **Correct** because the pension expense would be the sum of the expense for the defined contribution plan and the defined benefit plan:

Plan	Expense Components under US GAAP	Amount
Defined contribution plan	Contributions	1,000
Defined benefit plan	Employees' service cost	1,400
	Interest expense accrued on the beginning pension obligation	200
	Expected return on plan assets	(400)
Total Expense		2,200

Share-based compensation. Treated as an expense, so reduces earnings even if no cash outlay. Can also dilute EPS. Shares will make employees risk-averse, stock options will make them risk-seeking (but can also make them risk-averse). 'Unrecognized' share-based compensation usually means they were granted but not yet vested, i.e. employees cannot yet use them and not yet recognized as compensation expenses.

Stock grants. If equity is normally issued, it increases assets (cash) and equity in the BS (and nothing in IS; recorded as investment CF inflow). If equity is issued to an employee, equity rises but the asset is expensed (at fair market value) in the IS as it is lost in the BS (and nothing in CF). Outright, restricted, performance stock grants estimated at fair value, allocated over the current period ('employee service period', i.e. between the grant date and vesting date) unless it's vested in the future.

Stock options. Employee stock options can differ from stock options in the market, so have to value. Accounting standards don't prescribe a pricing model, just if it is reasonable. At time of grant, no money changes hands, so it's just a change in RE in the BS. Vesting date = date that employee can first exercise option. When an option is exercised, the market price of the option at the time of exercise is not relevant. The amount of expense is determined based on the fair value of the option at the grant date. The fair value amount is recognized as

compensation expense over the vesting period. Upon exercise, company credits (i.e. reduces) common stock and increases additional paid-in capital in liabilities.

Cash-settled share-based compensation, e.g. stock appreciation rights (SARs), phantom stock. Employee gets awarded for changes in stock price but no stocks or options awarded. Estimated at fair value like other share-based compensation.

Notes. (1) Stock grants are expensed over the service period (i.e. from grant to vesting date) based on the FMV of the shares as of the GRANT DATE. (2) Share-based compensation (SBCs) are to align employee interests with SHAREHOLDER interests, NOT those of management.

8.12 Taxes

Accounting profit is pre-tax income, and from this a 'book' tax or 'accounting' tax recorded under IFRS or GAAP financial reports. However, this is different from actual tax paid to governments, based on tax accounting - this leads to taxable incomes and then actual taxes.

Taxable income (tax base) is the income base that will be actually taxed. It can differ from accounting pre-tax income because of temporary (timing; e.g. method of depreciation) and permanent (e.g. fines; based on regulation) differences, which then lead to deferred tax assets and liabilities. Note deferred tax assets are only allowed if reasonable expectation of future profits against which the asset or liability can be recovered or settled, i.e. temporary not permanent (permanent = penalties, subsidies). If circumstances change and the deferred tax ASSET might not be realized in full, apply a valuation allowance (a balance sheet contra-asset, which can later be reversed).

Note: a valuation allowance can only be applied to deferred tax assets, not to deferred tax liabilities.

Balance Sheet Item	Carrying Amount vs. Tax Base	Results in Deferred Tax Asset/Liability
Asset	Carrying amount > tax base	Deferred tax liability
Asset	Carrying amount < tax base	Deferred tax asset
Liability	Carrying amount > tax base	Deferred tax asset
Liability	Carrying amount < tax base	Deferred tax liability

Tax expense = tax payable + changes in net deferred tax liabilities - changes in net deferred tax assets. E.g. if corporate tax rate declines, BOTH deferred tax liabilities and assets would decline.

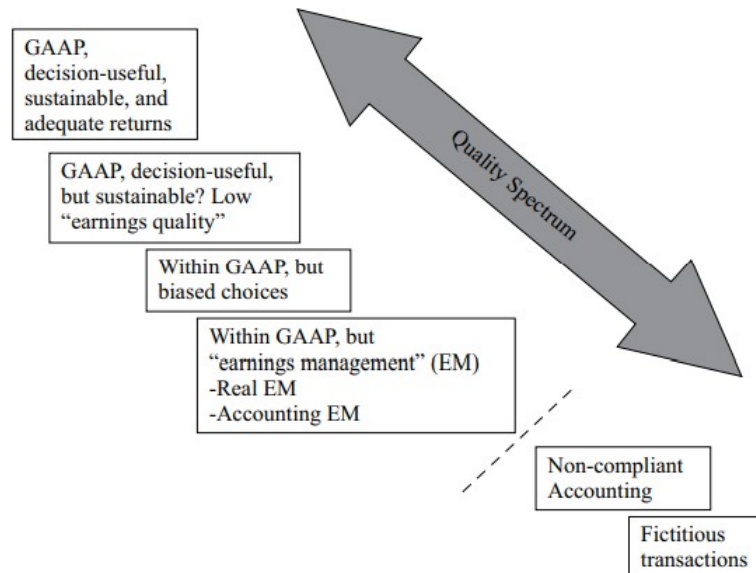
Corporate income taxes. Cash tax rate = actual tax paid divided by pretax income; used to forecast cashflows. Effective tax rate = tax provision (includes deferred taxes) divided by pretax income; used to forecast income.

Classify deferred tax liabilities as debt if the liability is expected to reverse, if not expected to reverse, there is no expectation of a cash outflow and classify as equity. Exclude from both debt and equity when both amounts and timing of reversals are uncertain.

8.13 Financial reporting quality

Earnings quality is different from financial reporting quality.

GAAP Conceptual Framework: high-quality if relevant and faithful information. Spectrum:



Non-GAAP metrics - sometimes companies include these in FRs. Non-GAAP financial metrics ('as-adjusted' measures, compared to GAAP 'as-reported' measures): adjusts GAAP compliant figures (usually earnings) to exclude items required by GAAP and/or include items prohibited by GAAP. This includes EBIT/EBITDA. Sometimes called underlying earnings, adjusted earnings, recurring earnings, core earnings, or similar. Must explain why this measure is useful, and what else it is used for (similar rules exist for IFRS, to prevent distraction from GAAP or IFRS measures). Non-GAAP operating metrics: subscriber numbers, occupancy rates, and other non-financial metrics. These aren't standardized, so hard to compare.

Management emphasis on non-GAAP financial measures to deflect attention from less-than-desirable GAAP financial results is an example of an aggressive presentation choice. Since 2003, if a company uses a non-GAAP financial measure in an SEC filing, it is required to display the most directly comparable GAAP measure with equal prominence and to provide a reconciliation between the non-GAAP measure and the equivalent GAAP measure. In other words, a company is not allowed to give more prominence to a non-GAAP financial measure in an SEC filing. ESMA also has guidelines similar to GAAP.

Non-GAAP measures. SEC prohibits (1) excluding cash settlements or liabilities other than EBIT and EBITDA, and (2) using non-recurring items to smooth +/- 2 years. However, (3) can exclude impairment losses.

Earnings management - biasing earnings to make them look better. Intentional and deliberate. Earnings can be managed upwards by real actions ('real earnings management'), e.g. delaying R&D to get a tax benefit, or making deliberate accounting choices ('accounting earnings management').

Conservatism. Helps avoid litigation costs. In general, conservatism means that revenues may be recognized once a verifiable and legally enforceable receivable has been generated and that losses need not be recognized until it becomes "probable" that an actual loss will be incurred. Examples of conservatism: Research costs - the future benefit of research costs is uncertain at the time the costs are incurred, both US GAAP and IFRS require immediate expensing instead of capitalization; Litigation losses - when it becomes "probable" that a cost will be incurred, both US GAAP and IFRS require expense recognition, even though a legal liability may not be incurred until a future date.; Insurance recoverables - a company that receives payment on an insurance claim may not recognize a receivable until the insurance company acknowledges the validity of the claimed amount.

Biased accounting is often more about intent. Big bath restructuring - take an exaggerated loss now so that future profits look better. Both US GAAP and IFRS provide for accrual of future costs associated with restructurings, and these costs are often associated with and presented along with asset impairments. But in some instances, companies use the accounting provisions to estimate "big" losses in the period under review so that performance in future periods will appear better.

Cookie jar accounting - intentionally overstating earnings in good times and understating them in bad times to smooth out earnings and give impression of consistent profitability.

Special items. Often cookie jars are hidden here. A large, one-time expense or source of income that a company does not expect to recur in future years. Special items are reported on the income statement and are separated out from other

categories of income and expenses so investors can more accurately compare the company's numbers across accounting periods. Examples of special items include extraordinary expenses, restructuring charges, gains from the elimination of debt, and earnings from discontinued operations.

Typically, three conditions exist when low-quality financial reports are issued: opportunity, pressure or motivation, and rationalization — sometimes referred to as the fraud triangle.

Market disciplines reporting: biased reporting isn't good for reducing cost of capital. Other disciplinary forces: market regulatory authorities, auditors, and private contracts. Market regulators enforce registration requirements and disclosures. Private sector standard agencies like IASB and FASB often set the standards and public regulatory agencies like the SEC enforce them or make their own - e.g. auditing requirements, MD&A. Auditors are not present to detect fraud as they only look at what is presented; if companies deliberately conceal things auditors usually will not find it. Auditors also only sample, and auditors are paid by the companies which need auditing.

To address noise from different accounting systems and treatments for e.g. depreciation and amortization, there is EBITDA and adjustments to it to get adjusted EBITDA. Lenders in loan covenants might also use this.

The SEC prohibits the exclusion of charges or liabilities requiring cash settlement from any non-GAAP liquidity measures, other than EBIT and EBITDA. Also prohibited is the calculation of a non-GAAP performance measure intended to eliminate or smooth items tagged as non-recurring, infrequent, or unusual when such items are very likely to occur again. The SEC views the period within two years of either before or after the reporting date as the relevant time frame for considering whether a charge or gain is a recurring item.

Depreciation methods. Straight-line, double-declining (or other speed) balance methods, units of production. Units of production example: the amount to be depreciated is USD480,000 (USD500,000 minus USD20,000). Dividing the USD480,000 by the machine's useful life of 240,000 units, the depreciation will be USD2 per unit.

Areas where IFRS or GAAP measures can be manipulated. Deferred tax assets (e.g. manipulating the valuation allowance), depreciation methods (e.g. using double-declining balance method or other accelerated depreciation method over straight-line), whether to capitalize or expense immediately (e.g. capitalizing and depreciating increases net income as WorldCom did), assigning value to different assets in an acquisition (incentive to assign low fair value to depreciable assets to have more leftover for goodwill which is only tested for impairment - but then future incentive to artificially inflate asset prices to keep it high because impairment leads to an expense).

Accounting choices and the income statement. How is revenue recognized - e.g. FOB or CFR, bill-and-hold (deferred revenue). Channel-stuffing, e.g. lots of discounts to temporarily boost revenue.

Accounting choices and the cashflow statement. Many believe operating CF is a better measure than BS/IS which are subject to accrual accounting which can be manipulated. However, operating CF can be manipulated, e.g. stretching or limiting the AP credit dates or changing the way it is calculated (e.g. FOB/CFR - so analysts should examine working capital patterns) or they can move some into financing and/or investing CFs. Another way is to buy low for the financial year and then buy higher from the same seller to smooth, but mark-to-market accounting rules mean the purchase has to be valued at market prices (see the example on Dynergy). Interest can be part-capitalized and part-immediately expensed; if these are investing and operating CFs as in IFRS, the proportions can be manipulated to make operating CFs look better. Interest paid and received can also be manipulated to make operating CFs better by classifying interest paid as financing and interest received as operating.

8.14 Financial analysis techniques

Net debt = Debt minus cash. Ability to pay down debt hypothetically now.

Common size analysis.

Activity (asset utilisation, operating efficiency) ratios: measure efficiency of company's operations. How well working capital and long-term assets are being used. Numerator usually from IS, denominator usually from BS.

Liquidity ratios. Defensive interval ratio: cash + marketable securities + AR divided by daily cash expenditures (daily cash can be obtained from expenses + depreciation/amortization divided by 365).

Activity Ratios	Numerator	Denominator
Inventory turnover	Cost of sales or cost of goods sold	Average inventory
Days of inventory on hand (DOH)	Number of days in period	Inventory turnover
Receivables turnover	Revenue	Average receivables
Days of sales outstanding (DSO)	Number of days in period	Receivables turnover
Payables turnover	Cost of sales or cost of goods sold	Average trade payables
Number of days of payables	Number of days in period	Payables turnover
Working capital turnover	Revenue	Average working capital
Fixed asset turnover	Revenue	Average net fixed assets
Total asset turnover	Revenue	Average total assets

Current (working capital) ratio. Current assets divided by current liabilities.

Quick ratio is more conservative than current ratio. Some current assets such as prepaid expenses, some taxes, INVENTORIES and employee-related prepayments—represent costs of the current period that have been paid in advance and cannot usually be converted back into cash.

Payables turnover. Net credit purchases divided by average AP. A high ratio indicates prompt payment is being made to suppliers for purchases on credit.

Solvency ratios. Debt and coverage ratios are most common. To calculate debt in the numerator of debt ratios, some analysts use solvency ratios that deduct cash, cash equivalents, and marketable securities from interest-bearing short-term and long-term debt to calculate net debt (or net cash if the former exceeds the latter), which is the 'true debt'.

Solvency Ratios	Numerator	Denominator
Debt Ratios		
Debt-to-assets ratio ^a	Total debt ^b	Total assets
Debt-to-capital ratio	Total debt ^b	Total debt ^b + Total shareholders' equity
Debt-to-equity ratio	Total debt ^b	Total shareholders' equity
Financial leverage ratio ^c	Average total assets	Average total equity
Debt-to-EBITDA	Total or net debt	EBITDA
Coverage Ratios		
Interest coverage	EBIT	Interest payments
Fixed charge coverage	EBIT + Lease payments	Interest payments + Lease payments

The fixed charge coverage ratio relates fixed charges, or obligations, to the cash

flow generated by the company. It measures the number of times a company's earnings (before interest, taxes, and lease payments) can cover the company's interest and lease payments. Similar to the interest coverage ratio, a higher fixed charge coverage ratio implies stronger solvency, offering greater assurance that the company can service its debt (i.e., bank debt, bonds, notes, and leases) from normal earnings. The ratio is sometimes used as an indication of the quality of the preferred dividend, with a higher ratio indicating a more secure preferred dividend.

Profitability ratios. Return on sales (e.g. denominator is revenue) or return on investment (e.g. denominator is assets), return on invested capital (denominator is all debt + equity).

Profitability Ratios	Numerator	Denominator
Return on Sales^a		
Gross profit margin	Gross profit	Revenue
Operating profit margin	Operating income ^b	Revenue
Pretax margin	EBT (earnings before tax but after interest)	Revenue
Net profit margin	Net income	Revenue
Return on Investment		
Operating ROA	Operating income	Average total assets
ROA	Net income	Average total assets
Return on invested capital	EBIT × (1 - Effective Tax Rate)	Average total short- and long-term debt and equity
ROE	Net income	Average total equity
Return on common equity	Net income – Preferred dividends	Average common equity

DuPont analysis (ROE decomposition).

$$ROE = \frac{\text{Net income}}{\text{Av equity}} = ROA \times Leverage$$

So, ROE is decomposed to profitability and solvency ratios. If the company is able to borrow at a rate lower than the marginal rate it can earn investing the borrowed money, the company is making an effective use of leverage and ROE would increase as leverage increases. If a company's borrowing cost exceeds the marginal rate it can earn on investing in the business, ROE would decline as leverage increased because the effect of borrowing would be to depress ROA.

$$ROE = \text{Net profit margin} \times \text{Total asset turnover} \times \text{Leverage}$$

So, ROE is now decomposed to profitability, activity (efficiency), and solvency ratios.

$$ROE = \frac{NI}{EBT} \times \frac{EBT}{EBIT} \times \frac{EBIT}{Revenue} \times \text{Total asset turnover} \times \text{Leverage}$$

$ROE = \text{Tax burden} \times \text{Interest burden} \times \text{EBIT margin} \times \text{Total asset turnover} \times \text{Leverage}$

The EBIT margin is an operating margin. An alternative equation is:

$$ROE = \frac{NI}{EBT} \times \frac{EBT}{R} \times \text{Total asset turnover} \times \text{Leverage}$$

where the second term is the 'pretax margin'.

Alternative ROIC definition:

$$ROIC = \frac{\text{Net operating profit} - \text{Adjusted taxes}}{\text{Operating assets} - \text{Operating liabilities}}$$

8.15 Financial statement modelling

When forecasting, start with income statement variables - revenues and expenses. Then use the starting BS + IS to get cash flows. Then finish with BS.

Five key behavioral biases that influence analyst forecasts are overconfidence, illusion of control, conservatism, representativeness, and confirmation bias. Overconfidence - confidence intervals should be higher than what most analysts believe and analysts are usually overconfident when arguing against expectations. Illusion of control - more information, granular methods may be better but diminishing returns and can get overfitting. Conservatism - anchoring and adjustment; anchor adjustment tends to be too small; not taking into account enough information (opposite of illusion of control in this respect). Representativeness bias - base-rate neglect: neglect the 'outside view' of the industry to which the company belongs; should use both outside and inside (company-specific) views.

Porter's five forces: threat of substitutes, rivalry, supplier bargaining power, buyer bargaining power, threat of new entrants. Government (and tax power) is not a force, but affects the five forces.

Inflation and deflation. Inflation varies depending on where it is, even revenue inflation and expense inflation are different. Inflation is slower depending on industrial structure, e.g. US retailers have less power than European retailers so when there is a rise in costs, US retailers have less power and there is greater inflation versus Europe. Inflation-sales relationship depends on elasticity - if inelastic, price increase leads to revenue increase; if elastic the opposite.

In inflation/deflation, first company to act makes some gains and others make successively less gains. In an inflationary environment, raising prices too late (compared to competitors) will result in a profit margin squeeze but acting too soon could result in volume losses. In a deflationary environment, lowering prices too soon will result in a lower gross margin, but waiting too long will result in volume losses.

To address inflation, companies can only partially pass on the costs to consumers, cut back on other parts of their costs like marketing, and use substitutes for their inputs (e.g. different coffee beans if some coffee bean prices are rising).

Forecast horizon. Investment time horizon of x years implies annual portfolio turnover of $1/x$, i.e. the number of people leaving or entering the portfolio. Extending a forecast for the long-term allows better normalized earnings forecast, like an 'average forecast'. Similarly, normalized free cash flow can be defined as the expected level of mid-cycle cash flow from operations adjusted for unusual items just described less recurring capital expenditures.

COGS. Usually forecast based on revenue or sales or gross profit margin.

Working capital. Usually forecast using efficiency ratios. Then use it to forecast current assets and liabilities. Then from this combine with sales and costs to get AR, AP, inventories.

SG&A. Usually forecast as percentage of sales, as they often have a variable cost component.

Terminal values in DCF. An analyst should consider whether the terminal year free cash flow projection should be normalized before that cash flow is incorporated into a long-term projection. For example, if the explicitly forecasted terminal year free cash flow is "low" (e.g., because of business cycle reasons or capital investment projects), an adjustment to normalize the amount might be warranted. Second, an analyst should consider whether and how the future long-term growth rate will differ from the historical growth rate.

Long-term factors. Economics, business cycles, technological changes, regulatory changes, long-term growth. Note payment types = 1 (beginning of year)

9 Portfolio Management

9.1 Risk and return I

Expected return.

$$1 + E(R) = (1 + r_f)[1 + E(\pi)] \times [1 + E(RP)]$$

Assume: (1) normally-distribution means, and (2) efficient (operational and informational) markets. But these assumptions can be violated e.g. skew and kurtosis. Market operational efficiency: the cost of trading has three main components—brokerage commission, bid-ask spread, and price impact; liquidity affects the latter two.

There are other market-related characteristics that affect investment decisions because they might instill greater confidence in the security or might affect the costs of doing business. These include analyst coverage, availability of information, firm size, etc. These characteristics about companies and financial markets are essential components of investment decision making.

Portfolio with riskfree asset and security i:

$$R_p = w_1 R_f + (1 - w_1) E(R_i)$$

$$\sigma_p^2 = w_1^2 \sigma_f^2 + (1 - w_1)^2 \sigma_i^2 + 2w_1(1 - w_1) \text{Cov}(r_i, r_f)$$

$$\sigma_p = (1 - w_1) \sigma_i$$

The last is the market price of risk. Solve for w_1 :

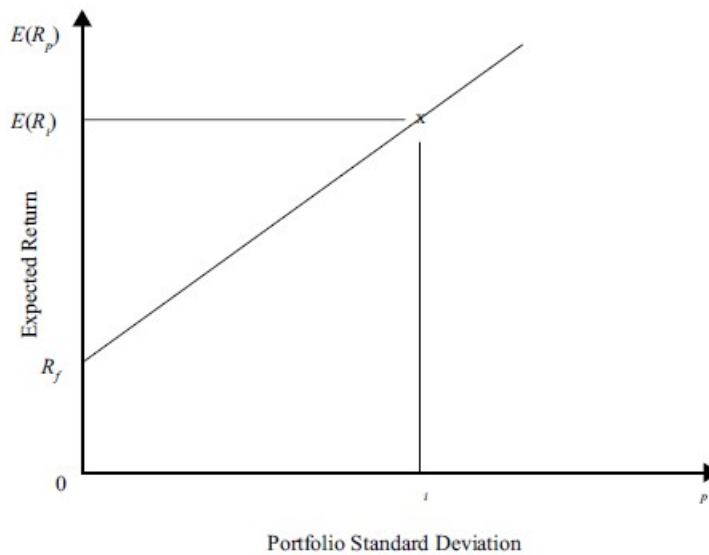
$$w_1 = 1 - \frac{\sigma_p}{\sigma_i}$$

Then substitute into above:

$$E(R_p) = \left(1 - \frac{\sigma_p}{\sigma_i}\right) R_f + \left(\frac{\sigma_p}{\sigma_i}\right) E(R_i)$$

$$E(R_p) = R_f + \frac{E(R_i) - R_f}{\sigma_i} \sigma_p$$

which is the capital allocation line - the best line for investors based on the riskfree asset and the securities available on the market:



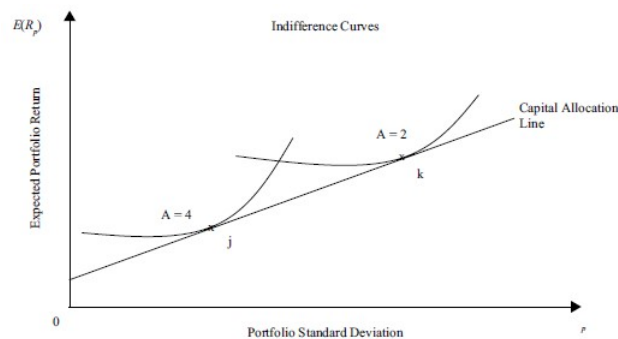
Can move beyond $w_1 = 0$ by borrowing at the riskfree rate and then paying back (and the extension will be a perfectly linear continuation), i.e. $w_1 = -0.5$ in

$$E(R_p) = w_1 R_f + (1 - w_1) E(R_i)$$

$$E(R_p) = -0.5 R_f + 1.5 E(R_i)$$

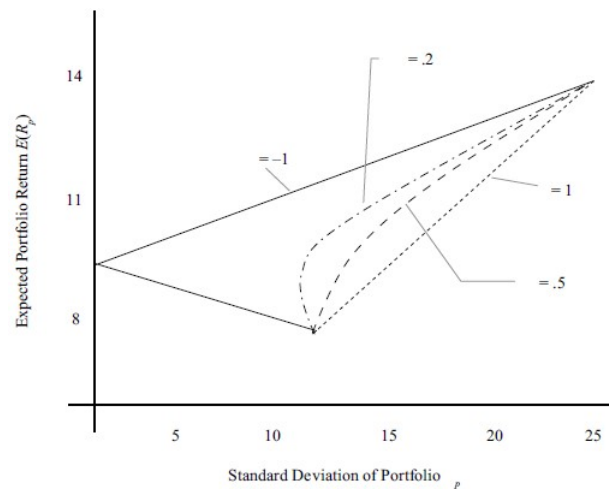
and note that w_1 can be anything because it doesn't appear in

$$E(R_p) = R_f + \frac{E(R_i) - R_f}{\sigma_i} \sigma_p$$



Risk diversification. Note $Cov(1, 2) = \rho \sigma_1 \sigma_2$. A $\rho = 1$ means if you have two securities with the same standard deviations, the portfolio will also have the same standard deviation, if $\rho = -1$ then 0 standard deviation, and somewhere in between (but not necessarily the average) if $\rho = 0$. For a portfolio comprised of securities i and j,

$$\sigma_p^2 = w_1^2 \sigma_1^2 + (1 - w_1)^2 \sigma_2^2 + 2w_1(1 - w_1)Cov(r_i, r_j)$$



The portfolio risk is less than the weighted average of risks when the correlation of asset returns is less than one.

Two asset-portfolio where one is in a foreign currency, e.g. UK equity and UK gilt turns into UK equity and US treasury in USD. The foreign currency asset (e.g. US Treasuries from the UK perspective), will have a return on the treasury and the currency:

$$r_{UK} = (1 + r_{US})(1 + r_{FX}) - 1$$

This is treated as one security in the portfolio, though it introduces more covariances (previously, there was covariance between the UK equity and UK gilt, now it's UK equity, US treasury, and FX rate - so three covariances are needed).

$$r_{UK} = r_{US} + r_{FX} + r_{US}r_{FX}$$

Assume $r_{US}r_{FX}$ is negligible. Then

$$Var(r_{UK}) = Var(r_{US}) + Var(r_{FX}) + 2Cov(r_{US}, r_{FX})$$

, which means that the covariance or ρ between just the US treasury and the FX rate return is needed. Note that weights here are effectively 1 for each variance term. In regular two-asset portfolios, there are weights - e.g.:

$$r_p = w_1r_1 + (1 - w_1)r_2$$

$$Var(r_p) = w_1^2Var(r_1) + (1 - w_1)^2Var(r_2) + 2w_1(1 - w_1)Cov(r_1, r_2)$$

When the two-asset (effectively three-asset) portfolio is combined, the return is

$$r_p = w_1r_1 + (1 - w_1)r_2$$

However,

$$r_p = w_1r_1 + (1 - w_1)((1 + r_{US})(1 + r_{FX}) - 1)$$

$$r_p = w_1r_1 + (1 - w_1)(r_{US} + r_{FX} + r_{US}r_{FX})$$

$$r_p = w_1r_1 + (1 - w_1)(r_{US} + r_{FX} + 0)$$

$$r_p = w_1r_1 + (1 - w_1)r_{US} + (1 - w_1)r_{FX}$$

as for the two-asset portfolio. However, the variance is

$$Var(r_p) = w_1^2Var(r_1) + (1 - w_1)^2Var(r_{US}) + (1 - w_1)^2Var(r_{FX}) +$$

$$2w_1(1 - w_1)Cov(r_1, r_{US}) + 2w_1(1 - w_1)Cov(r_1, r_{FX}) + 2(1 - w_1)^2Cov(r_{FX}, r_{US})$$

Many-asset portfolio.

$$E(r_p) = \sum w_i r_i$$

$$\sum w_i = 1$$

$$\sigma_p^2 = \sum w_i^2 \sigma_i^2 + 2 \sum w_i w_j \text{Cov}(i, j)$$

where the last can be simplified for large N by assuming equal weights, so that

$$w_i = \frac{1}{N}$$

then,

$$\sigma_p^2 = N w_i^2 + w_i w_j N(N-1) \text{Cov}(i, j)$$

$$\sigma_p^2 = N \left(\frac{1}{N} \right)^2 + \frac{N(N-1)}{N^2} \text{Cov}(i, j)$$

$$\sigma_p^2 = \frac{\bar{\sigma}^2}{N} + \frac{N-1}{N} \overline{\text{Cov}(r_i, r_j)}$$

$$\sigma_p \approx \sqrt{\overline{\text{Cov}(r_i, r_j)}}$$

so as N increases, the variance term becomes unimportant.

Historical stats. Historical returns are not a good guide to future returns. However, historical risk has been relatively stable through time, even despite business cycles. Correlations among assets WITHIN a country have been stable. Correlations of assets BETWEEN countries has been rising. Correlation between stocks and bonds is low, often negative.

Diversification. Different asset classes, assets from different countries, use index funds (if costly to diversify), don't own employer's stock, insurance has a negative correlation, alternative investments (e.g. gold). The addition of a security does this (from CAPM):

$$E(r_{new}) = r_f + \left(\frac{\sigma_{new} \rho_{new,p}}{\sigma_p} \right) (E(r_p) - r_f)$$

which can be re-arranged to a Sharpe ratio form:

$$\frac{E(r_{new}) - r_f}{\sigma_{new}} = \frac{E(r_p) - r_f}{\sigma_p} \times \rho_{new,p}$$

Sharpe ratio. Sharpe Ratio = (Expected Return - Risk-Free Rate) / Standard Deviation. Higher means better.

Plotting the portfolio, the Markowitz efficient portfolio frontier emerges. Note portfolios to the upper right have diminishing returns per increase in risk. Connect riskfree bond to the Markowitz frontier and then extend the line - the top of this line is what the risk-neutral investor would choose, because they only care about the return as per A=0 in the utility equation:

$$U = E(r) - \frac{1}{2} A \sigma^2$$

To find the slope of the capital allocation line (CAL), maximize the slope from

$$E(R_p) = R_f + \frac{E(R_{rp}) - R_f}{\sigma_{rp}} \sigma_p$$

where rp is the risky portfolio (i.e. strictly on the Markowitz frontier) while p is the portfolio along the CAL. First specify the returns and standard deviations of the individual assets for the risky portfolio, which will be a function of the weight, then take derivative of the slope and set it to 0, solving for the weight that maximizes the CAL slope.

The two-fund separation theorem states that all investors regardless of taste, risk preferences, and initial wealth will hold a combination of two portfolios or funds: a risk-free asset and an optimal portfolio of risky assets. So, identify the Markowitz frontier and then the CAL.

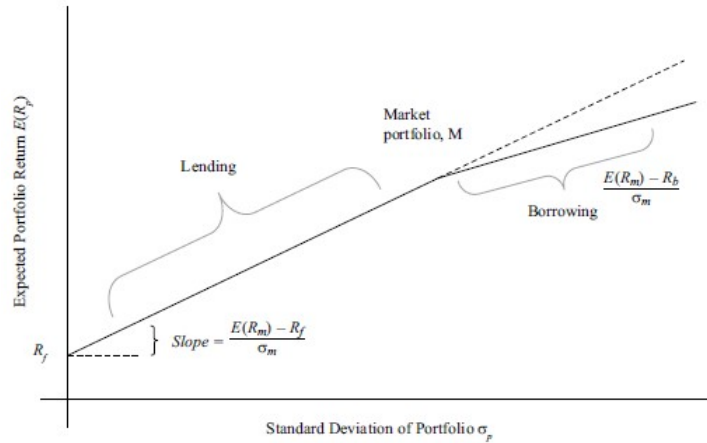
9.2 Risk and return II

Assume homogeneity of expectations for optimal risky portfolio, but if there are different expectations as in reality, this isn't possible.

Capital market line (CML) is a special instance of the CAL where the optimal risky portfolio is replaced by the market portfolio, e.g. S&P 500. The capital market line (CML) does not apply to all securities or assets but only to portfolios on the efficient frontier.

$$E(r_p) = r_f + \left(\frac{E(r_m) - E(r_f)}{\sigma_m} \right) \sigma_p$$

Moving beyond the market portfolio means leveraging. If leveraging (returns = equity return minus borrowing cost, all divided by investor's equity) using the riskless rate, the straight line continues upward with the same slope. However, if the borrowing occurs at a higher rate, then the line becomes kinked or curved:



The slopes are different, where $r_b > r_f$:

$$E(r_p) = r_b + \left(\frac{E(r_m) - r_b}{\sigma_m} \right) \sigma_p$$

Because of the gentler slope, less risk-averse investors will invest; more risk-averse investors will not.

Systematic risk can't be diversified but non-systematic risk can be. Total variance = Systematic variance + Nonsystematic variance. Investors are theoretically only compensated for systematic risk; non-systematic risk can be diversified away - if investors were compensated for non-systematic risk, they would keep buying this until returns are eventually gone because the share price is so high.

Return generating models, e.g. Fama-French multi-factor model. This is different from the CAL and CML models which blend the risky or market portfolio with the riskfree asset, and depending on the LEVEL OF PORTFOLIO RISK chosen along the x-axis, a corresponding return is calculated. In RGMs, we calculate the return on a security i depending on the RETURN OF THE MARKET PORTFOLIO. In RGMs, we use β , which considers the COVARIATION between the security i and the risky portfolio or market portfolio (and that's also why the β includes the correlation coefficient) - in the CAL and CML, the slope is the risk-return tradeoff and the covariation is already calculated in the market portfolio. For a security i,

$$E(r_i) - r_f = \sum_j \beta_{ij} E(factor_j)$$

A single-index model (CAPM) with the security being the market portfolio:

$$E(r_i) - r_f = \beta_{i1} (E(r_m) - r_f)$$

However, it can be shown that this is equivalent to the CML. The CML is:

$$E(r_i) - r_f = \frac{E(r_m) - r_f}{\sigma_m} \sigma_i$$

Start with the single factor model:

$$E(r_i) - r_f = \beta_i (E(r_m) - r_f)$$

$$r_i - r_f = \beta_i (r_m - r_f) + e_i$$

where e_i is the NON-SYSTEMATIC RISK.

$$r_i = r_f + \beta_i (r_m - r_f) + e_i$$

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_e^2$$

Since non-systematic risk should be uncompensated,

$$\sigma_i = \beta_i \sigma_m$$

$$\beta_i = \frac{\sigma_i}{\sigma_m}$$

which is substituted back into the CML equation to get

$$E(r_i) - r_f = \beta_i(E(r_m) - r_f)$$

Note return-generating models are of the form

$$r_i = \alpha + \beta_i r_m + e_i$$

and for the riskfree asset,

$$r_f = \alpha + \beta_i r_f$$

so,

$$\alpha = (1 - \beta) r_f$$

Calculating beta.

$$r_i = r_f + \beta_i r_m + e_i$$

$$Cov(r_i, r_m) = Cov(r_f + \beta_i r_m + e_i, r_m)$$

$$= \beta_i Cov(r_m, r_m) = \beta_i \sigma_m^2$$

$$\beta_i = \frac{Cov(r_i, r_m)}{\sigma_m^2} = \frac{\rho_{i,m} \sigma_i \sigma_m}{\sigma_m^2} = \frac{\rho_{i,m} \sigma_i}{\sigma_m}$$

Beta is SYSTEMATIC RISK, so CAPM is about systematic risk. The beta includes an adjustment for the correlation between asset i and the market because the market model covers all assets whereas the CML works only for fully diversified portfolios.

Beta estimates - based on calculation like above or regression. Regression - short time regression estimates are close to current systematic risk but might be affected by current events and less accurate; longer time regressions lead to more accuracy but bad future expectations which can be affected by current events.

The e_i is what is β cannot explain and divergence between the security return and the market; it is the leftover and shrinks when building up a diversified portfolio; it shrinks to almost nothing with the market portfolio. So the non-systematic risk (=standard deviation of e_i) gets diversified away as the portfolio builds. The standard deviation of $\beta \times (E(r_m) - r_f)$ is the systematic risk because it is positive and doesn't 'shrink' - in fact, the β can increase with a more diversified portfolio.

CAPM is single-factor mode:

$$E(r_i) = r_f + \beta_i(E(r_m) - r_f)$$

In CAPM, the primary determinant of expected return for a security is its beta, or how well the security correlates with the market. In certain cases, assets may require a return less than the risk-free return, e.g. if an asset's beta is negative, the required return will be less than the risk-free rate. When combined with the market, the asset reduces the risk of the overall portfolio, which makes the asset very valuable. Insurance is one such asset.

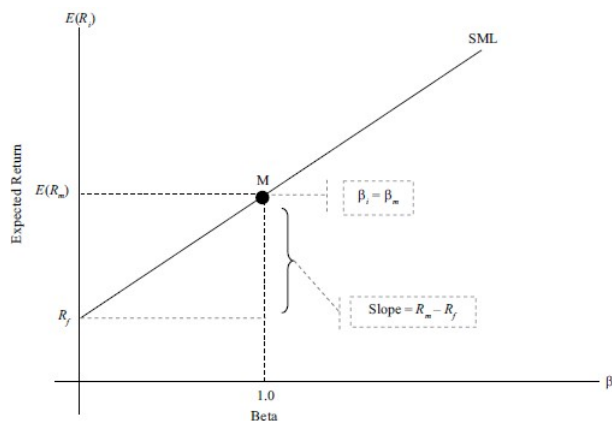
Notes: (1) The average of all betas in the market is 1 by definition because from above:

$$\beta_i = \frac{\sigma_i}{\sigma_m} = \frac{\sigma_m}{\sigma_m}$$

(2) The 'market risk premium' is $E(r_m) - r_f$. (3) The 'market risk' (systematic risk) is β_i .

CAPM assumptions. Individuals are rational, utility-maximizing, risk-averse. Homogeneity of beliefs. No transaction costs or taxes. One holding period (no multi-level periods, no multi-level strategies). Securities are infinitely divisible. Investors are price takers.

Security market line (SML). Regress returns of many different securities against their betas. Applies to any security, efficient or not. Total risk and systematic risk are equal only for efficient portfolios because those portfolios have no diversifiable risk remaining. It charts returns against BETAs on the x-axis.



Portfolio betas.

$$\begin{aligned} E(r_1) &= r_f + \beta_1(E(r_m) - r_f) \\ E(r_2) &= r_f + \beta_2(E(r_m) - r_f) \\ E(r_p) &= w_1 E(r_1) + w_2 E(r_2) \\ &= w_1 r_f + w_1 \beta_1 (E(r_m) - r_f) + w_2 r_f + w_2 \beta_2 (E(r_m) - r_f) \\ &= r_f + (w_1 \beta_1 + w_2 \beta_2) (E(r_m) - r_f) \end{aligned}$$

so the portfolio beta is

$$\sum w_i \beta_i$$

CAPM gives the required rate of return by regressing returns of a PARTICULAR security against $r_m - r_f$.

CAPM limitations. (1) Only looks at market returns, and nothing else. Limited to systematic risk. (2) Single-period. (3) The true market portfolio according to the CAPM includes all assets, financial and nonfinancial, which means that it also includes many assets that are not investable, such as human capital and assets in closed economies. Arbitrage pricing theory (APT). Adds more factors to CAPM - more variables, but it doesn't specify the variables.

Fama-French with Carhart's momentum. Empirical model which doesn't have market equilibrium theory that CAPM does. Better explains returns. MKT is same as CAPM (beta close to 0); SMB is returns difference between small-cap and large-cap stocks (smaller companies outperform larger companies); HML is the difference in returns between high-book-to-market stocks and low-book-to-market stocks (value versus growth); UMD = the difference in returns of the prior year's winners and losers (momentum). When these further variables are added, the MKT regression coefficient is very small. The Fama-French 3-factor model suggests that investors should consider not only the overall market risk but also the size and value characteristics of stocks when assessing expected returns and risk.

$$E(r_i, t) = \alpha_i + \beta_{i,MKT}MKT_t + \beta_{i,SMB}SMB_t + \beta_{i,HML}HML_t + \beta_{i,UMD}UMD_t$$

CAPM and portfolio appraisal and performance evaluation.

$$E(r_i) = \alpha + \beta(E(r_m) - r_f) + e_i$$

$$\sigma_i^2 = \beta^2 \sigma_m^2 + \sigma_e^2$$

where the RHS is systematic risk + unsystematic (idiosyncratic) risk.

Sharpe ratio on the portfolio, compared to the riskless return.

$$Sharpe = \frac{E(r_p) - r_f}{\sigma_p}$$

The Sharpe ratio, also called the reward-to-variability ratio, is simply the slope of the capital allocation line. The ratio uses the TOTAL risk of the portfolio σ_p , not its systematic risk. Sharpe ratio can be used for historical returns as well as expected. However, it uses total risk.

Treynor ratio addresses the Sharpe ratio's limitation of systematic risk by using beta (systematic risk) instead of total risk:

$$Treynor = \frac{E(r_p) - r_f}{\beta_p}$$

Neither Sharpe nor Treynor ratios are quantitatively significant; they are ordinal, not cardinal.

M2 alpha. Quantifies (cardinal) differences. 'Rescales' Sharpe ratio. However, it's for TOTAL risk.

$$M^2 = \frac{E(r_p) - r_f}{\sigma_p} \sigma_m + r_f = SR \times \sigma_m + r_f$$

Recall that CAL is

$$\frac{E(r_i) - r_f}{\sigma_i} \sigma_p + r_f = E(r_p)$$

and CML is

$$\frac{E(r_i) - r_f}{\sigma_m} \sigma_p + r_f = E(r_i)$$

where i is a portfolio (could have easily been called p) on the CML. So, M2 alpha calculates $E(r_m)$:

$$M^2 = \frac{E(r_p) - r_f}{\sigma_p} \sigma_m + r_f$$

because if $p = m$,

$$M^2 = \frac{E(r_m) - r_f}{\sigma_m} \sigma_m + r_f = E(r_m)$$

The above shows M2. M2 alpha is $M^2 - E(r_m)$, and it allows use of the Sharpe ratio by comparing different market portfolio returns that would arise depending on which portfolio p chosen. It quantifies the differences in how the market average would have been in percentage terms. A higher Sharpe ratio translates quantitatively into better fund performance. So, compare M^2 with $E(r_m)$. Another way to interpret M2 is that it is the market portfolio risk on the CML. It effectively uses the CML at the market portfolio and compares to see if the M2 is higher than $E(r_m)$. Alternatively, the M2 alpha equation is just the CAL inverted, so it estimates the market portfolio based on the risky portfolio:

$$\frac{E(r_i) - r_f}{\sigma_i} \sigma_p + r_f = E(r_p)$$

$$\frac{E(r_i) - r_f}{\sigma_i} = \frac{E(r_p) - r_f}{\sigma_p}$$

$$E(r_i) = r_f + \frac{E(r_p) - r_f}{\sigma_p} \sigma_i$$

$$E(r_m) = r_f + \frac{E(r_p) - r_f}{\sigma_p} \sigma_m$$

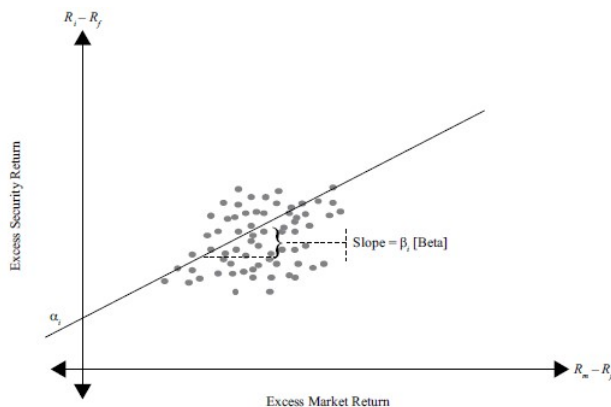
Jensen's alpha. Also quantifies differences to address shortcomings of Sharpe and Treynor ratios.

$$\alpha_p = r_p - (r_f + \beta_p[E(r_m) - r_f])$$

If $\alpha_p > 0$, portfolio outperformed the market. Jensen's alpha is the maximum amount that you should be willing to pay the manager to manage your money.

Security characteristic line (SCL). Uses Jensen's alpha.

$$r_i - r_f = \alpha_i + \beta_i(r_m - r_f)$$



When do we use total risk performance measures like the Sharpe ratio and M2, and when do we use beta risk performance measures like the Treynor ratio and Jensen's alpha? Total risk is relevant for an investor when he or she holds a portfolio that is not fully diversified, which is not a desirable portfolio. In such cases, the Sharpe ratio and M2 are appropriate performance measures.

Using advanced portfolio optimization, the share of a security in a portfolio should be proportional to its performance versus nonsystematic variance. A security is more valuable if its returns are high and nonsystematic variance is low.

$$\frac{\alpha_i}{\sigma_{e,i}^2}$$

The total weight of these added securities should be

$$\frac{\sum w_i \alpha_i}{\sum w_i^2 \sigma_{e,i}^2}$$

The information ratio,

$$\frac{\alpha_i}{\sigma_{e,i}}$$

(i.e., alpha divided by nonsystematic risk), measures the abnormal return per unit of risk added by the security to a well-diversified portfolio. The larger the

information ratio is, the more valuable the security.

Total variance = systematic variance + nonsystematic variance.

$$r_i = r_f + \beta_i(E(r_m) - r_f) + e_i$$
$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{e,i}^2$$

Add non-market security to portfolio if

$$\frac{\alpha_i}{\sigma_{e,i}^2} > 0$$

Note: a market-neutral strategy is when beta is 0.

9.3 Smart betas

Smart beta: rules using e.g. volatility, liquidity, momentum on ETFs. Higher management fees + higher turnover compared to passive market-cap based strategies.

9.4 Portfolio management overview

Diversification ratio. Standard deviation of portfolio divided by standard deviation of average of individual securities in portfolio.

$$DivRatio = \frac{\sigma_p}{\bar{\sigma}_i}$$

Diversification reduces risk, but will not be very effective during market turmoil.

Portfolio management. Planning - figuring out what the client wants, including drafting an investment policy statement. Execution - security analysis, asset allocation, build portfolio. Monitoring - rebalancing, review.

Asset management industry. Active or passive. New trends: big data, passive investing, robo-advisors.

Mutual funds. Calculate per share using net asset value divided by number of shares. Pooled funds with PM. Open-ended or closed-ended. Load (fee to buy in) or no-load (fixed percentage fee) funds. Types: (1) money market funds - invest in liquid instruments with maturities less than 90 days generally (note: in US some can be used in lieu of banks but deposits not protected); (2) bond mutual funds - instruments with maturities of 1 year to 30 years; (3) stock mutual funds - active or passive - active has research and trading costs and can incur more capital gain taxes because it distributes more proceeds to investors; (4) hybrid.

Open-end funds trade at their net asset value per share, whereas closed-end

funds and exchange traded funds can trade at a premium or a discount.

Exchange traded funds do not have capital gain distributions. If an investor sells shares of an ETF (or open-end mutual fund or closed-end mutual fund), the investor may have a capital gain or loss on the shares sold; however, the gain (or loss) from the sale is not a distribution.

A fund management service for institutions or individual investors with substantial assets is the separately managed account (SMA), which is also commonly referred to as a “managed account,” “wrap account,” or “individually managed account.”

Hedge funds are private investment vehicles that typically use leverage, derivatives, and long and short investment strategies.

9.5 Portfolio planning and construction

Investment policy statement. Most important are objectives and constraints.

Risks. Absolute, e.g. not to lose X over a 12-month period. Relative, e.g. not be X percent below the NASDAQ within a 12-month period.

Risk tolerance is a function of ability to bear risk and willingness to take risk. When both the ability and willingness to take risk are consistent, the investment adviser’s task is the simplest. When ability to take risk is below average and willingness to take risk is above average, the investor’s risk tolerance should be assessed as below average overall. When ability to take risk is above average but willingness is below average, the portfolio manager or adviser may seek to counsel the client and explain the conflict and its implications.

Risk assessment questionnaires. Most useful for assessing WILLINGNESS to accept risk.

Constraints. Liquidity, time horizon, tax concerns, legal and regulatory factors, and unique circumstances. Can use value-at-risk and tracking error (risk relative to a benchmark). LDI - liability driven investment where have to meet financial obligations, e.g. pension funds and insurance companies.

Strategic asset allocation - divide securities into asset classes, look at correlations among them, plot risk-returns and find utility-maximizing portfolio.

Risk-budgeting. Beyond strategic asset allocation risks, there are tactical asset allocation risks and security selection risks. Tactical asset allocation is using other information or forecasts to change the strategic asset allocation. Tactical asset allocation may deliberately deviate from the policy portfolio. Security selection is using other information or forecasts to pick stocks within an asset class.

Drift. Over time, the weights will breach the policy weights, bandwidth or IPS thresholds; then rebalance.

Risk parity investing. Capital markets theory is destabilizing, e.g. too much equities in portfolios. Risk parity investment strategies seek to diversify sources of risk. Each source of risk carries a risk premium, which is a potential source of return. To achieve this diversification, these strategies assume that asset classes (such as stocks and bonds) should contribute equally to the level of risk in a portfolio. Reweight portfolio according to risk, to give less funds to equities.

9.6 Behavioral biases of individuals

Two types: cognitive errors and emotional biases. Cognitive errors can often be corrected or eliminated through better information, education, and advice.

Cognitive errors: belief perseverance bias, processing errors.

Belief perseverance is the tendency to cling to one's previously held beliefs by committing statistical, information-processing, or memory errors in the face of cognitive dissonance. The belief perseverance biases discussed are conservatism, confirmation, representativeness, illusion of control, and hindsight.

Conservatism - difficulty processing new information. First recognize it because it has a cognitive cost. Then ask how forecast changes. Overweighting prior probability in Bayes' Rule.

Confirmation bias, e.g. hold their employer's stock, underdiversify (believing in one or few stocks), consider only reinforcing information about a stock instead of seeking a different opinion.

Representativeness. Classify new information based on past experiences. Base-rate neglect, e.g. looking too much into one stock and ignoring the characteristics of its asset class. Sample-size neglect. NOT associated with underdiversified portfolios.

Illusion of control. Underdiversify (as they feel they have control over certain stocks), trade a lot more than necessary (incurring more costs, and this strategy shown to correlate with negative returns), construct overly-detailed financial models (believing these details make a difference).

Hindsight bias. People tend to remember their predictions as being more accurate than they were because of the new information and hindsight bias. Good results from poor decisions celebrated and vice versa. Unfairly assess past (e.g. portfolio) performance.

Processing errors describe how information may be processed and used illogically or irrationally in financial decision making. The processing errors discussed are anchoring and adjustment, mental accounting, framing, and availability.

Anchoring and adjustment. Start with initial guess but glued to that guess. Similar to conservatism bias. Ask oneself whether forecast is referenced to a 'default' value.

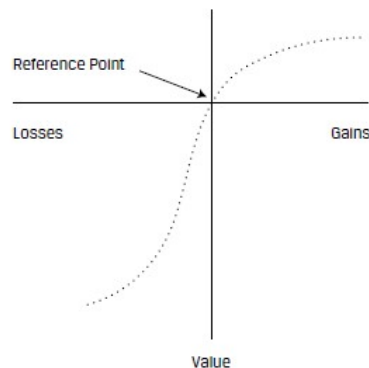
Mental accounting bias. Though money is fungible, people tend to construct different accounts, e.g. offsetting money unexpectedly earned from luck recklessly because it was money unexpected. Underdiversify, not distinguish between income and principal.

Framing bias. For example, a situation may be presented within a gain context (one in four start-up companies succeed) or within a loss context (three out of four start-ups fail). Often manifests as 'narrow framing', e.g. chasing details instead of the big picture. Miscalculate risks (because it depends on how it was presented), focus on short-term rather than long-term. Ask oneself what is the net gain or loss position.

Availability bias is when people estimate the probability of an outcome or the importance of a phenomenon based on how easily information is recalled. Various sources of availability bias exist; the four most applicable to FMPs are retrievability (first information to be retrieved is weighted more correct), categorization, narrow range of experience, and resonance (biased based on personal experiences and familiarity). These people may limit their research universe, search only those that have a lot of news coverage or advertising, underdiversify (because based on personal experiences). Ask oneself reason for the limited choices.

Emotional biases. Emotional biases, on the other hand, are harder to correct because they stem from impulses and intuitions. It is often possible only to recognize an emotional bias and adapt to it.

Loss aversion bias. Loss aversion leads one to hold their losers to avoid recognizing losses and sell their winners to lock in profits (and avoid disappearing profits). In practice, it manifests as the disposition effect: the holding of investments that have experienced losses too long, and the selling of investments that have experienced gains too quickly (i.e., holding on to losers and selling winners). The resulting portfolio may be riskier than the optimal portfolio based on the investor's risk-return objectives. See utility function below:



To address, use a disciplined, quantitative approach to investing.

Overconfidence bias. Poorly-diversified portfolios. Unwarranted faith in their own abilities. Overconfidence may be intensified when combined with self-attribution bias, in which people take too much credit for successes (self-enhancing) and assign responsibility to others for failures (self-protecting). Overconfidence is both a cognitive and emotional error but more emotional. Two forms: prediction overconfidence (confidence intervals too narrow than justified by fundamental analysis) and certainty overconfidence (probabilities to outcomes are too high). Both types demonstrate faulty reasoning combined with “gut feel” and hope. Underestimate risk and underdiversify. Look at oneself objectively.

Self-control bias. Lose long-term goals because of short-term demands.

Status quo bias. People choose to do nothing when something should be done. Generally subconscious and inertia. People tend to leave portfolios as-is and not look for new opportunities. Education and periodic review might help.

Endowment bias. Often leads to indecision. Psychologists have found that people tend to state minimum selling prices for a good that exceed maximum purchase prices that they are willing to pay for the same good. Effectively, ownership “endows” the asset with added value. E.g. overvaluing shares already in their own portfolios. Fail to sell certain assets and replace them. Ask oneself ‘would you buy these shares at this price?’.

Regret-aversion bias. Regret-aversion bias has two dimensions: actions that people take and actions that people could have taken. Regret is more intense when the unfavorable outcomes are the result of an action taken versus the result of an action not taken. Thus, no action becomes the default decision. They are too conservative and engage in herding behavior.

Behavioral finance. Anomalies are apparent deviations from the efficient market hypothesis, identified by persistent abnormal returns that differ from zero and

are predictable in direction. If use 'normal return' from an asset pricing model (e.g. CAPM) then perhaps it has to do with the model. Or perhaps it's just due to more risk. Small samples, survivorship, data mining, temporary disequilibria.

Momentum or trending effects - often last 2-5 years. Can be explained by loss of risk aversion, hindsight (regret on not riding on momentum), and availability (based on traders' recent experiences and therefore recency bias; many traders just extrapolate though some others expect mean reversion) biases.

In bubbles, investors often exhibit symptoms of overconfidence; overtrading, underestimation of risks, failure to diversify, and rejection of contradictory information. The overconfidence and excessive trading that contribute to a bubble are linked to confirmation bias and self-attribution bias. Selling for a gain appears to validate a good decision in an original purchase and may confer a sense of pride in locking in the profit. This dynamic fuels overconfidence that can lead to poor decisions. Regret aversion can also encourage investors to participate in a bubble, believing they are "missing out" on profit opportunities as stocks appreciate. As a bubble unwinds, markets may underreact because of anchoring when investors do not sufficiently update their beliefs. The early stages of unwinding a bubble can involve investors in cognitive dissonance, who ignore losses and attempt to rationalize flawed decisions. Eventually, investors capitulate, which accelerates price declines.

Growth versus value stocks. Value stocks are typically characterized by low price-to-earnings ratios, high book-to-market equity, and low price-to-dividend ratios. Growth stock characteristics are generally the opposite of value stock characteristics. For example, growth stocks are characterized by low book-to-market equity, high price-to-earnings ratios, and high price-to-dividend ratios.

Other concepts. Halo effect within a company. Home effect on domestic stocks.

9.7 Introduction to risk management

Risk management is the process by which an organization or individual defines the level of risk to be taken, measures the level of risk being taken, and adjusts the latter toward the former, with the goal of maximizing the company's or portfolio's value or the individual's overall satisfaction, or utility. Not about eliminating risk, it's about managing it. Good risk management does not prevent losses, but provides a full top-to-bottom framework that rigorously informs the decision-making process—before, during, and after a risk event.

Risk governance is the foundation for risk management. Top-down process and guidance that directs risk management activities to align with and support the goals of the overall enterprise. It typically emanates from a board of directors with fiduciary obligations and risk oversight and who prescribe goals and authorities. Sets risk context.

Risk governance is not about specifying methods to mitigate risk at the business line level. Rather, it is about establishing an appropriate level of risk for the entire enterprise. Specifics of dealing with risk fall under risk management and the risk infrastructure framework.

Risk tolerance. Set at risk governance level. Looks at what the business is trying to do (returns, liquidity) and identify risk drivers and then tolerance. Looks at BOTH inside view of what level of loss would leave the organization unable to meet critical objectives AND the outside view of what the sources of risks are. Pursue core goals but peripheral business risks need to be limited.

Risk budgeting. Determines how the risk tolerance is taken concretely. Four well-known single-dimension measures that are often used are standard deviation, beta, value at risk (VaR), and scenario loss, but there are many others. The budgeting of risk should result in an approach, whether explicit or not, of choosing to invest where the return per unit of risk is the highest. Or compare to market risk-return intensity somehow. Risk budgeting does not require nor prohibit hedging, although hedging is available as an implementation tool to support risk budgeting and overall risk governance.

The risk management committee is a part of the risk governance structure at the operational level—as such, it does not approve the governing body’s policies.

Risk infrastructure refers to the people and systems required to track risk exposures and perform most of the quantitative risk analysis to allow an assessment of the organization’s risk profile. The risk management infrastructure identifies, measures, and monitors risks (among other things).

Risk identification. The three primary types of financial risks are market risk, credit risk, and liquidity risk. Market risk is the risk that arises from movements in interest rates, stock prices, exchange rates, and commodity prices.

Non-financial risks. Default, settlement risks, legal/regulatory/reputational risks, model (e.g. forecast) risk, tail risk, operational risk (e.g. rogue trader, accounting fraud, weather disasters, cybersecurity, poor accounting practices), solvency risk.

Interactions between risks. Risks interact. E.g. market risk begets credit risk. Buying a financial option, e.g. on equity stock, which is somehow correlated with the counterparty is ‘wrong-way risk’. The uncertainty about death creates two risks: mortality risk and longevity risk.

Measuring and modifying risk.

The Greeks. The sensitivity of the derivative price to a small change in the

value of the underlying asset is called the delta. Yet delta is limited to capturing only small changes in the value of the underlying. Large changes are captured by the concept of gamma. Whereas delta is a first-order risk, gamma is considered a second-order risk because it reflects the risk of changes in delta. Some derivatives, such as options, are also sensitive to changes in the volatility of the underlying. This risk is captured by a concept called vega, which is a first-order measure of the change in the derivative price for a change in the volatility of the underlying. Derivatives are also sensitive to changes in interest rates, which are reflected in a measure called rho. Most options have relatively low sensitivity to interest rates.

Other asset classes have other risks: bond durations.

Value at risk or VaR is a measure of the size of the tail of the distribution of profits on a portfolio or for an organization. A VaR measure contains three elements: an amount stated in units of currency, a time period, and a probability. VaR is an extreme loss metric, with, e.g. a 5 percent MINIMUM loss of 30 million USD over 12 months. The maximum loss is the entire equity of an organization or the entire value of a portfolio. Conditional VaR or CVaR is a common tail loss measure, defined as the weighted average of all loss outcomes in the statistical distribution that exceed the VaR loss. Another tail risk metric in the credit risk space that is analogous to CVaR is expected loss given default, which answers the question for a debt security, "If the underlying company or asset defaults, how much do we lose on average?". Scenario analysis and stress testing used to complement VaR.

Risk modification. Four broad categories of risk modification: risk prevention and avoidance, risk acceptance, risk transfer, and risk shifting. If the risk measurement process shows that the risk exceeds the acceptable level, there are three approaches to managing the risk: self-insuring, risk transfer, and risk shifting.

Risk acceptance is similar to self-insurance. An organization choosing to self-insure may set up a reserve fund to cover losses. Buying insurance is a form of risk transfer and using derivatives is a form of risk-shifting, not risk acceptance.

Risk modification. Risk transfer: e.g. insurance. Insurance works by pooling risk - including rare risks through surety, fidelity bonds. Risk shifting: e.g. derivatives. Derivatives shift through probability distributions and across parties.

Example. If a UK investor buys US securities at a certain price and then 100% hedges the FX through a forward, this still will not eliminate the risk completely. As time progresses, the US security will fluctuate in value and by the end of the tenor of the forward contract will be worth less or more in USD than the amount of USD you have sold. You have ended up under- or overhedged.

In fact, because the value of the US security is to increase with your expected return, it is logical to expect to be underhedged. The amount to which you are under- or overhedged is subject to currency risk.

10 Ethical and professional standards

10.1 Ethics and trust

Ethics encompasses a set of moral principles and rules of conduct that provide guidance for our behavior. Moral principles or ethical principles are beliefs regarding what is good, acceptable, or obligatory behavior and what is bad, unacceptable, or forbidden behavior.

Codes of ethics - principles. Standards of conducts - more specific. Some organizations will adopt only a code of ethics, which communicates the organization's values and overall expectations regarding member behavior. Others may adopt both a code of ethics and standards of conduct. Standards of conduct identify specific behavior required of community members and serve as benchmarks for the minimally acceptable behavior of community members.

Modern professional standards distinguish professions from the craft guilds and trade bodies that were established in many countries. In particular, the requirement for members of professions to uphold high ethical standards is one clear difference. Another difference is that trade bodies do not normally have a mission to serve society or to set and enforce professional conduct rules for practitioners. Code and standards not enough; need good judgment and practice.

How professions establish trust. Normalize practitioner behavior, provide service to society, client-focused, high competency, monitor conduct, etc. At a minimum, professionals must act in the best interest of the client, exercising a reasonable level of care, skill, and diligence. The obligation to deliver a high standard of care when acting for the benefit of another party is called fiduciary duty.

CFA Institute candidates and charterholders must meet the highest standards among those established by CFA Institute, regulators, or the employer.

Investment management professionals should be interdependent with firms to maintain trust. Employers and regulators have their own standards and practices, which may differ from regulations and standards set by professional bodies.

CFA updates knowledge through practice analysis.

Environment or social factors are a major determinant why some people for-

get the long term and focus on the short term. Social psychologists find moral people can act unethically based on the situation.

Ethical decision-making framework. Long-term, look at issue from multiple perspectives; look at issue from perspective of multiple stakeholders to benefit them.

- **Identify:** Relevant facts, stakeholders and duties owed, ethical principles, conflicts of interest
- **Consider:** Situational influences, additional guidance, alternative actions
- **Decide and act**
- **Reflect:** Was the outcome as anticipated? Why or why not?

Situational factors - indirect, not direct, e.g. loyalty to employer, prestige/money rewards. Duties - client first, unless market integrity is at stake, then employer and/or others.

10.2 Professional Conduct Program

The 2 principles of the Rules of Procedure for disciplinary proceedings are (1) Confidentiality of proceedings, and (2) Fairness to the member.

In a Professional Conduct Program investigation, there is only a disciplinary review committee (DRC) when there is contention. The CFAI can get information from media for a PCP investigation.

The CFAI board of directors has oversight but only oversight over the PCP. In the PCP, Rules of Procedure + Bylaws are used.

Can't disclose confidential information, unless specifically permitted by law, even to the PCP.

CFAI professional conduct penalties aren't monetary. But they can include public censure, revocation.

10.3 Code of Ethics and Standards of Professional Conduct

Members and Candidates have a duty of loyalty to their clients and must act with reasonable care and exercise prudent judgment. Members and Candidates must act for the benefit of their clients and place their clients' interests before their employer's or their own interests.

Members and Candidates must act for the benefit of their employer and not deprive their employer of the advantage of their skills and abilities, divulge confidential information, or otherwise cause harm to their employer.

Members and Candidates must not accept gifts, benefits, compensation, or consideration that competes with or might reasonably be expected to create a conflict of interest with their employer's interest unless they obtain written consent from all parties involved.

Members and Candidates must make reasonable efforts to ensure that anyone subject to their supervision or authority complies with applicable laws, rules, regulations, and the Code and Standards.

Disclosures. Must disclose to client prospects of investment, risks, costs, process for selecting securities.

10.4 SPC I: Professionalism

Competence.

Conflicts of interest. Should not pay others - even if personal funds aligned with other causes (see page 255). Gifts from clients (those who hire you) is deemed to be less of a concern, but pay attention to whether this means you favor that client over others. Don't be pressured by general market trends - don't let analysis be affected by this.

Misrepresentation. Plagiarism - don't have to cite the conduit of information; better to just look at primary source in case the conduit conducted an error. Inaccurate promotion of one's products/services - The Code and Standards stress protecting the reputation of the firm and the sustainability and integrity of the capital markets. Misrepresenting the quality and risks associated with the investment pool may lead to negative consequences for others well beyond the direct investors. Don't provide investment guarantees or 'expectations'.

Misconduct. Although CFA Institute discourages any sort of unethical behavior by members and candidates, the Code and Standards are primarily aimed at conduct and actions related to a member's or candidate's professional life, e.g. environmental civil disobedience actions in personal life even if the person works in ESG professionally and has an employment contract that allows termination if arrested. Further, the Code and Standards are not automatically violated with every violation of an employer's employment contract unless the violations relate to conduct addressed in the Code and Standards.

Recommended procedures. I(A) Knowledge and the law

- Stay informed, review procedures, maintain current files.
- Dissociate from conduct - and document
- Use legal counsel and compliance department

- Get firm to: adopt code of ethics, provide information on applicable laws, establish procedure for violations.
- Even though this standard is about 'knowledge of the LAW', it includes regulations and rules. If a colleague uses CFA credentials without being CFA before a client, then must dissociate.

Recommended procedures. I(B) Independence and objectivity

- Establish policies to ensure analytical independence
- Ensure integrity for analyst compensation
- Create restricted list for companies that don't want negative coverage
- Cost arrangements - issuers shouldn't pay for trips
- Limit gifts, consider limit
- Adopt policies on employee purchases of equity, IPOs, with prior approval for IPOs.
- Strict limits on employee purchases of private placements.
- Review procedures so that PMs and analysts follow policies
- Policies so that analysts aren't controlled by departments that could compromise independence
- Appoint officer to oversee code and regulations
- Tender offer information can be material nonpublic information

Recommended procedures. I(C) Misrepresentation

- Designate people to speak on behalf of the firm
- Firm can provide examples of presentations and qualifications to show how it can be ethically described
- Each member should prepare qualification summary of capabilities
- Develop policy to verify third party information
- Maintain webpages
- No plagiarism: keep records, attribute

Recommended procedures. I(D) Misconduct

- Adopt a firm code of ethics
- List violations and disciplinary actions
- Check potential employees for prior infractions of law

Recommended procedures. I(E) Competence

- Continuing education, conferences, webinars, joining industry organizations, etc.

10.5 SPC II: Integrity of capital markets

Trading on material non-public information. Information which COULD - not just does - affect security value; even information which is of very general tenor or atmosphere descriptions which are not public and material. Information is material if it would significantly alter the total mix of information currently available about a security in such a way that the price of the security would be affected. The specificity of the information, the extent of its difference from public information, its nature, and its reliability are key factors in determining whether a particular piece of information fits the definition of material. Also consider reliability of source, and whether the information will have a clear impact on the price. Information disclosed by a company to a select number of investors or analysts doesn't make the information public.

A prohibition on all types of proprietary activity when a firm comes into possession of material nonpublic information is not appropriate. Firms that continue market-making activity while in the possession of material nonpublic information should, however, instruct their market makers to remain passive with respect to the market—that is, to take only the contra side of unsolicited customer trades. In risk-arbitrage trading, the case for a trading prohibition is more compelling than it is in the case of market making. The impetus for arbitrage trading is neither passive nor reactive, and the potential for illegal profits is greater than in market making.

Note: new or changing equity or debt ratings issued by a third party (e.g., sell-side recommendations and credit ratings) can be material non-public information.

Mosaic theory. Material public and non-material non-public information.

There are often rumors and whisper numbers before a release of any kind. A text message from another trader would most likely be considered market noise. Unless the trader had an ongoing business relationship with the public firm.

Market manipulation includes (1) the dissemination of false or misleading information and (2) transactions that deceive or would be likely to mislead market participants by distorting the price-setting mechanism of financial instruments.

Pump-priming is market manipulation, though not in all cases, e.g. an exchange enters into contracts that it wants to sell to show it has liquidity.

Recommended procedures. II(A) Material public information.

- Make reasonable efforts to achieve public dissemination - persuade company; if not possible, then tell supervisory and compliance personnel; don't induce others to act on the private information

- Adopt compliance procedures. Procedures should fit nature and size of company.
- Disclose information to all clients fairly
- Issue press releases prior to meetings; script remarks to prevent unintended disclosures; if unintended disclosure, then issue press release to make publicly available.
- Adopt firewall: control between departments (or use legal/compliance dept), review employee trading based on watch lists, documentation of procedures, heightened review if possess material nonpublic information
- Physical documents separated, personnel separated
- A single compliance officer to look over the walls if information needs to cross the wall
- Procedures to watch over employee trading, esp. if beneficiary, employees should report
- Keep records
- Updated training for employees on procedures
- Prop traders - no blanket ban. E.g. if market maker for small firm, withdrawing is itself a tip to the market. Instead, just passively take contra trades.
- Overhearing information about a tender offer

Recommended procedures. II(B) Market manipulation

- Information-based manipulation
- Transaction-based manipulation

10.6 SPC III: Duties to clients

Client interests are priority. Duty of loyalty and duty to exercise reasonable care. If in 'custody' of funds (direct or indirect access to client funds or assets), then heightened duties of care. If only execution trading, duty is less, and have to consider integrity of markets (e.g. liquidate if not enough margins - see page 266).

Fiduciary duty = loyalty, care, prudence. When the manager is responsible for the portfolios of pension plans or trusts, however, the client is not the person or entity who hires the manager but, rather, the beneficiaries of the plan or trust. The duty of loyalty is owed to the ultimate beneficiaries. The duty can be to the investing public at large, instead of identifiable individuals.

Soft commissions and fiduciary duty. Conflicts may arise when an investment manager uses client brokerage to purchase research services, a practice commonly called “soft dollars” or “soft commissions.” A member or candidate who pays a higher brokerage commission than he or she would normally pay to allow for the purchase of goods or services, without corresponding benefit to the client, violates the duty of loyalty to the client.

Directed brokerage is not a violation of fiduciary duty, but must seek best price and best execution. A client will direct a manager to use the client’s brokerage to purchase goods or services for the client, a practice that is commonly called “directed brokerage.”

Fair dealing. Must fairly allocate trades to fund accounts.

Suitability. Make reasonable inquiries into client’s situation and only make recommendations after, and update regularly. Judge in total context. Execution traders, sell-side analysts might not need to consider this. Sometimes a client will ask for a trade that is inconsistent with the IPS - if small impact, then go ahead; if larger impact, discuss and update IPS; if the client insists on not changing IPS, may need to reconsider client relationship. If managing a fund, e.g. ETF, then no client but there is an investment mandate - follow the mandate.

Performance presentation. When presenting performance information, reasonable efforts to make sure it is fair, accurate and complete. Use GIPS if applicable (Global Investment Performance Standards).

Preservation of confidentiality. Duty extends even if no longer client. Follow law - if applicable law requires members and candidates to maintain confidentiality, even if the information concerns illegal activities on the part of the client, members and candidates should not disclose such information.

Recommended procedures. III(A) LPS

- Inform clients of all forms of manager compensation
- Disclose all conflicts
- Keep client accounts updated - quarterly
- If in doubt, ask for client approval
- Regular account reviews
- Vote proxies in best interests of ultimate beneficiaries
- Seek best execution

Recommended procedures. III(B) Fair dealing.

- Can also provide discount if referred client (+ pay for referrals is allowed too).
- Have written compliance guidances
- Keep lists of clients
- Disseminate recommendations to customers + firm members simultaneously.
- Don't have to disseminate recommendations to all customers, but this decision should be based on suitability and interest, not favoritism.
- Guidelines for pre-recommendation publication behavior within the firm
- Disclose to clients how trade is allocated (e.g. block trades)
- Disclose to clients levels of service based on fees, not favoritism

Recommended procedures. III(C) Suitability

- Write IPS and review
- Develop and apply suitability tests in written policies
- PMs should invest in accordance with fund mandate.
- Those with advisory relationships need to know about client constraints, etc.

Recommended procedures. III(D) Performance presentation

- Use different performance presentations for different audiences - different audiences have different sophistication
- State when simulated results used

Recommended procedures. III(E) Confidentiality

- Encourage employers to develop confidentiality procedures
- Review regularly because technology changes
- Let client know not all firm-sponsored resources are good means for communication

10.7 SPC IV: Duties to employers

Loyalty. Clients before employers but must consider employer interests. Employers not obligated to follow CFA code and standards. Should not enter into activities that compete with employer. If going to run competitive business, get consent from employer - types of services they will render to prospective independent clients, the expected duration of the services, and the compensation for the services.

Additional compensation arrangements. Conflicts of interest from gifts for e.g. Must obtain WRITTEN consent.

Responsibilities of supervisors. Must be able to handle job before taking it rather than being significantly trained on it on the job.

Recommended procedures. IV(A) Loyalty

- Develop procedures
- Member should understand constraints and non-compete agreements if any
- Follow employer procedures
- Loyalty to employer - only if there is a conflict or deprives the employer
- Taking something from an employer - e.g. copying files as an intern violates this SPC

Recommended procedures. IV(B) Additional compensation

- When disclosing, note the details (e.g. terms, and confirmation by other party).
- Teaching an economics course (paid) in spare time doesn't require permission from employer. No conflict.
- Additional compensation - only if there is a conflict or deprives the employer
- ACAs recommended to be reported to firm + supervisor
- ACAs recommended to be reported quarterly

Recommended procedures. IV(C) Supervisors

- Encourage giving code to clients
- Draft procedures and expected conduct and disseminate
- If violation, respond promptly and investigate and review procedures

- Supervisors should conduct professional conduct review as part of employee performance
- Provide code of ethics to clients
- Supervisors AND firms must look closely at their incentive structure to determine whether the structure encourages profits and returns at the expense of ethically appropriate conduct

10.8 SPC V: Investment analysis

Diligence and reasonable basis. Primary research - done by the person; secondary research - someone else in team; third-party.

Communication with clients. Disclose outsourced fund managers, disclose leverage because it's high risk. Disclose limiting factors such as liquidity and capacity.

Record retention. The retention requirement applies to decisions to buy or sell a security as well as reviews undertaken that do not lead to a change in position. In the absence of regulatory guidance or firm policies, CFA Institute recommends maintaining records for at least seven years. The responsibility to maintain records that support investment action generally falls with the firm rather than individuals.

Recommended procedures. V(A) Diligence and reasonable basis.

- Policy to ensure diligence and reasonable basis. Have someone appointed to review before publication.
- Detailed guidelines for what is diligent and reasonable
- Develop measurable criteria, e.g. accuracy of recommendations
- Detailed guidelines for computer models
- Measurable criteria for outside information
- Criteria for selecting outside managers

Recommended procedures. V(B) Client communications

- Case-by-case basis
- Maintain records to backup research

Recommended procedures. V(C) Record retention

- Keep records for at least 7 years
- Recommended procedure is that the investment firm, not the analysts, retain records

10.9 SPV VI: Conflicts of interest

Full and fair disclosures of conflicts of interest - must be prominent. Payments FROM and TO clients.

For the purposes of Standard VI(A), members and candidates beneficially own securities or other investments if they have a direct or indirect pecuniary interest in the securities, have the power to vote or direct the voting of the shares of the securities or investments, or have the power to dispose or direct the disposition of the security or investment.

A change in bonus structure (e.g. from annual to quarterly performance) - clients should be informed under the SPC.

Although conflicts of interest exist, nothing is inherently unethical about individual managers, advisers, or mutual fund employees making money from personal investments as long as (1) the client is not disadvantaged by the trade, (2) the investment professional does not benefit personally from trades undertaken for clients, and (3) the investment professional complies with applicable regulatory requirements. Consider investment professional reporting requirements, blackout or restriction periods, allowing clients to trade first.

Referral fees. Must disclose the nature of the consideration or benefit—for example, flat fee or percentage basis, one-time or continuing benefit, based on performance, benefit in the form of provision of research, or other noncash benefit—together with the estimated dollar value. Consideration includes all fees, whether paid in cash, in soft dollars, or in kind.

According to Standard III(A) Loyalty, Prudence, and Care, the CFA Program would be considered a personal or firm expense and should not be paid for with the fund's brokerage commissions. Soft dollar accounts should be used only to purchase research services that directly assist the investment manager in the investment decision-making process, not to assist the management of the firm or to further education.

Recommended procedures. VI(A) Avoid or disclose conflicts

- avoid or fully disclose to clients, potential clients, and employers all actual and potential conflicts of interest.
- Disclose in prominent and in plain language
- If own beneficially stocks, best to avoid but might be too burdensome, so disclose.

Recommended procedures. VI(B) Priority of transactions

- Limited participation in IPOs - preclear if employee to purchase

- Restrictions on private placements
- Blackout periods
- Reporting requirements: disclose beneficial ownership, provide duplicate broker transactions, preclear if potential conflict
- Complete disclosure to client

VI(C) Referral fees.

- Must disclose both the nature of the consideration and the estimated dollar value
- Must disclose to their employer, clients, and prospective clients, as appropriate, ANY compensation, consideration, or benefit received from or paid to others for the recommendation of products or services.
- Must disclose even if referred from a brokerage that provides best price and execution when the investment advisor refers those clients back to that brokerage for trades.

10.10 SPV VII: CFA member and candidate responsibilities

Members and Candidates must not engage in any conduct that compromises the reputation or integrity of CFA Institute or the CFA designation or the integrity, validity, or security of CFA Institute programs.

When referring to CFA Institute, CFA Institute membership, the CFA designation, or candidacy in the CFA Program, Members and Candidates must not misrepresent or exaggerate the meaning or implications of membership in CFA Institute, holding the CFA designation, or candidacy in the CFA Program.

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Recommended procedures.

- Don't use or tag with CFA designation if anonymous
- Someone else using the CFA designation falsely is not a VI violation - it's a I(A) violation and must dissociate.

10.11 Relationships between SPC sections

II(B) Independence and objectivity are about NOT doing something potentially against clients. VI(A) Avoid or disclose is about how to address independence and objectivity if it happens, i.e. avoid or disclose. III(A) LPC is about DOING something positive for clients.

III(B) Fair dealing is a general rule to be fair to clients, while VI(B) Priority of transactions is a specific fair dealing rule.

10.12 Introduction to GIPS

GIPS for firms, asset owners, and verifiers. Read GIPS for firms for Level 1 exam. Any firm that MANAGES ASSETS is eligible. Asset owners can comply with GIPS same way as firms if they compete for business.

Consultants cannot make a claim of compliance unless they actually manage the assets for which they are making a claim of compliance. They can claim to endorse the GIPS standards and/or require that their investment managers comply with the GIPS standards. Similarly, software (and the vendors who supply software) cannot be “compliant.” Software can assist firms in achieving compliance with the GIPS standards (e.g., by calculating performance in a manner consistent with the calculation requirements of the GIPS standards), but only a firm managing assets can claim compliance once the firm has satisfied all applicable requirements of the GIPS standards.

Compliance is a firm-wide process that cannot be achieved on a single product or composite. Must place all products or accounts in one of the composites.

Composites. Portfolios managed according to a similar investment mandate, objective, or strategy.

Firm. The GIPS standards state “The firm should adopt the broadest, most meaningful definition of the firm. The scope of this definition should include all geographical (country, regional, etc.) offices operating under the same brand name, regardless of the actual name of the individual investment management company.”

Discretion. The firm’s definition of discretion establishes criteria to judge which portfolios must be included in a composite and is based on the firm’s ability to implement its investment strategies. If documented client-imposed restrictions interfere with the implementation of the intended strategy to the extent that the portfolio is no longer representative of the strategy, the firm may determine that the portfolio is non-discretionary. Non-discretionary portfolios must not be included in a firm’s composites.

Verification. Voluntary. Independent third-party only. Verification is performed with respect to an entire firm, not on specific composites or pooled funds. Verification does not ensure the accuracy of any specific performance report.

Chapters:

- Fundamentals of compliance: firm-wide compliance, laws and regulations, etc.
- Input data: emphasizes importance of accurate data input
- Calculation methodology: How to calculate time-weighted and money-weighted returns
- Composite and pooled fund maintenance: definition and how to create composites (fee paying discretionary accounts) and pooled funds (e.g mutual funds, ETFs)
- Composite time-weighted report
- Composite money-weighted report
- Pooled fund time-weighted report
- Pooled fund money-weighted report
- GIPS advertising: how to advertise performance

Dividends in the binomial model. Escrow method. Security price without dividends:

$$\hat{S} = S - \gamma$$

and at expiration

$$\hat{S}_T = S_T$$

Most option contracts don't provide protection against dividends. With dividend stocks, often advantageous to exercise BEFORE the ex-dividend date for American options. That's one reason why American options can be worth more than European options.

Interest rate options. From the bond value, get the interest rate. The strike (exercise) price is based on the interest rate. This is different from standard 2-period binomial model. Calculate $c^+ = PV(\pi c^{++} + (1 - \pi)c^{+-})$, etc. where the PV is based on the BOND VALUE, e.g. 0.96, for c^+ .

BSM model. Extend multiperiod model to infinitesimally continuous time. European options. No discontinuities. Security price RETURNS are lognormally distributed - e.g. if a stock price drops from 100 to 90, it is the log return $\ln(0.9) < 0$. Volatility σ is annualized standard deviation of continuously compounded percentage change in the underlying, i.e. log return.